

Advanced Microeconomics Theory

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Fall 2023

Contents

1	Choice Theory	2
1.1	Preference-Based Approach	2
1.2	Choice-Based Approach	5
2	Consumer Theory	8
2.1	Setups	8
2.2	Utility Representation	9
2.3	Utility Maximizing Problem	13
2.4	Expenditure Minimization Problem	24
2.5	Duality and Comparative Statics	28
2.6	Consumer Welfare	33
3	Production Theory	37
3.1	Setups	37
3.2	Profit Maximization and Rationalizability	40
3.3	Profit Maximization Problem	47
3.4	Cost Minimization Problem	51
4	Comparative Statics Analysis	56
4.1	Univariate Comparative Statics	56
4.2	Multivariate Comparative Statics	61
5	Uncertainty	65
5.1	Setups	66
5.2	Expected Utility Representation	67
5.3	Measures of Risk	70
5.4	Comparison of Risky Prospects	74
5.5	Comparative Statics Under Risk	77
6	General Equilibrium	82
6.1	Pure Exchange Economy	84
6.2	Allocation	87
6.3	General Equilibrium with Production	90

Chapter 1

Choice Theory

The utility-maximization approach to choice has several characteristics that help account for its long and continuing dominance in economic analysis:

- Normative usefulness: policy-making; individuals' choices v.s. government's welfare criterion; modern democratic values.
- Positive predictions: comparative statics predictions.
- Wide scope.
- Compactness: make empirical predictions from a relatively sparse model of the choice problem, just a description of the chooser's objectives and constraints.

1.1 Preference-Based Approach

Rational choice theory starts with the idea that individuals have preferences and optimize their utilities according to preferences. The primary task is to formalize what that means and precisely what it implies about the pattern of decision-making we should observe.

1.1.1 Preference Relation

Definition 1.1.1: (Weak) Preference Relation

Let X be a set of possible choices. Consider a *weak preference relation* $\succeq$ over the set X , as a binary relationship:

$$x \succeq y \iff \text{"}x \text{ is at least as good as } y\text{"}$$

Remark.

The weak preference relation $\succeq$ implies the associated strict preference relation $\succ$ and indifference relation $\sim$:

- x is *strictly preferred* to y , or $x \succ y$, if $x \succeq y$ but not $y \succeq x$.
- x is *indifferent* to y , or $x \sim y$, if $x \succeq y$ and $y \succeq x$.

After the definition of weak preference relation, we need to make assumptions about rationality, or definition of rational preference.

The first is *completeness*, which means an agent would never be clueless when faced with two choices.

Definition 1.1.2: Completeness

A preference relation $\succeq$ on X is *complete*, if for all $x, y \in X$, either $x \succeq y$, or $y \succeq x$, or both.

The second is *transitivity*, which means that an agent's weak preference cannot cycle unless among choices that are indifferent.

Definition 1.1.3: Transitivity

A preference relation $\succeq$ on X is *transitive*, if whenever $x \succeq y$ and $y \succeq z$, $x \succeq z$.

From completeness and transitivity, we have the following corollaries:

Corollary 1.1.4

1. While the definition of transitivity involves only 3-choice cycles, it also extends to all n -object cycles, i.e., that it implies that for any n choices $x_1, x_2, \dots, x_n \in X$ such that $x_1 \succeq x_2, x_2 \succeq x_3, \dots, x_{n-1} \succeq x_n$, we must have $x_1 \succeq x_n$. (Hint: use induction on n .)
2. Transitivity of a weak preference relation $\succeq$ implies transitivity of the associated strict preference relation $\succ$ and the indifference relation $\sim$.

Remark.

1. Preference relation is defined upon $\succsim$ because it is natural to assume the weak preference relation $\succsim$ is complete.
2. The assumption of transitivity is inconsistent with certain "framing effects".

Completeness and transitivity allow us to formalize what we mean by rationality.

Definition 1.1.5: Rationality

A preference relation $\succsim$ on X is *rational* if it is both complete and transitive.

1.1.2 Choice Rule

Given preferences, we should define the agent's *choice rule* (given an "opportunity set" $B \subseteq X$) induced by preference relation $\succeq$ as

$$C_{\succeq}(B) = \{x \in B \mid x \succeq y, \forall y \in B\},$$

which is the set of items in B the agent likes at least as much as any of the other alternatives in B ; or say, most preferred.

Remark.

1. $C_{\succeq}(B)$ may contain more than one element.
2. $C_{\succeq}(B)$ might be empty. If B is finite and non-empty, then $C_{\succeq}(B)$ is non-empty.
 - As an example for $C_{\succeq}(B)$ to be empty, define a preference relation $x \succeq y \iff x \geq y$. Take $X = [0, +\infty)$ and $B = (0, 1)$. Then clearly there is no “most preferred” option.
 - To guarantee the existence of choices from infinite choice sets, we will later make some technical assumptions (such as compactness of choice sets and continuity of preference relation).

Proposition 1.1.6

Suppose $\succeq$ is complete and transitive. Then, for every finite non-empty set B ,

$$C_{\succeq}(B) \neq \emptyset.$$

Proof for Proposition.

Proceed by mathematical induction on the number of elements of B .

- $|B| = 1$, say $B = \{x\}$.
 - By completeness, $x \succeq x$, so $x \in C_{\succeq}(B)$. $C_{\succeq}(B) \neq \emptyset, \forall |B| = 1$.
- Fix $n \geq 1$ and suppose that for all sets B with exactly n elements, $C_{\succeq}(B) \neq \emptyset$. Next we move on to examine the case of $|B| = n + 1$.
 - Consider any B_n such that $|B_n| = n$. Since $C_{\succeq}(B_n) \neq \emptyset$, say $C_{\succeq}(B_n) = x^*$. Let $B = B_n \cup \{x_{n+1}\}$.
 - By completeness, we have only two (not mutually-exclusive) possibilities:
 - * If $x^* \succeq x_{n+1}$, then by definition $x^* \in C_{\succeq}(B)$, so $C_{\succeq}(B) \neq \emptyset$.
 - * If $x_{n+1} \succeq x^*$. Since $x^* \in C_{\succeq}(B_n)$, by definition, $x^* \succeq y, \forall y \in B_n$. By transitivity, this implies $x_{n+1} \succeq y, \forall y \in B$. Therefore, $x_{n+1} \in C_{\succeq}(B)$, so $C_{\succeq}(B) \neq \emptyset$.
- Hence, for every set B with exact $n + 1$ elements, $C_{\succeq}(B) \neq \emptyset$. By the principle of mathematical induction, it follows that for every finite set B that is non-empty, $C_{\succeq}(B) \neq \emptyset$.

1.2 Choice-Based Approach

Much empirical work reasons in the reverse way contrary to economic theories (in preference-based approach): it looks at individuals' choices and tries to “rationalize” those choices. That is, to figure out whether the choices are compatible with preference maximization and, if so, what they imply about those preferences. Under choice-based approach, choice rule is the primitive object of the theory.

Definition 1.2.1: Choice Rule

Let $\mathcal{B}$ be the set of all nonempty subsets of X ($\mathcal{B} = 2^X \setminus \emptyset = \{B \neq \emptyset : B \subset X\}$). A *choice rule* is a function $C : \mathcal{B} \rightarrow \mathcal{B}$ with the property that for all $B \in \mathcal{B}$, $C(B) \subseteq B$.

Remark.

- $\mathcal{B}$ is the set of all nonempty **subsets** of X , which means all possible set of available choice(s) the agent is facing. And the choice rule C is a mapping from $\mathcal{B}$ to $\mathcal{B}$, which means that the agent is choosing from a set of available choice(s) to pick his most preferred choice(s), also a subset of X .
- Here we assume that we can see the agent choose from *all* possible subsets of X , and that the agent reports *all* of his optimal choices from a given opportunity set.

By definition of choice rule, we do not impose any assumptions or restrictions on the rule or underlying preference relation. We are then interested in two questions:

- If this rule comes from maximizing some underlying preferences, what can we infer about these preferences?
- Is this choice rule consistent with the maximization of *some* complete and transitive preference relation (i.e., *rationalizable*)?

Consider the first question first. Suppose that choice rule C is consistent with the maximization of some preference relation $\succsim$, i.e., $C(\cdot) = C_{\succsim}(\cdot)$. Then, observing for some $A \subseteq X$ that $y \in A$ and $x \in C(A)$ (i.e., x is chosen when y is available) allows us to infer that $x \succsim y$. This implies that for any $B \subseteq X$ such that $x \in B$ and $y \in C(B)$, we must also have $x \in C(B)$ (Indeed, we have $x \succsim y$ and $y \succsim z$ for all $z \in B$, and so by transitivity $x \succsim z$ for all $z \in B$). By a symmetric argument, we should then have $y \in C(A)$. Thus, any rationalizable choice rule must have the following property (as a *necessary* condition):

Definition 1.2.2: HARP

A choice function $C : \mathcal{B} \rightarrow \mathcal{B}$ satisfies *Houthaker's Axiom of Revealed Preference (HARP)* if, whenever $x, y \in A \cap B$, and $x \in C(A)$ and $y \in C(B)$, we have $x \in C(B)$ and $y \in C(A)$.

In words, HARP says that if choices x, y are both available in two choice experiments, and x is chosen in one experiment and y in the other, then both x and y must be chosen in both experiments. HARP guarantees that there is no (obvious) inconsistency in the agent's choices.

It has been argued that HARP is a necessary condition for a choice rule to be rationalizable. Turns out that HARP is also sufficient for rationalizability:

Proposition 1.2.3

Suppose $C : \mathcal{B} \rightarrow \mathcal{B}$ is nonempty-valued. Then there exists a rational (complete and transitive) preference relation $\succsim$ on X such that $C(\cdot) = C_{\succsim}(\cdot)$ if and only if C satisfies HARP.

Proof for Proposition.

- “Only if” part: As argued before, HARP is a necessary condition for rational relation’s existence.
- “If” part
 - Definition of revealed preference relation (by choice rule).
 Suppose choice rule C satisfies HARP. Construct the “revealed preference relation” $\succsim_C$ as follows: say that $x \succsim_C y$ if and only if there exists some $A \subseteq X$ such that $y \in A$ and $x \in C(A)$.
 - Prove that $\succsim_C$ is complete.
 Since C is nonempty-valued, pick any $x, y \in X$, we have either $x \in C(\{x, y\})$, in which case $x \succsim_C y$, or $y \in C(\{x, y\})$, in which case $y \succsim_C x$ (or both, in which case $x \sim_C y$).
 - Prove that $\succsim_C$ is transitive.
 Suppose $x \succsim_C y$ and $y \succsim_C z$, and consider $C(\{x, y, z\})$, which by hypothesis is non-empty. There are three possibilities (which are though not mutually exclusive):
 - * $x \in C(\{x, y, z\})$. Then by construction of $\succsim_C$, $x \succsim_C z$.
 - * $y \in C(\{x, y, z\})$. Then by HARP, since $x \succsim_C y$, we must also have $x \in C(\{x, y, z\})$, and so by case 1 we have $x \succsim_C z$.
 - * $z \in C(\{x, y, z\})$. Then by HARP, since $y \succsim_C z$, we also have $y \in C(\{x, y, z\})$, and so by case 2 $x \succsim_C z$.
 - Prove that $C(\cdot) = C_{\succsim_C}(\cdot)$.
 It is equivalent to show that, for all $x \in X$ and $A \in \mathcal{B}$, $x \in C(A)$ if and only if $x \in A$ and $x \succsim_C y$ for all $y \in A$.
 - * “Only if” part holds by construction of $\succsim_C$.
 - * “If” part: Take any $x \in C_{\succsim_C}(A)$. Then, $x \succsim_C y$ for all $y \in A$. Since $C(\cdot)$ is nonempty-valued, $\exists y_0 \in A$ such that $y_0 \in C(A)$. By the definition of $\succsim_C$, $\exists A_0$ such that $x, y_0 \in A_0$ and $x \in C(A_0)$. It follows that $x, y_0 \in A_0 \cap A$ and $x \in C(A_0)$ and $y_0 \in C(A)$. By HARP, $x \in C(A)$.

Remark.

1. It is *HARP* that plays a fundamental role in the proof of “if” part, that is, to endow the revealed preference relation $\succsim_C$ with characteristics of “rationality”.

2. The properties of rational choice are guaranteed only when the entire choice function $C(A)$ is observed, i.e., (i) for any given choice set, all of the agent's optimal choices are observed, not just some of them; (ii) the agent's optimal choices over all choice sets are observed. However,
 - Real data is commonly less comprehensive like that. For example, in consumer choice problems, the relevant sets A may be only budget sets that consist of affordable choices given income and prices, which is a particular subcollection of $\mathcal{B}$.
 - The first aspect of incomplete observation is virtually insolvable. For the second, to develop a theory based on more limited observations, other “axioms of revealed preference” have been developed. For example, the *weak axiom of revealed preference (WARP)* is an equivalent of HARP for choice sets restricted to budget sets described by linear prices and for choice rules restricted to be single-valued (so the conclusion of HARP can be strengthened to $x = y$). WARP is necessary for rationalizability but proves insufficient. A stronger version, *Generalized Axiom of Revealed Preference*, has been shown both necessary and sufficient for rationalizability.

Chapter 2

Consumer Theory

Choice theory assumes that a rational decision-maker selects their most preferred option from their choice set. Consumer theory can be viewed as an application of choice theory. We will focus on the *preference-based approach*, with three goals in mind:

1. Formalize the “choice set” and “preferences” of a consumer.
 - Set of alternatives: consumption set.
 - Choice set: budget set.
 - Preferences: utility representation of a preference relation.
2. Derive a consumer’s optimal choice(s) based on the information of their choice set and the preference relation.
3. Analyze the properties of optimal choices (demand).

2.1 Setups

At the beginning, a few assumptions are needed in consumer theory:

Assumption 2.1.1

1. *Perfect* information.
2. Consumers are *price takers*.
3. Prices are *linear*.
4. Goods are *divisible*.

Remark.

- By assuming consumers to be price takers, we assume that prices $\mathbf{p}$ are taken as known, fixed and exogenous, and there is no searching or bargaining for discounts.
- Linear price means that no quantity discount is offered.
- The idea that goods are divisible can be formally represented by the condition of

$x \in \mathbb{R}_+^n$. Note that the divisibility assumption does not prevent us from applying the model to situations with discrete and indivisible goods.

For simplicity, we will view the entire positive quadrant as the consumption set.

Definition 2.1.2: Consumption Set

With n goods, the *consumption set* is given by:

$$X = \mathbb{R}_+^n = \{x \in \mathbb{R}^n : x_i \geq 0 \text{ for } i = 1, 2, \dots, n\}.$$

The budget set differs from consumption set in that the former is additionally defined by the agent's income (and thus the price vector).

Definition 2.1.3: Budget Set

With n goods and income of m , given a price vector $p = (p_1, \dots, p_n)' \geq 0, p \neq 0$, the *budget set* is given by:

$$B(p, m) = \{x \in \mathbb{R}^n : p \cdot x \leq m, x \geq 0\}.$$

By reinterpreting the consumption goods and the budget, the derivation of the budget set can be extended to encompass other economic problems. For example, consumption-leisure choice, and inter-temporal choice.

2.2 Utility Representation

The preference relation is intuitive, but difficult to work with. Having a utility representation for preferences is convenient because it transforms the problem of preference maximization into a relatively familiar mathematical problem.

Definition 2.2.1: Utility Representation

A preference relation $\succsim$ on X is represented by a *utility function* $u : X \rightarrow \mathbb{R}$ if

$$x \succsim y \iff u(x) \geq u(y).$$

Under this definition, the utility function is an *ordinal* representation. The utility level per se has no economic meaning and is just a convenient mathematical tool to represent the consumer's *relative rankings* of different options.

If u represents $\succsim$, then the choice rule defined upon budget set is:

$$C(B(p, m); \succsim) = \left\{ x \mid x \text{ solves } \max_{x \in B(p, m)} u(x) \right\}.$$

An equivalent notation of such choice rule is $C_{\succsim}(B)$.

A natural question is that, given a preference relation $\succsim$, whether or not we can always find a function $u(\cdot)$ to represent $\succsim$. Luckily, in cases where preference relation is defined upon a *finite* consumption set X , the answer is yes.

Proposition 2.2.2

If X is finite, then any rational (complete and transitive) preference relation $\succsim$ on X can be represented by a utility function $u : X \rightarrow \{1, \dots, n\}$, where $n = |X|$.

Proof for Proposition.

- For $|X| = n = 1$, say $X = \{x\}$, let $u(x) = 1$. . The conclusion is trivial.
- Next, suppose that any preference can be represented as described for any set with at most n elements. We move on to consider a set X with $n + 1$ elements.
 - Since $C_{\succsim}(X) \neq \emptyset$, the set $Y = X \setminus C_{\succsim}(X)$ has no more than n elements. By doing so, we can limit our discuss and deduction back to the hypothesized case. Thus, any preference restricted to that set can be represented by a utility function $u : Y \rightarrow \{1, 2, \dots, n\}$.
 - We extend the domain of u to X by setting $u(x) = n + 1$ for each $x \in C_{\succsim}(X)$. By construction, we have $u(x) \in \{1, \dots, n, n + 1\}$ for all $x \in X$.
 - Now we show that the constructed u represents $\succsim$, i.e., for any $x, y \in X$, $x \succsim y$ if and only if $u(x) \geq u(y)$. Suppose $x \succsim y$. There are three possibilities:
 - * $x \in C_{\succsim}(X), y \in Y \iff u(x) = n + 1 \geq u(y)$.
 - * $x, y \in C_{\succsim}(X) \iff u(x) = u(y) = n + 1$, also, $u(x) \geq u(y)$.
 - * $x, y \in Y$. Then, since by construction u represents $\succsim$ on Y , $x \succsim y$ if and only if $u(x) \geq u(y)$.

However, if X is infinite, things are a bit more complicated. In general, given any rational preference relation $\succsim$ on $\mathbb{R}_+^n$, it cannot always be represented by a utility function $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$.

Example.

Consider the lexicographic preferences on the square $X = \mathbb{R}^2$, where $(x_1, x_2) \succ (y_1, y_2)$ if either (1) $x_1 > y_1$ or (2) $x_1 = y_1$ and $x_2 > y_2$. These preferences cannot be represented by a utility function, and this is also an example for which “indifference curves” do not exist, because the agent is never indifferent between any two choices.

To see this, suppose by contradiction there exists a utility representation $u(x, y)$. For any $x \in \mathbb{R}_+$, consider the interval $I(x) = (\inf_y u(x, y), \sup_y u(x, y))$. $I(x)$ is not degenerate, and $I(x_1), I(x_2)$ do not overlap for $x_1 \neq x_2$. Since rational numbers are dense, we construct a function r to pick a rational number inside $I(x)$, i.e., $r(x) \in I(x)$. Since $I(x_1), I(x_2)$ do not overlap, $r(x_1) \neq r(x_2)$ for $x_1 \neq x_2$. Thus, $r : \mathbb{R}_+ \rightarrow \mathbb{Q}_+$ is an injective function. Since $\mathbb{R}_+$ is uncountable and $\mathbb{Q}_+$ is countable, this mapping is impossible.

However, note that if we replace $X = \mathbb{R}_+^2$ with $X' = \mathbb{Q}_+^2$, the lexicographic preference relation would admit a utility representation. More generally, this idea is formalized into the proposition in countable set case.

Take a step forward. If a preference relation $\succsim$ is defined upon a countable set X (possibly infinite), then $\succsim$ admits a utility representation. We can always achieve this by construction, as it goes in the following proposition.

Proposition 2.2.3

If $X \neq \emptyset$ is countable, then any rational (complete and transitive) preference relation $\succsim$ on X can be represented by a utility function $u : X \rightarrow (0, 1)$.

Proof for Proposition.

- First, construct a mapping of utility function $u : X \rightarrow (0, 1)$.
 - Let $(x_n)_{n=1}^\infty$ be a countable enumeration of X .
 - Let $u(x_1) = \frac{1}{2}$. Consider x_{n+1} , $n \geq 1$.
 - * If $x_{n+1} \sim x_i$ for some $1 \leq i \leq n$, then define $u(x_{n+1}) = u(x_i)$.
 - * Otherwise, define

$$M_n = \max\{\max\{u(x_i) : x_{n+1} \succ x_i, 1 \leq i \leq n\}, 0\}$$

$$m_n = \min\{\min\{u(x_i) : x_i \succ x_{n+1}, 1 \leq i \leq n\}, 1\}$$

By construction $M_n > m_n$. Define $u(x_{n+1}) = \frac{M_n + m_n}{2}$.

- Second, prove that $u(\cdot)$ is a utility representation of preference relation $\succsim$.
 - Take any $y, z \in X$. Since X is countable, $\exists i, j$ such that $y = x_i$ and $z = x_j$. Without loss of generality, suppose $i \leq j$.
 - By construction, $u(x_i) = u(x_j)$ if and only if $x_i \sim x_j$. For $x_j \succ x_i$, $u(x_j) > u(x_i)$ if and only if $x_j \succ x_i$.

Intuitively, the problem with the lexicographic preference relation is that there is sudden preference reversals. For instance, we know $(3, 3) \succ (3, 2)$, but $(3, 2) \succ (x, 3)$ for x slightly less than 3. This motivates the following additional (technical) restriction on preference relations. The “continuity” restriction on preferences is a condition motivated from the simple and compelling reason that any *finite* set of observed choices that is consistent with HARP is also consistent with continuity.

Definition 2.2.4: Continuity

A preference relation $\succsim$ on $X \subseteq \mathbb{R}^n$ is *continuous* if for any sequence $\{(x^n, y^n)\}_{n=1}^\infty$ with $x^n \rightarrow x$, $y^n \rightarrow y$, and $x^n \succsim y^n$ for all n , we have $x \succsim y$.

Continuity condition implies not only that a utility representation exists, but that a *continuous* representation exists.

Proposition 2.2.5

Any complete, transitive and continuous preference relation $\succsim$ on X on $X \subseteq \mathbb{R}^n$ can be represented by a continuous utility function $u : X \rightarrow \mathbb{R}$.

Proof for Proposition.

In order to have a simple, constructive proof, we prove the proposition only for the case of a monotone preference relation $\succsim$ on $X = \mathbb{R}_+^n$.

Let $e = (1, \dots, 1)$ and consider bundles of the form $\alpha e = (\alpha, \dots, \alpha)$ where $\alpha \geq 0$. For each $x \in \mathbb{R}_+^n$, we construct a utility number as follows: $u(x) = \max A(x)$, where $A(x) = \{\alpha \in \mathbb{R}_+ : \alpha e \preceq x\}$. To see that the set $A(x)$ has a maximal point, note that the set is

- Nonempty, since $0 \in A(x)$ by monotonicity of $\succsim$;
- Closed, by the continuity of $\succsim$;
- Bounded, since by monotonicity of $\succsim$, $\alpha \leq \max\{x_1, \dots, x_n\}$ for each $\alpha \in A(x)$.

Now we show that we must have $u(x)e \sim x$.

1. $u(x)e \preceq x$. This is satisfied by construction of $u(x) \in A(x)$.
2. $u(x)e \succsim x$. For each $n \geq 1$, we have $u(x) + \frac{1}{n} \notin A(x)$, hence $(u(x) + \frac{1}{n})e \not\preceq x$, therefore by completeness of $\succsim$ we have $(u(x) + \frac{1}{n})e \succsim x$, which by continuity of $\succsim$ implies

$$\lim_{n \rightarrow \infty} (u(x) + \frac{1}{n})e = u(x)e \succsim x.$$

Now it remains to show that the constructed utility function $u(\cdot)$ has

1. Ability to represent the preference relation $\succsim$, and
2. Continuity.

For representation part, note that by transitivity $x \succsim y$ if and only if $u(x)e \succsim u(y)e$ (since $u(x)e \sim x \succsim y \sim u(y)e$), and by monotonicity of $\succsim$, this holds if and only if $u(x) \geq u(y)$. For continuity part, this is more subtle and not covered here. ■

Remark.

1. The construction in the proof specifies the utility of any bundle x by finding the point on the 45° line on the indifference curve passing through x .
 - This specification is, of course, completely arbitrary; just for mathematical convenience.
 - To reflect this arbitrariness, utility representation of preferences is *ordinal*, i.e., only the induced preference ordering of choices is meaningful, instead of the exact utility numbers assigned to them.
 - In fact, if u represents $\succsim$, then $U(\cdot) = v(u(\cdot))$ also represents $\succsim$ so long as $v : \mathbb{R} \rightarrow \mathbb{R}$ is an increasing function.

2. The thought process and the introduction of the additional assumption are the most important for this part.
 - Recall that our motivation is to come up with a tractable framework to analyze the consumer's problem. One possibility is to attach a utility level to each consumption bundle (ordinal utility representation).
 - We ask whether any preference relation can be represented by a utility function. The answer is yes, if the consumption set X is finite (or countable), but no if $X = \mathbb{R}_+^n$. We notice that the problem with the counter-example (e.g., the lexicographic preference relation) is that there are sudden preference reversals.
 - We then impose the restriction of “continuity” of preferences to rule out those unfavorable scenarios and show that any rational and continuous preference relation on $X = \mathbb{R}_+^n$ can be represented by a continuous utility function.
3. The continuity of preference relation and continuity of its utility representation is not interdependent. If a preference relation can be represented by a continuous utility function, then such preference relation must be continuous. On the other hand, a continuous preference relation can be represented by a discontinuous utility function, if you like. (That is not convenient for mathematical issues though.)

2.3 Utility Maximizing Problem

Apart from the general problem of choice theory, what makes consumer consumption decisions worthy of separate studies is its particular structure that allows us to derive economically meaningful results. The structure arises because the consumer's choice sets are assumed to be defined by certain prices and the consumer's income or wealth. With this in mind, when the preference relation can be represented by a utility function $u(\cdot)$, the Utility Maximization Problem (UMP), or Consumer Problem (CP) is reasonably defined.

Definition 2.3.1: Utility Maximization Problem

$$\begin{aligned}
 & \max_{\mathbf{x} \in \mathbb{R}_+^n} u(\mathbf{x}) \\
 & \text{s.t. } \mathbf{p} \cdot \mathbf{x} \leq m \\
 & \quad x_i \geq 0, \forall i = 1, 2, \dots, n
 \end{aligned}$$

The idea is that, the consumer chooses a vector of goods $\mathbf{x} = (x_1, \dots, x_n)' \geq \mathbf{0}$ to maximize her utility, subject to a budget constraint that given a price vector $\mathbf{p} = (p_1, \dots, p_n)' \geq \mathbf{0}$, $\mathbf{p} \neq \mathbf{0}$, she cannot spend more than her total wealth m . In other words, the CP can be formulated as:

$$C(B(\mathbf{p}, m); \succsim) = \left\{ \mathbf{x} \mid \mathbf{x} \text{ solves } \max_{\mathbf{y} \in B(\mathbf{p}, m)} u(\mathbf{y}) \right\}.$$

2.3.1 Existence of Optimal Choice(s)

We are then natural to seek for the existence of an optimal choice for a utility-maximizing consumer. Luckily, the answer is yes.

Proposition 2.3.2

Suppose that the consumer has a rational (complete and transitive) and continuous preference relation and makes rational decisions, then for $(\mathbf{p}, m) \gg \mathbf{0}$, they have (at least) one optimal choice.

Proof for Proposition.

- Since the consumer has a complete, transitive and continuous preference relation, then her preferences can be represented by a continuous utility function $u(\cdot)$.
- When $(\mathbf{p}, m) \gg \mathbf{0}$, the budget set $B(\mathbf{p}, m)$ is closed and bounded and hence compact.
- A continuous function on a compact set has at least one maximizer.

2.3.2 Further Assumptions

Now that existence of optimal choice is guaranteed (under rational and continuous preference relations, and $(\mathbf{p}, m) \gg \mathbf{0}$, we still need to answer:

- How to find the optimal choice(s)?
- In particular, what (additional) restrictions on the consumer's preference relation can simplify our analysis?

Locally Non-Satiation One possible simplification is to replace the inequality in the budget constraint with the equality: $\mathbf{p} \cdot \mathbf{x} = m$.

Intuitively, if a consumer thinks “more of a good is good”, her preference relation should be monotone, and she would definitely exhaust all her wealth when maximizing her utility.

Definition 2.3.3: Monotonicity

A preference relation $\succsim$ on $X = \mathbb{R}_+^n$ is *monotone* if for any $\mathbf{x}, \mathbf{y} \in X$, we have:

- $\mathbf{x} \geq \mathbf{y} \implies \mathbf{x} \succsim \mathbf{y}$;
- $\mathbf{x} \gg \mathbf{y} \implies \mathbf{x} \succ \mathbf{y}$.

Indeed, the monotonicity condition proves to be a bit strong; the weaker condition of *locally non-satiated* preferences instead serves our purpose.

Definition 2.3.4: Locally Non-Satiation

A preference relation $\succsim$ on $X = \mathbb{R}_+^n$ is *locally non-satiated* if for any $\mathbf{x} \in X$ and $\varepsilon > 0$, there exists $\mathbf{y} \in \text{Ball}(\mathbf{x}, \varepsilon) \cap X$ such that $\mathbf{y} \succ \mathbf{x}$.

Intuitively, a preference relation is locally non-satiated if there is no *bliss* point. Moreover, a monotone preference relation must be locally non-satiated.

Proposition 2.3.5

Suppose that the consumer has a complete, transitive, continuous and locally non-satiated preference relation and makes rational decisions, then for $(\mathbf{p}, m) \gg \mathbf{0}$, the budget constraint must hold with equality at any optimal choice $\mathbf{x}^*$, i.e., $\mathbf{p} \cdot \mathbf{x}^* = m$.

Proof for Proposition.

Intuitively, if the consumer does not exhaust her budget, then there must be a nearby affordable bundle which is strictly more preferred, by locally non-satiation.

- Consider any bundle $\mathbf{x} \in X = \mathbb{R}_+^n$ where $\mathbf{p} \cdot \mathbf{x} < m$. Since the preference relation $\succsim$ is locally non-satiated, for any $\varepsilon > 0$, there exists $\mathbf{y} \in \text{Ball}(\mathbf{x}, \varepsilon) \cap X$ such that $\mathbf{y} \succ \mathbf{x}$.
- We claim that $\mathbf{y} \in B(\mathbf{p}, m)$, i.e., the alternative bundle falls within the consumer's budget set for $\varepsilon > 0$ small enough, so that $\mathbf{x}$ cannot be an optimal choice.

Let $\mathbf{y} = \mathbf{x} + \mathbf{z}$. By construction, $\|\mathbf{z}\| < \varepsilon$. In particular, $|z_i| < \varepsilon$, for any $i = 1, 2, \dots, n$. It follows that $\mathbf{p} \cdot \mathbf{y} = \mathbf{p} \cdot \mathbf{x} + \mathbf{p} \cdot \mathbf{z} \leq \mathbf{p} \cdot \mathbf{x} + n \cdot \varepsilon \cdot \max_i p_i$. As ε take appropriately small values, we finish our proof. ■

Convexity Another possible simplification is the case of *unique* optimal choice. Notice that the consumer's budget set $B(\mathbf{p}, m)$ is convex. Based on this, when the utility function $u(\cdot)$ is *strictly quasi-concave*, there is a *unique* global maximizer. This translates into the following condition of strictly convex preference relations.

Definition 2.3.6: Convexity

A preference relation $\succsim$ on $X = \mathbb{R}_+^n$ is *convex* if for any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in X$ such that $\mathbf{x} \succsim \mathbf{z}$ and $\mathbf{y} \succsim \mathbf{z}$, we have $t\mathbf{x} + (1-t)\mathbf{y} \succsim \mathbf{z}$, for any $t \in [0, 1]$. If $t\mathbf{x} + (1-t)\mathbf{y} \succ \mathbf{z}$ for any $\mathbf{x} \neq \mathbf{y}$ and $t \in (0, 1)$, then the preference relation is *strictly convex*.

Remark.

1. Convexity can be equivalently defined as: For any $\mathbf{x}, \mathbf{y} \in X$ such that $\mathbf{y} \succsim \mathbf{x}$, we have $t\mathbf{y} + (1-t)\mathbf{x} \succsim \mathbf{x}$ for any $t \in [0, 1]$.
2. An equivalent way to describe convexity involves indifference curve and *upper contour set* of choice bundles, Upper Contour Set of $\mathbf{y} = \{\mathbf{x} \in X : \mathbf{x} \succsim \mathbf{y}\}$, graphically the area sitting upper-right above the indifference curve (included). Convexity of preferences amounts to the assumption that the upper contour set of any $\mathbf{y} \in X$, is a *convex* set.
3. Convexity is fundamental in the standard model of competitive economics. When consumer preferences are convex, market clearing prices exist; otherwise this may not exist. Convexity is also needed to be able to recover consumer preferences from choices from various budget sets. Convexity is often described as capturing the idea that the agent like diversity. However, whether convexity makes sense often depends on the interpretation of the goods space, in particular on the level of aggregation (e.g., over time or categories).

Proposition 2.3.7

Suppose that the consumer has a complete, transitive, continuous and strictly convex preference relation and makes rational decisions, then for $(\mathbf{p}, m) \gg \mathbf{0}$, there is exactly one optimal choice.

Proof for Proposition.

- The rational and continuous preference relation has guaranteed the existence of at least one optimal choice.
- Suppose to the contrary that the consumer has at least two optimal choices $\mathbf{x}^*$ and $\mathbf{y}^*$, and $\mathbf{x}^* \neq \mathbf{y}^*$, then by optimality $\mathbf{x}^* \sim \mathbf{y}^*$. Construct a bundle $\mathbf{w} = \frac{1}{2}\mathbf{x}^* + \frac{1}{2}\mathbf{y}^*$. Since the consumer's preference relation is strictly convex, $\mathbf{w} \succ \mathbf{x}^* \sim \mathbf{y}^*$. Moreover, since the budget set $B(\mathbf{p}, m)$ is convex when $(\mathbf{p}, m) \gg \mathbf{0}$, so $\mathbf{w} \in B(\mathbf{p}, m)$. So neither $\mathbf{x}^*$ nor $\mathbf{y}^*$ can be an optimal choice, which is a contradiction.

Not only do these assumptions of properties of preference relation play an important role in shaping the desired characteristics of the solutions to utility maximization problem, but each has a corresponding property in a specific preference-representing utility function u .

Proposition 2.3.8

Suppose the preference relation $\succsim$ on X can be represented by $u : X \rightarrow \mathbb{R}$. Then,

1. $\succsim$ is monotone if and only if u is non-decreasing.
2. $\succsim$ is locally non-satiated if and only if u has no local maxima in X .
3. $\succsim$ is (strictly) convex if and only if u is (strictly) quasi-concave.

2.3.3 Marshallian Demand & Indirect Utility Function**Definition 2.3.9: Indirect Utility Function**

Given $(\mathbf{p}, m) \gg \mathbf{0}$, the *indirect utility function* is defined to give the optimal utility level:

$$v(\mathbf{p}, m) = \sup_{\mathbf{x} \in B(\mathbf{p}, m)} u(\mathbf{x}).$$

Here we rigorously use “sup” to define $v(\mathbf{p}, m)$ instead of “max”, because sometimes $u(\mathbf{x})$ does not behave well to have a maximum.

Definition 2.3.10: Marshallian Demand Correspondence

The *Marshallian demand correspondence* is defined as the consumer's optimal choice(s):

$$\mathbf{x}^M(\mathbf{p}, m) = \{\mathbf{x} \in B(\mathbf{p}, m) : u(\mathbf{x}) = v(\mathbf{p}, m)\}.$$

$\mathbf{x}^M(\mathbf{p}, m)$ is, in general, a set of utility-maximizing consumption bundles.

Assume throughout the analysis of utility maximization problem that the consumer has a rational preference relation $\succsim$ and makes rational decisions given $(\mathbf{p}, m) \gg \mathbf{0}$. Then Marshallian demand and indirect utility function must satisfy the following properties.

Proposition 2.3.11: Properties of Marshallian Demand Correspondence and Indirect Utility Function

- **Existence of optimal choice(s):** If $\succsim$ is continuous, then the Marshallian demand $\mathbf{x}^M(\mathbf{p}, m) \neq \emptyset$.
- **Structure of Marshallian demand:** If $\succsim$ is convex, then $\mathbf{x}^M(\mathbf{p}, m)$ is a convex set. If $\succsim$ is strictly convex, then $\mathbf{x}^M(\mathbf{p}, m)$ is a singleton.
- **Homogeneity:** Both $v(\mathbf{p}, m)$ and $\mathbf{x}^M(\mathbf{p}, m)$ are homogeneous of degree 0 in $(\mathbf{p}, m)$, that is, for any $t > 0$, $v(t\mathbf{p}, tm) = v(\mathbf{p}, m)$ and $\mathbf{x}^M(t\mathbf{p}, tm) = \mathbf{x}^M(\mathbf{p}, m)$.
- **Monotonicity of $v(\mathbf{p}, m)$:** $v(\mathbf{p}, m)$ is non-increasing in $\mathbf{p}$ and non-decreasing in m . If $\succsim$ is locally non-satiated, then $v(\mathbf{p}, m)$ is strictly increasing in m .
- **Walras' Law:** If $\succsim$ is locally non-satiated, then $\mathbf{p} \cdot \mathbf{x} = m$, for any $\mathbf{x} \in \mathbf{x}^M(\mathbf{p}, m)$.

2.3.4 Derivation of Utility Maximization Problem

The consumer's utility maximization problem is give by:

$$\begin{aligned} & \max_{\mathbf{x} \in \mathbb{R}_+^n} u(\mathbf{x}) \\ & \text{s.t. } \mathbf{p} \cdot \mathbf{x} \leq m \\ & \quad x_i \geq 0, \forall i = 1, 2, \dots, n \end{aligned}$$

For $(\mathbf{p}, m) \gg \mathbf{0}$, the constraint qualification is always satisfied, and we can apply the necessary KKT conditions when $u(\cdot)$ is continuously differentiable.

The Lagrangian is given by:

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = u(\mathbf{x}) + \lambda \left(m - \sum_{i=1}^n p_i x_i \right) + \sum_{i=1}^n \mu_i x_i,$$

where λ is the Lagrange multiplier on the budget constraint and, for each i , μ_i is the multiplier on the constraint that $x_i \geq 0$. The UMP is then transformed into:

$$\max_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda, \mu) = \max_{\mathbf{x}} u(\mathbf{x}) + \lambda \left(m - \sum_{i=1}^n p_i x_i \right) + \sum_{i=1}^n \mu_i x_i.$$

The first-order conditions are given by:

- w.r.t. x_i : $\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial u}{\partial x_i} - \lambda p_i + \mu_i = 0$.
- Inequality constraints: $m - \sum_{i=1}^n p_i x_i \geq 0, x_i \geq 0, \lambda \geq 0, \mu_i \geq 0$.
- Complementary slackness: $\lambda(m - \sum_{i=1}^n p_i x_i) = 0, \mu_i x_i = 0$.

Remark.

- In contrast to equality constraints, the direction of each inequality constraint determines the way in which we set up the Lagrangian.

Intuitively, we make sure each multiplier is non-negative and penalize constraint violations. For instance, when the budget constraint is violated, that is, $m - \sum_{i=1}^n p_i x_i < 0$, the value of the Lagrangian strictly decreases.

- Despite little economic meaning, λ represents the *shadow price* of wealth, that is, marginal utility of an additional unit of income.

However, note that utility only has ordinal meanings. Nothing in the consumer theory developed so far suggests any basis for using the shadow price of wealth to guide redistribution policies.

- If the preference relation is well-behaved (i.e., locally non-satiated and strictly convex) and the non-negativity constraints are not binding, then $\frac{\partial u}{\partial x_i} = \lambda p_i$, and we are back to the familiar “tangency conditions”, that is, for all i, j :

$$MRS_{ij} = \frac{\frac{\partial u}{\partial x_i}}{\frac{\partial u}{\partial x_j}} = \frac{p_i}{p_j}.$$

The “tangency conditions” say that at the consumer’s maximum, relative marginal utility of any two choices equals their relative price, that is, all possible inner “gains from trade” has been realized (thus no more inner “gain from trade”).

To facilitate the derivation of the consumer’s problem, we should **first check whether the preference relation is locally non-satiated and strictly convex**.

- Case 1: If *locally non-satiated* and *strictly convex*, then we proceed in two steps
 1. Directly apply the “tangency conditions”: $\frac{MU_i}{p_i} = \frac{MU_j}{p_j} = \lambda$, for any i, j .
 2. Check the *non-negativity constraints* and apply the complementary slackness condition(s) if necessary.

Note that in the previous step we first assume that the non-negativity holds and that we have interior solution.
- Case 2: If neither locally non-satiated nor strictly convex, then use logic or economic intuition to tackle the problem.

Cobb-Douglas Utility Example.

Suppose that the consumer's preference relation can be represented by the following utility function:

$$u(x_1, x_2, x_3) = (x_1 + a)(x_2 + b)(x_3 + c)$$

where $a, b, c \geq 0$ are non-negative constants. Moreover, the consumer faces constant prices p_1, p_2, p_3 and has income $m \geq 0$.

1. First suppose that $a = b = c = 0$. Solve for the consumer's Marshallian demand correspondence $\mathbf{x}^M(x_1, x_2, x_3)$ and indirect utility function $v(p_1, p_2, p_3, m)$.
2. Next suppose that $a, b, c > 0$. Solve for the consumer's Marshallian demand correspondence $\mathbf{x}^M(p_1, p_2, p_3, m)$ and indirect utility function $v(p_1, p_2, p_3, m)$.

Claim: Cobb-Douglas Utility

Cobb-Douglas utility representation for a preference relation $\succsim$ can be written as

$$u(x_1, x_2, \dots, x_n) = x_1^{\alpha_1} x_2^{\alpha_2} \dots x_n^{\alpha_n}.$$

Then the consumer would spend her income on each good according to its share,

$$x_i = \frac{\alpha_i}{\sum_{j=1}^n \alpha_j} \cdot \frac{m}{p_i} \iff p_i x_i = \frac{\alpha_i}{\sum_{j=1}^n \alpha_j} \cdot m$$

Solution.

1. To solve the first question, we need to rigorously follow the steps:
 - Step 1: Apply the positive monotonic transformation for computation-convenience: $v = \ln u$.
 - Step 2: Check the locally non-satiation and convexity of preference relation for simplification of optimal solution.
 - Check that the preference relation is monotonic and hence locally non-satiated.
 - Check that the preference relation is strictly convex, that is, the utility function is strictly quasi-concave.
 - Step 3: Now that both non-satiation and convexity are satisfied, apply the “tangency conditions”:

$$\frac{MU_1}{p_1} = \frac{MU_2}{p_2} = \frac{MU_3}{p_3}$$

- Step 4: Together with the budget constraint, so we have the Marshallian demand correspondence $\mathbf{x}^M(\mathbf{p}, m)$.

$$\mathbf{x}^M(p_1, p_2, p_3, m) = \left(\frac{m}{3p_1}, \frac{m}{3p_2}, \frac{m}{3p_3} \right) \geq \mathbf{0}.$$

2. The first three steps are quite similar to the previous part and thus omitted. In

step 4, this time we have

$$\begin{aligned}x_1^M(p_1, p_2, p_3, m) &= \frac{m + p_1a + p_2b + p_3c}{3p_1} - a \\x_2^M(p_1, p_2, p_3, m) &= \frac{m + p_1a + p_2b + p_3c}{3p_2} - b \\x_3^M(p_1, p_2, p_3, m) &= \frac{m + p_1a + p_2b + p_3c}{3p_3} - c\end{aligned}$$

Note that depending on the parameter values, we may or may not have $\mathbf{x}^M(\mathbf{p}, m) \geq \mathbf{0}$. In other words, the non-negativity constraints may be binding and are for us to check. For simplicity, suppose that

$$2p_1a - p_2b - p_3c \geq 2p_2b - p_1a - p_3c \geq 2p_3c - p_1a - p_2b$$

The other five symmetric cases are similar. In our assumption, $p_1a \geq p_2b \geq p_3c$.

- (a) $m \geq 2p_1a - p_2b - p_3c$, then the non-negativity constraints are not binding, and we have the interior solution described above.
- (b) $2p_1a - p_2b - p_3c \geq m$, then at optimum, $x_1^* = 0$. The constraint $x_1^M \geq 0$ is binding, so the consumer only purchases goods 2 and 3, and we have

$$\mathbf{x}^M(p_1, p_2, p_3, m) = \left(0, \frac{m + p_2b + p_3c}{2p_2} - b, \frac{m + p_2b + p_3c}{2p_3} - c\right)$$

- i. $m \geq p_2b - p_3c$, then $\mathbf{x}^M$ is as above.
- ii. $p_2b - p_3c > m$, then the two constraints $x_1^M \geq 0$ and $x_2^M \geq 0$ are both binding, so the consumer only purchases good 3, and

$$\mathbf{x}^M(p_1, p_2, p_3, m) = \left(0, 0, \frac{m}{p_3} - c\right)$$

Inter-Temporal Choice Example.

You have a saving $s > 0$ to spend for this year and next year. Since you are now in graduate school, you will not earn any additional income over the two years. Suppose your utility is time-separable and is given by $v(c_1, c_2) = u(c_1) + \beta u(c_2)$, where $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ is a twice continuously differentiable function with $u'(c) > 0$ and $u''(c) < 0$ for any $c \in \mathbb{R}_+$ and $0 < \beta < 1$ is the discount factor. Further suppose that the going interest rate is $0 < r < 1$, which will remain constant. Let c_1^* and c_2^* be your optimal consumption choices for this year and next year.

1. Give a (necessary and sufficient) condition for $(c_1^*, c_2^*) \gg \mathbf{0}$.
2. Suppose that $(c_1^*, c_2^*) \gg \mathbf{0}$, compare c_1^* with c_2^* .

Solution.

1. The utility maximization problem is given by:

$$\begin{aligned} \max_{c_1, c_2} v(c_1, c_2) &= u(c_1) + \beta u(c_2) \\ \text{s.t. } c_1 + \frac{c_2}{1+r} &\leq s \\ c_1, c_2 &\geq 0 \end{aligned}$$

Since $u(\cdot)$ is strictly increasing and strictly concave, $v(\cdot)$ is also strictly increasing and strictly concave (in particular, strictly quasi-concave), and we can directly apply the “tangency condition”:

$$\frac{u'(c_1)}{1} = \frac{\beta u'(c_2)}{\frac{1}{1+r}}$$

By strictly monotonicity, $c_1 + \frac{c_2}{1+r} = s$. For $(c_1^*, c_2^*) \gg \mathbf{0}$, we need:

$$\frac{u'(s)}{u'(0)} < \beta(1+r) < \frac{u'(0)}{u'((1+r)s)}$$

2. At the optimum,

$$\frac{u'(c_1)}{u'(c_2)} = \beta(1+r)$$

- If $\beta(1+r) > 1$, then $\frac{u'(c_1)}{u'(c_2)} > 1$ and $c_1^* < c_2^*$.
- If $0 < \beta(1+r) < 1$, then $\frac{u'(c_1)}{u'(c_2)} < 1$ and $c_1^* > c_2^*$.
- If $\beta(1+r) = 1$, then $\frac{u'(c_1)}{u'(c_2)} = 1$ and $c_1^* = c_2^*$.

Intuitively, $\beta(1+r)$ represents how your tradeoff between next year and this year compares with that of the market when $c_1 = c_2$.

Utility Representation Example.

Let $X = \mathbb{R}_+ \times \mathbb{N}$, where (x, t) is interpreted as receiving x yuan at time t . Consider the following six properties of preference relations on X :

- Rationality (completeness and transitivity).
- Continuity.
- There is indifference between receiving 0 yuan at time 0 and receiving 0 yuan at any other time.
- It is (strictly) better to receive any positive amount of money as soon as possible.
- Money is always desirable.
- The preference between (x, t) and $(y, t+1)$ is independent of t .

Consider the following questions:

1. Use precise mathematical language to formally define the six properties.

2. Suppose a preference relation on X can be represented by the utility function $v(x, t) = u(x)\beta^t$, where $0 < \beta < 1$ and $u(\cdot)$ is continuous, strictly increasing and $u(0) = 0$. Check whether this preference relation satisfies each of the six properties.
3. Suppose a preference relation on X satisfies all of the six properties. Show that this preference relation must admit a utility representation.
4. Use precise mathematical language to formalize the idea that "one preference is more patient than another".
5. Based on your definition in part (4), prove or disprove the following statement: A preference relation represented by $v_1(x, t) = u_1(x)\beta_1^t$ is more patient than another preference relation represented by $v_2(x, t) = u_2(x)\beta_2^t$ if $0 < \beta_2 < \beta_1$ (where $u_1(\cdot)$ and $u_2(\cdot)$ are both continuous, strictly increasing and $u_1(0) = u_2(0) = 0$).

Solution.

1. Mind yourself that the time t is not continuous here, so take caution when you try to define a "limit" with regard to t .
 - Continuity: For any $t, t' \in \mathbb{N}$, and any pair of sequences $\{x(n)\}_{n=1}^{\infty}$ and $\{y(n)\}_{n=1}^{\infty}$ from $\mathbb{R}^+$ with $x(n) \rightarrow x^*$, $y(n) \rightarrow y^*$. If $(x(n), t) \succsim (y(n), t')$ for all n , we have $(x^*, t) \succsim (y^*, t')$.
 - The preference between (x, t) and $(y, t+1)$ is independent of t : For any $x, y \in \mathbb{R}^+$, and $t, t' \in \mathbb{N}$, we have $(x, t) \succsim (y, t+1) \iff (x, t') \succsim (y, t'+1)$.
2. A continuous preference relation can be represented by a discontinuous function. However, if a preference relation can be represented by a continuous function, then the preference relation must be continuous.
3. Recall the proof in our lecture of existence of utility representation for any rational and continuous preference relation.
 - Claim 1: For any pair (x, t) , there is a unique number $u(x, t) \in \mathbb{R}^+$ such that $(x, t) \sim (u(x, t), 0)$.
Proof: First, by indifference when receiving nothing, we have $(0, t) \sim (0, 0)$. For any pair (x, t) , we have $(x, t) \succsim (0, 0)$ and $(x+1, t) \succsim (x, t)$. Then by continuity, there is y for which $(x, t) \sim (y, 0)$, and we then define $u(x, t) := y$.
 - Claim 2: The preference relation is represented by $u(x, t)$.
 By claim 1 and property that money is more desirable:

$$u(x, t) \geq u(y, t') \iff (u(x, t), 0) \succsim (u(y, t'), 0) \iff (x, t) \succsim (y, t')$$

4. The definition should be clearly based on preference relations and try to be somewhat math-irrelevant.

$\succsim_1$ is more patient than $\succsim_2$ if for any (x, t) and any (y, t') with $t' > t$, $y > x$:

$$(y, t') \succsim_2 (x, t) \implies (y, t') \succsim_1 (x, t)$$

The definition means that if I prefer to wait from t to t' under $\succsim_1$, then I will also prefer to wait from t to t' when I'm more patient (say under $\succsim_2$).

5. Intuitively, if the two preference relations value money differently at the baseline level (i.e., simply in terms of money), they would generate different preference over combinations of money and receiving time.

Example.

Let $\succsim$ be a rational (complete and transitive) preference relation on $X = \mathbb{R}_+^2$. Consider the following three properties:

- Additivity: If $(x_1, x_2) \succsim (y_1, y_2)$, then for any t, s such that $(x_1 + t, x_2 + s), (y_1 + t, y_2 + s) \in \mathbb{R}_+^2$, $(x_1 + t, x_2 + s) \succsim (y_1 + t, y_2 + s)$.
- Strong monotonicity: If $x_1 \geq y_1$ and $x_2 \geq y_2$, then $(x_1, x_2) \succsim (y_1, y_2)$. If in addition, $x_1 > y_1$ or $x_2 > y_2$, then $(x_1, x_2) \succ (y_1, y_2)$.
- Standard continuity: For any two sequences $\{x_n\}_{n=1}^\infty$ and $\{y_n\}_{n=1}^\infty$, if $x_n \succsim y_n$ for any n , and $\lim_{n \rightarrow \infty} x_n = x^*$ and $\lim_{n \rightarrow \infty} y_n = y^*$, then $x^* \succsim y^*$.

Consider the following questions:

1. Show that if $\succsim$ has a linear utility representation, i.e., $u(x_1, x_2) = ax_1 + bx_2$, for some $a, b > 0$, then this preference relation satisfies the above three properties.
2. Show that these three properties are necessary for the preference relation $\succsim$ to have a linear utility representation, i.e., show that for any pair of the three properties, there is a preference relation that does not satisfy the third property.
3. Show that if $\succsim$ satisfies the three properties, then this preference relation admits a linear utility representation, i.e., there exists $a, b > 0$ such that $u(x_1, x_2) = ax_1 + bx_2$, for any $(x_1, x_2) \in \mathbb{R}_+^2$. (Hint: Think about the indifference curves/sets of this preference relation.)

Solution.

1. Easy to verify. Notice that if a preference relation can be represented by a continuous function, then the preference relation must be continuous.
2. This question means the three properties are “parallel” from a preference relation to have a linear utility representation.
 - (i)(ii) $\xrightarrow{\times}$ (iii): The lexicographic preference: $(x_1, x_2) \succsim (y_1, y_2)$ if either $x_1 > y_1$ or $x_1 = y_1$ and $x_2 \geq y_2$.
 - (i)(iii) $\xrightarrow{\times}$ (ii): $u(x_1, x_2) = x_1 - x_2$; $u(x_1, x_2) = x_1 - \frac{1}{x_2}$.
 - (ii)(iii) $\xrightarrow{\times}$ (i): $u(x_1, x_2) = x_1^2 + x_2^2$; $u(x_1, x_2) = x_1$.
3. Starting from possible intuitions from linear utility representation, we need to establish the following two properties of the indifference curve:

- Property 1: The indifference curves are linear.
- Property 2: The indifference curves are parallel, downward sloping and not thick.

In order to establish the two properties, we first prove the following two lemmas:

- Lemma 1: For $\mathbf{x} \neq \mathbf{y}$, if $\mathbf{x} \sim \mathbf{y}$, then for $\mathbf{z} = \frac{1}{2}\mathbf{x} + \frac{1}{2}\mathbf{y}$ and $\mathbf{z}' = 2\mathbf{y} - \mathbf{x}$, we have $\mathbf{x} \sim \mathbf{y} \sim \mathbf{z} \sim \mathbf{z}'$.
- Lemma 2: For $\mathbf{x} \neq \mathbf{y}$, if $\mathbf{x} \sim \mathbf{y}$, then for any $\mathbf{w} = t\mathbf{x} + (1-t)\mathbf{y}$, $0 \leq t \leq 1$, we have $\mathbf{x} \sim \mathbf{y} \sim \mathbf{w}$.

Here we omit the technical proofs and move on with establishment of property 1. Pick any point on the horizontal axis $\mathbf{x} = (x, 0)$, $x > 0$. By the proof of the utility representation in lecture and strong monotonicity, $\exists 0 < w < x$ such that $\mathbf{x} \sim \mathbf{w} = (w, w)$. Connect $\mathbf{x}$ and $\mathbf{w}$ and extend it to the vertical axis. Denote the intersection of the ray $\mathbf{xw}$ with the vertical axis as $\mathbf{y} = (0, y)$. Jointly from Lemma 1 and 2 we can say the points on the line $\mathbf{xy}$ are indifferent to each other. Finally, by strong monotonicity, the indifference curves must be downward sloping and for any $x \neq x'$, we cannot have $(x, 0) \sim (x', 0)$. Moreover, if $(x, 0) \sim (w, w)$, then for any $t \geq -w$, we have $(x+t, 0) \sim (w+t, w)$, so the indifference curves are parallel.

2.4 Expenditure Minimization Problem

In order to disentangle the price effect and the income effect, we introduce the following expenditure minimization problem, holding the consumer's utility at or above a certain level.

Definition 2.4.1: Expenditure Minimization Problem

$$\begin{aligned} & \min_{\mathbf{x}} \mathbf{p} \cdot \mathbf{x} \\ & \text{s.t. } u(\mathbf{x}) \geq u \\ & x_i \geq 0, \forall i = 1, 2, \dots, n \end{aligned}$$

Definition 2.4.2: Expenditure Function; Hicksian Demand Correspondence

Let $F(\mathbf{p}, u) = \{\mathbf{x} \in \mathbb{R}_+^n : u(\mathbf{x}) \geq u\}$ be the feasible set. Define the optimal (minimal) value function as the *expenditure function*:

$$e(\mathbf{p}, u) = \inf_{\mathbf{x} \in F(\mathbf{p}, u)} \mathbf{p} \cdot \mathbf{x},$$

and the consumer's optimal choice(s) as the *Hicksian demand correspondence*:

$$\mathbf{x}^H(\mathbf{p}, u) = \{\mathbf{x} \in F(\mathbf{p}, u) : \mathbf{p} \cdot \mathbf{x} = e(\mathbf{p}, u)\}.$$

Since the consumer's utility is held at or above a certain level, when there is a price change, the change in the Hicksian demand only captures the substitution effect.

Similar to the analysis of the consumer's utility maximization problem, before seeking the optimal solution(s), we approach the expenditure minimization problem from two aspects:

- When can we simplify the problem (e.g., existence of solution(s), binding utility level and uniqueness of solution)?
- Properties of expenditure function and Hicksian demand correspondence.

2.4.1 Existence of Solution

Proposition 2.4.3

Suppose $u(\cdot)$ represents a continuous preference relation and that $\mathbf{p} \gg \mathbf{0}$ and $F(u) \neq \emptyset$, then the expenditure minimization problem has at least one minimizer, i.e.,

$$\mathbf{x}^H(\mathbf{p}, u) \neq \emptyset.$$

Proof for Proposition.

- Pick any $\mathbf{x}_0 \in F(u)$ and consider the alternative feasible set:

$$\tilde{F} = \{\mathbf{x} \in \mathbb{R}_+^n : u(\mathbf{x}) \geq u \text{ and } \mathbf{p} \cdot \mathbf{x} \leq \mathbf{p} \cdot \mathbf{x}_0\}.$$

It is easy to see that the expenditure minimization problem with the feasible set $\tilde{F}$ has the same solution as the original problem. Moreover, the key motivation for this construction is that $\tilde{F}$ is closed and bounded, and hence compact.

- Boundedness is crafted by picking $\mathbf{x}_0$ purposefully but without loss of generality.
- Closedness is guaranteed by continuity of $u(\cdot)$.
 - * Note that $u(\cdot)$ need not be continuous, because we can always find an alternative continuous function $v(\cdot)$ that represents the same preference relation regardless of the continuity of $u(\cdot)$. Thus, $\tilde{F}$ is closed.
 - * Note that the continuity of preference relation has no direct relation to the continuity of its utility representation.
- The objective function $\mathbf{p} \cdot \mathbf{x}$ is continuous. We know that a continuous function on a compact set has at least one minimizer, which ends our proof.

2.4.2 Biding Utility Level

Proposition 2.4.4

Suppose $u(\cdot)$ represents a continuous preference relation and that $\mathbf{p} \gg \mathbf{0}$, $u \geq u(\mathbf{0})$ and $F(u) \neq \emptyset$, then at any minimizer $\mathbf{x}^*$,

$$u(\mathbf{x}^*) = u.$$

Proof for Proposition.

- By the argument in the previous proposition, we can assume without loss that $u(\cdot)$ is continuous. Suppose to the contrary that a minimizer $\mathbf{x}^*$, we have $u(\mathbf{x}^*) > u$. Since $u \geq u(\mathbf{0})$, $\mathbf{x}^* \neq \mathbf{0}$.
- By the continuity of $u(\cdot)$, we know for some $\varepsilon > 0$, $u((1 - \varepsilon)\mathbf{x}^*) > u$. It follows that $(1 - \varepsilon)\mathbf{x}^* \in F(u)$ and that $\mathbf{p} \cdot (1 - \varepsilon)\mathbf{x}^* < \mathbf{p} \cdot \mathbf{x}^*$, which is a contradiction to the optimality of $\mathbf{x}^*$.

Remark.

The binding condition here in EMP is much weaker than that in UMP, where we do not put “any” additional condition on preference relation. One can understand this as the objective function in EMP is itself locally non-satiated.

2.4.3 Unique Minimizer**Proposition 2.4.5**

Suppose $u(\cdot)$ represents a continuous and strictly convex preference relation and that $\mathbf{p} \gg \mathbf{0}$ and $F(u) \neq \emptyset$. Then the expenditure minimization problem has exactly one minimizer, i.e., $\mathbf{x}^H(\mathbf{p}, u)$ is a singleton.

Proof for Proposition.

- By the preceding proposition, *at least one minimizer exists*.
- Suppose to the contrary that there are two minimizers $\mathbf{x}^* \neq \mathbf{y}^*$. Then by feasibility, $u(\mathbf{x}^*) \geq u$ and $u(\mathbf{y}^*) \geq u$. Since the preference relation is strictly convex, $u(\cdot)$ is strictly quasi-concave, so for the bundle $\mathbf{w} = \frac{1}{2}\mathbf{x}^* + \frac{1}{2}\mathbf{y}^* \neq \mathbf{0}$, we have $u(\mathbf{w}) > \min\{u(\mathbf{x}^*), u(\mathbf{y}^*)\} \geq u$. By continuity, for some small $\varepsilon > 0$, $u((1 - \varepsilon)\mathbf{w}) \geq u$. Moreover, $\mathbf{p} \cdot (1 - \varepsilon)\mathbf{w} = \frac{1}{2}(1 - \varepsilon)\mathbf{p} \cdot \mathbf{x}^* + \frac{1}{2}(1 - \varepsilon)\mathbf{p} \cdot \mathbf{y}^* < \mathbf{p} \cdot \mathbf{x}^* = \mathbf{p} \cdot \mathbf{y}^*$, which is a contradiction.

2.4.4 Summary of Properties

Proposition 2.4.6: Properties of Expenditure Function and Hicksian Demand Correspondence

Suppose $\mathbf{p} \gg \mathbf{0}$ and $F(u) \neq \emptyset$.

- **Existence of minimizer:** If $u(\cdot)$ represents a continuous preference relation and that, then the Hicksian demand $\mathbf{x}^H(\mathbf{p}, u) \neq \emptyset$.
- **Structure of Hicksian demand:** If $u(\cdot)$ represents a convex preference relation, then $\mathbf{x}^H(\mathbf{p}, u)$ is a convex set. If $u(\cdot)$ represents a continuous and strictly convex preference relation, then $\mathbf{x}^H(\mathbf{p}, u)$ is a singleton.
- **Homogeneity:** $e(\mathbf{p}, u)$ is homogeneous of degree 1 in $\mathbf{p}$, that is, for any $t > 0$, $e(t\mathbf{p}, u) = te(\mathbf{p}, u)$.
- **Monotonicity of $e(\mathbf{p}, u)$:** $e(\mathbf{p}, u)$ is non-decreasing in $\mathbf{p}$ and u . If $u(\cdot)$ represents a continuous preference relation, then $e(\mathbf{p}, u)$ is strictly increasing in u when $u \geq u(\mathbf{0})$.
- **Binding utility level:** Suppose $u(\cdot)$ represents a continuous preference relation and $u \geq u(\mathbf{0})$. Then at any minimizer $\mathbf{x}^*$, $u(\mathbf{x}^*) = u$.

Since $\min \mathbf{p} \cdot \mathbf{x}$ is equivalent to $\max -\mathbf{p} \cdot \mathbf{x}$, EMP can be solved in an analogous manner to UMP. EMP and UMP share the same “tangency condition” for interior solutions:

$$\frac{MU_i}{MU_j} = \frac{p_i}{p_j}$$

Example.

Suppose a consumer’s preference relation can be represented by the following utility function:

$$u(x_1, x_2) = \ln x_1 + x_2.$$

Moreover, the consumer faces constant prices $(p_1, p_2) \gg \mathbf{0}$ and has income $m > 0$.

1. Solve the consumer’s utility maximization problem to derive the Marshallian demand $\mathbf{x}^M(p_1, p_2, m)$ and indirect utility function $v(p_1, p_2, m)$.
2. Solve the consumer’s expenditure minimization problem to derive the Hicksian demand $\mathbf{x}^H(p_1, p_2, u)$ and expenditure function $e(p_1, p_2, u)$.

Solution.

1. It is easy to check that the preference relation is monotonic and strictly convex. The utility maximization can then be simplified as:

$$\max_{x_1, x_2 \geq 0} u(x_1, x_2) = \ln x_1 + x_2 \quad \text{s.t.} \quad p_1 x_1 + p_2 x_2 = m.$$

The “tangency condition” is: $|MRS| = \frac{MU_1}{MU_2} = \frac{1}{x_1} = \frac{p_1}{p_2}$.

Together with the binding budget constraint, we have $x_1^* = \frac{p_2}{p_1}$ and $x_2^* = \frac{m}{p_2} - 1$.

Note that the constraint $x_2 \geq 0$ may be binding. Marshallian demand is given by

$$\mathbf{x}^M(p_1, p_2, m) = \begin{cases} \left(\frac{p_2}{p_1}, \frac{m}{p_2} - 1 \right), & \text{if } m \geq p_2 \\ \left(\frac{m}{p_1}, 0 \right), & \text{if } 0 < m < p_2 \end{cases}$$

2. Similar to the previous part, the expenditure minimization problem can be simplified as:

$$\min_{x_1, x_2 \geq 0} p_1 x_1 + p_2 x_2 \quad \text{s.t.} \quad u(x_1, x_2) = \ln x_1 + x_2 = u.$$

The “tangency condition” is $|MRS| = \frac{MU_1}{MU_2} = \frac{1}{x_1} = \frac{p_1}{p_2}$.

Together with binding utility level, we have $x_1^* = \frac{p_2}{p_1}$ and $x_2^* = u - \ln \frac{p_2}{p_1}$.

Notice the constraint $x_2 \geq 0$ may be binding, so the Hicksian demand is given by:

$$\mathbf{x}^H(p_1, p_2, m) = \begin{cases} \left(\frac{p_2}{p_1}, u - \ln \frac{p_2}{p_1} \right), & \text{if } u \geq \ln \frac{p_2}{p_1} \\ (e^u, 0), & \text{if } u < \ln \frac{p_2}{p_1} \end{cases}$$

Remember to check if the utility function is monotone and quasi-concave beforehand!

2.5 Duality and Comparative Statics

2.5.1 Duality between UMP and EMP

Recall consumer’s utility maximization problem (UMP):

$$\begin{aligned} & \max_{\mathbf{x}} u(\mathbf{x}) \\ & \text{s.t.} \quad \mathbf{p} \cdot \mathbf{x} \leq m \\ & \quad x_i \geq 0, \forall i = 1, 2, \dots, n \end{aligned}$$

Consumer’s expenditure minimization problem (EMP):

$$\begin{aligned} & \min_{\mathbf{x}} \mathbf{p} \cdot \mathbf{x} \\ & \text{s.t.} \quad u(\mathbf{x}) \geq u \\ & \quad x_i \geq 0, \forall i = 1, 2, \dots, n \end{aligned}$$

Notice that the objective function in the UMP is precisely the constraint in the EMP and vice versa. In mathematics, the two optimization problems are called *dual problems*. By duality, Marshallian and Hicksian demand have a special relationship.

Proposition 2.5.1

Suppose $u(\cdot)$ is a utility function that represents a continuous and locally non-satiated preference relation on $X = \mathbb{R}_+^n$, then for any $\mathbf{p} \gg \mathbf{0}$, we have:

1. For any $m \geq 0$, $\mathbf{x}^M(\mathbf{p}, m) = \mathbf{x}^H(\mathbf{p}, v(\mathbf{p}, m))$ and $e(\mathbf{p}, v(\mathbf{p}, m)) = m$.
2. For any $u \geq u(\mathbf{0})$, $\mathbf{x}^H(\mathbf{p}, u) = \mathbf{x}^M(\mathbf{p}, e(\mathbf{p}, u))$ and $v(\mathbf{p}, e(\mathbf{p}, u)) = u$.

Proof for Proposition.

- Fix $m > 0$ and any $\mathbf{x}_0^M \in \mathbf{x}^M(\mathbf{p}, m)$, we have:

$$e(\mathbf{p}, v(\mathbf{p}, m)) \leq \mathbf{p} \cdot \mathbf{x}_0^M = m.$$

Fix any $u \geq u(\mathbf{0})$ and any $\mathbf{x}_0^H \in \mathbf{x}^H(\mathbf{p}, u)$, we have

$$v(\mathbf{p}, e(\mathbf{p}, u)) \geq u(\mathbf{x}_0^H) = u.$$

- Applying the first inequality to the wealth level $m = e(\mathbf{p}, u)$, we have:

$$e(\mathbf{p}, v(\mathbf{p}, e(\mathbf{p}, u))) \leq e(\mathbf{p}, u).$$

On the other hand, since the preference relation is continuous, $e(\mathbf{p}, u)$ is strictly increasing in u , so from the second inequality, we have

$$e(\mathbf{p}, v(\mathbf{p}, e(\mathbf{p}, u))) \geq e(\mathbf{p}, u).$$

- Again by strict monotonicity of $e(\mathbf{p}, u)$ in u , we have $v(\mathbf{p}, e(\mathbf{p}, u)) = u$. Similarly, $e(\mathbf{p}, v(\mathbf{p}, m)) = m$.
- Finally, since the preference relation is continuous and locally non-satiated, the budget constraint must bind at the UMP and the utility level must bind at the EMP. Correspondingly,

$$\mathbf{x}^M(\mathbf{p}, m) = \{\mathbf{x} \in \mathbb{R}_+^n : \mathbf{p} \cdot \mathbf{x} = m, u(\mathbf{x}) = v(\mathbf{p}, m)\} = \mathbf{x}^H(\mathbf{p}, v(\mathbf{p}, m))$$

$$\mathbf{x}^H(\mathbf{p}, u) = \{\mathbf{x} \in \mathbb{R}_+^n : u(\mathbf{x}) = u, \mathbf{p} \cdot \mathbf{x} = e(\mathbf{p}, u)\} = \mathbf{x}^M(\mathbf{p}, e(\mathbf{p}, u))$$

Intuitively, their relationship implies that, at the “right” level of wealth and utility, the Marshallian demand correspondence is identical to the Hicksian demand correspondence.

2.5.2 Envelope Theorem

Theorem 2.5.2: Envelope Theorem for Unconstrained Optimization

Let $f : X \times [\underline{\theta}, \bar{\theta}] \rightarrow \mathbb{R}$ and $V(\theta) = \sup_{\mathbf{x} \in X} f(\mathbf{x}, \theta)$. Suppose $f(\mathbf{x}, \cdot)$ is differentiable in θ for all $\mathbf{x} \in X$. Moreover, there exists an integrable function $b : [\underline{\theta}, \bar{\theta}] \rightarrow \mathbb{R}_+$ such that $|f_\theta(\mathbf{x}, \theta)| \leq b(\theta)$ for all $\mathbf{x} \in X$ and almost all $\theta \in [\underline{\theta}, \bar{\theta}]$. Then $V(\cdot)$ is absolutely continuous and hence differentiable almost everywhere. In addition, for any $\mathbf{x}^* \in \arg \max_{\mathbf{x} \in X} f(\mathbf{x}, \theta)$,

$$V'(\theta) = f_2(\mathbf{x}^*(\theta), \theta).$$

Remark.

- The intuition is that, only the direct effect matters; the indirect effect of $\mathbf{x}^*(\theta)$ on $V(\theta)$ can be ignored, since $f(\cdot)$ does not change with $\mathbf{x}$ at the optimum.
- May get some inspiration from the simplest version.
 - Suppose x is one-dimensional and that the optimizer is unique and differentiable in θ . Combined with F.O.C., we would have:

$$V'(\theta) = f_1(x^*(\theta), \theta) \cdot (x^*)'(\theta) + f_2(x^*(\theta), \theta) = f_2(x^*(\theta), \theta).$$

- For the envelope theorem to hold, $\mathbf{x}$ need not be one-dimensional, $f(\cdot, \theta)$ need not be differentiable in $\mathbf{x}$, and the optimal $\mathbf{x}^*(\theta)$ need not be unique or differentiable in θ .

Theorem 2.5.3: Envelope Theorem for Constrained Optimization

Suppose X is compact and convex. Let $f, g : X \times [\underline{\theta}, \bar{\theta}] \rightarrow \mathbb{R}$ and $V(\theta) = \sup_{\mathbf{x} \in X: g(\mathbf{x}, \theta) \geq 0} f(\mathbf{x}, \theta)$. The Lagrangian is given by $\mathcal{L}(\mathbf{x}, \theta; \lambda) = f(\mathbf{x}, \theta) + \lambda g(\mathbf{x}, \theta)$. Suppose f and g are continuous and concave in $\mathbf{x}$, $f_2(\mathbf{x}, \theta)$ and $g_2(\mathbf{x}, \theta)$ are continuous in $(\mathbf{x}, \theta)$, and there exists $\mathbf{x}_0 \in X$ such that $g(\mathbf{x}_0, \theta) > 0$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$. Then $V(\cdot)$ is absolutely continuous and hence differentiable (a.e.). In addition, for any $\mathbf{x}^*(\theta) \in \arg \max_{\mathbf{x} \in X: g(\mathbf{x}, \theta) \geq 0} f(\mathbf{x}, \theta)$,

$$V'(\theta) = \mathcal{L}_2(\mathbf{x}^*(\theta), \theta; \lambda^*).$$

Roy's identity and Shepard's lemma are two direct applications of envelope theorem.

Corollary 2.5.4: Roy's Identity

Suppose $u(\cdot)$ represents a locally non-satiated and strictly convex preference relation on $X = \mathbb{R}_+^n$. Then, for any $(\mathbf{p}, m) \gg \mathbf{0}$, the Marshallian demand for good i , $x_i^M(\mathbf{p}, m)$, is given by:

$$x_i^M(\mathbf{p}, m) = -\frac{\partial v(\mathbf{p}, m)/\partial p_i}{\partial v(\mathbf{p}, m)/\partial m}.$$

Proof for Corollary.

The Lagrangian of the utility maximization problem is given by:

$$\mathcal{L}(\mathbf{x}, \lambda, \mu; \mathbf{p}, m) = u(x_1, x_2, \dots, x_n) + \lambda \left(m - \sum_{i=1}^n p_i x_i \right) + \sum_{i=1}^n \mu_i x_i$$

By the envelope theorem, we have:

$$\begin{cases} \frac{\partial v(\mathbf{p}, m)}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial p_i} |_{(\mathbf{x}^*, \lambda^*, \mu^*)} = -\lambda^* x_i^M(\mathbf{p}, m) \\ \frac{\partial v(\mathbf{p}, m)}{\partial m} = \frac{\partial \mathcal{L}}{\partial m} |_{(\mathbf{x}^*, \lambda^*, \mu^*)} = \lambda^* \end{cases} \implies x_i^M(\mathbf{p}, m) = -\frac{\partial v(\mathbf{p}, m)/\partial p_i}{\partial v(\mathbf{p}, m)/\partial m}$$

Corollary 2.5.5: Shepard's Lemma

Suppose $u(\cdot)$ represents a locally non-satiated and strictly convex preference relation on $X = \mathbb{R}_+^n$. Then, for any $\mathbf{p} \gg \mathbf{0}$ and $F(u) \neq \emptyset$, the Hicksian demand for good i , $x_i^H(\mathbf{p}, u)$, is given by:

$$x_i^H(\mathbf{p}, u) = \frac{\partial e(\mathbf{p}, u)}{\partial p_i}.$$

Proof for Corollary.

The Lagrangian of the expenditure minimization problem is given by:

$$\mathcal{L}(\mathbf{x}, \lambda, \mu; \mathbf{p}, u) = \sum_{i=1}^n p_i x_i - \lambda(u(\mathbf{x}) - u) - \sum_{i=1}^n \mu_i x_i.$$

By the envelope theorem, we have:

$$\frac{\partial e(\mathbf{p}, u)}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial p_i} |_{(\mathbf{x}^*, \lambda^*, \mu^*)} = x_i^H(\mathbf{p}, u).$$

By strict convexity, both Marshallian demand and Hicksian demand are single-valued.

2.5.3 Slutsky Equation

Slutsky equation mathematically describes how Marshallian demand reacts to a price change, decomposing into income effect and substitution effect.

Theorem 2.5.6: Slutsky Equation

Suppose $u(\cdot)$ represents a continuous, locally non-satiated and strictly convex preference relation $\succsim$ on $X = \mathbb{R}_+^n$ and that $\mathbf{x}^M(\mathbf{p}, m)$ and $\mathbf{x}^H(\mathbf{p}, u)$ are both differentiable and single-valued. Then

$$\frac{\partial x_i^M(\mathbf{p}, m)}{\partial p_j} = \frac{\partial x_i^H(\mathbf{p}, v(\mathbf{p}, m))}{\partial p_j} - \frac{\partial x_i^M(\mathbf{p}, m)}{\partial m} x_j^M(\mathbf{p}, m).$$

Proof for Theorem

The proof proceeds with duality of Marshallian and Hicksian demand throughout.

$$x_i^H(\mathbf{p}, u) = x_i^M(\mathbf{p}, e(\mathbf{p}, u)), \text{ as long as } u \geq u(\mathbf{0})$$

Take partial derivatives with respect to p_j :

$$\frac{\partial x_i^H(\mathbf{p}, u)}{\partial p_j} = \frac{\partial x_i^M(\mathbf{p}, e(\mathbf{p}, u))}{\partial p_j} + \frac{\partial x_i^M(\mathbf{p}, e(\mathbf{p}, u))}{\partial m} \cdot \frac{\partial e(\mathbf{p}, u)}{\partial p_j}$$

By Shepard's lemma:

$$\frac{\partial e(\mathbf{p}, u)}{\partial p_j} = x_j^H(\mathbf{p}, u)$$

By duality,

$$\begin{cases} x_j^H(\mathbf{p}, v(\mathbf{p}, m)) = x_j^M(\mathbf{p}, m) \\ e(\mathbf{p}, v(\mathbf{p}, m)) = m \end{cases}$$

Evaluating the partial derivative equation at $u = v(\mathbf{p}, m)$, we have:

$$\frac{\partial x_i^H(\mathbf{p}, v(\mathbf{p}, m))}{\partial p_j} = \frac{\partial x_i^M(\mathbf{p}, e(\mathbf{p}, v(\mathbf{p}, m)))}{\partial p_j} + \frac{\partial x_i^M(\mathbf{p}, m)}{\partial m} \cdot x_j^M(\mathbf{p}, m)$$

Rearrange the terms to finish the proof of Slutsky Equation. ■

The interpretation for Slutsky Equation is that, the LHS represents how the Marshallian demand changes with respect to a price change, and that the first term on the RHS represents the substitution effect and the second term reveals the income effect.

Slutsky Equation allows us to develop a body of comparative statics based on it. The following definitions related to comparative statics are quite useful.

For comparative statics with regard to income m , we define normal good and inferior good. Intuitively, a good is normal if it loses its charm when its price increases.

Definition 2.5.7: Normal Good; Inferior Good

Good i is a *normal good* if $x_i^M(\mathbf{p}, m)$ is increasing in m , an *inferior good* if $x_i^M(\mathbf{p}, m)$ is decreasing in m .

Inferior good is purely a definition for goods whose demand is negatively related with the income level, but not informative of its quality.

Regular good and Giffen good are defined upon the comparative statics with regard to

the good's own price.

Definition 2.5.8: Regular Good; Giffen Good

Good i is a *regular good* if $x_i^M(\mathbf{p}, m)$ is decreasing in p_i , a *Giffen good* if $x_i^M(\mathbf{p}, m)$ is increasing in p_i .

A Giffen good has to be an inferior good first, because the substitution effect of a price increase will never favor the demand of that good; the income effect has to be not only positive but positive enough to counteract the negative impact of the substitution effect.

Another aspect of comparative statics grows in the view of analyzing a good's demand change given the price change of another good.

Definition 2.5.9: Substitute; Complement

Good i is a *substitute* for good j if $x_i^H(\mathbf{p}, u)$ is increasing in p_j , a *complement* for good j if $x_i^H(\mathbf{p}, u)$ is decreasing in p_j .

Good i being a complement for good j means that, any increase in p_j would shift part of the original share of consumption onto alternative goods other than good i and good j .

Definition 2.5.10: Gross Substitute; Gross Complement

Good i is a *gross substitute* for good j if $x_i^M(\mathbf{p}, m)$ is increasing in p_j , a *gross complement* for good j if $x_i^M(\mathbf{p}, m)$ is decreasing in p_j .

2.6 Consumer Welfare

Consumer surplus is a basic quantitative measure of consumer welfare. But here are some issues with it:

- What if more than one price would change at the same time?
- No equivalent or immediate interpretation in utility theory.

To address the two shortcomings, we introduce two additional measures to quantify changes in consumer welfare. That is, compensating variation and equivalent variation, on measures of “conceived wealth”.

Definition 2.6.1: Compensating Variation; Equivalent Variation

Suppose the initial price is $\mathbf{p}^0$ and $u^0 = v(\mathbf{p}, m)$, and that the final price is $\mathbf{p}'$ and $u' = v(\mathbf{p}', m)$. Compensating variation and equivalent variation are defined as:

1. *Compensating variation*: $CV = e(\mathbf{p}^0, u^0) - e(\mathbf{p}', u^0)$.
2. *Equivalent variation*: $EV = e(\mathbf{p}^0, u') - e(\mathbf{p}', u')$.

Notice that CV and EV have the same sign, and is positive for a price drop and negative for a price increase (though the two cases are not exhaustive).

Mathematically, by duality,

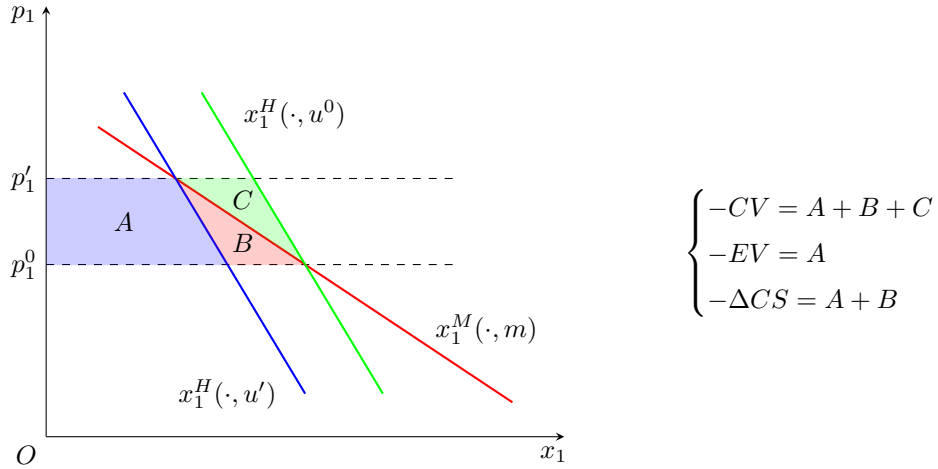
$$\begin{aligned} CV &= e(\mathbf{p}^0, u^0) - e(\mathbf{p}', u^0) = m - e(\mathbf{p}', u^0) \\ EV &= e(\mathbf{p}^0, u') - e(\mathbf{p}', u') = e(\mathbf{p}^0, u') - m \end{aligned}$$

Intuitively, $-CV$ measures how much we need to *compensate* the consumer for them to achieve the original level of utility at the new price vector, while EV measures what is the equivalent amount of money that the consumer values this price change if the price vector were fixed at the original level.

Suppose the price of a single good i changes from p_i^0 to p_i' , then

$$\begin{aligned} CV &= \int_{p_i'}^{p_i^0} \frac{\partial e(\mathbf{p}, u^0)}{\partial p_i} dp_i = \int_{p_i'}^{p_i^0} x_i^H(\mathbf{p}, u^0) dp_i \\ EV &= \int_{p_i'}^{p_i^0} \frac{\partial e(\mathbf{p}, u')}{\partial p_i} dp_i = \int_{p_i'}^{p_i^0} x_i^H(\mathbf{p}, u') dp_i \\ \Delta CS &= \int_{p_i'}^{p_i^0} x_i^M(\mathbf{p}, m) dp_i \end{aligned}$$

Suppose the price of good 1 increases from p_1^0 to p_1' . The change is depicted in the following graph, with Marshallian demand, Hicksian demands and the change of CV , EV and ΔCS . Try to understand the shaded areas in the graph.



Remark.

1. If the Marshallian demand curve is steeper than the Hicksian demand curve, it implies that the good is an inferior good. In the graph, the represented good is a normal good.
2. On any range where the good in question is either normal or inferior, then:

$$\min\{CV, EV\} \leq \Delta CS \leq \max\{CV, EV\}.$$

Notice that the two cases are not exhaustive. For example, a good may be normal good at some lower range of price, but reversed to inferior good at higher range

of price.

Example.

Suppose a consumer has a locally non-satiated and strictly convex preference relation on $\mathbb{R}_+^2$ that can be represented by a twice continuously differentiable utility function $u(x_1, x_2) \geq 0$. Moreover, for $(p_1, p_2) \gg \mathbf{0}$ and $u \geq 0$, the expenditure function is given by:

$$e(p_1, p_2, u) = \frac{p_1 p_2 u^2}{p_1 + p_2}$$

1. For $(p_1, p_2) \gg \mathbf{0}$ and $u > 0$, derive the Hicksian demand $\mathbf{x}^H(p_1, p_2, u)$.
2. For $(p_1, p_2, m) \gg \mathbf{0}$, derive the Marshallian demand $\mathbf{x}^M(p_1, p_2, m)$.
3. Now suppose $p_2 = 1$ and $m = 2$. Consider a price drop from $p_1^0 = 2$ to $p_1' = 1$. Calculate the compensating variation (CV), the equivalent variation (EV), and the change in consumer surplus (ΔCS) of this price change.

Solution.

1. By Shepard's Lemma,

$$\begin{aligned} x_1^H(p_1, p_2, u) &= \frac{\partial e(p_1, p_2, u)}{\partial p_1} = \frac{p_2^2 u^2}{(p_1 + p_2)^2} \\ x_2^H(p_1, p_2, u) &= \frac{\partial e(p_1, p_2, u)}{\partial p_2} = \frac{p_1^2 u^2}{(p_1 + p_2)^2} \end{aligned}$$

It follows that $\mathbf{x}^H(p_1, p_2, u) = \left(\frac{p_2^2 u^2}{(p_1 + p_2)^2}, \frac{p_1^2 u^2}{(p_1 + p_2)^2} \right)$.

2. By duality,

$$e(p_1, p_2, v(\mathbf{p}, m)) = \frac{p_1 p_2 v(\mathbf{p}, m)^2}{p_1 + p_2} = m.$$

We then have $v(\mathbf{p}, m)^2 = \frac{p_1 + p_2}{p_1 p_2} m$.

Again by duality,

$$\mathbf{x}^M(p_1, p_2, m) = \mathbf{x}^H(p_1, p_2, v(\mathbf{p}, m)) = \left(\frac{p_2 m}{p_1(p_1 + p_2)}, \frac{p_1 m}{p_2(p_1 + p_2)} \right).$$

3. Apply the definition of CV , EV and ΔCS :

$$\begin{cases} CV = e(p_1^0, p_2, u^0) - e(p_1', p_2, u') = m - e(p_1', p_2, u^0) \\ EV = e(p_1^0, p_2, u') - e(p_1^0, p_2, u') = e(p_1^0, p_2, u') - m \\ \Delta CS = \int_1^2 x_1^M(p_1, p_2, m) dp_1 = 2 \ln \frac{4}{3} = 0.58 \end{cases}$$

Notice that $u^0 = v(p_1^0, p_2, m) = \sqrt{3}$ and $u' = v(p_1', p_2, m) = 2$. Then the results

are:

$$\begin{cases} CV = \frac{1}{2} \\ EV = \frac{2}{3} \\ \Delta CS = 2 \ln \frac{4}{3} \approx 0.58 \end{cases}$$

Note that typically we have $\min\{CV, EV\} \leq \Delta CS \leq \max\{CV, EV\}$. This can be used to double check the “correctness” of your result.

There is one thing noteworthy about aggregation. While we have laid the foundation for individually analyzing consumer’s utility maximization problem, it may fail if we directly make the aggregation.

Example.

Consider two consumers 1 and 2, whose preferences can be represented by the following utility functions:

$$u^1(x_1, x_2) = \begin{cases} x_1 x_2^3 & \text{if } 0 \leq x_2 \leq 7.7 \\ (7.7)^3 x_1 & \text{if } x_2 \geq 7.7 \end{cases}$$

$$u^2(x_1, x_2) = \begin{cases} x_1^3 x_2 & \text{if } x_1 \geq 3x_2 \\ \frac{1}{3} x_1^4 & \text{if } 0 \leq x_1 \leq 3x_2 \end{cases}$$

Consider the following budget sets:

- Budget set A : $p_1 = p_2 = 2$, $m = 20$.
- Budget set B : $p_1 = 3$, $p_2 = 1$, $m = 20$.

Intuitively, the failure of aggregation is due to *diverse income effects*. For instance, in the example above, the price change has a positive income effect on consumer 1, but a negative income effect on consumer 2. Aggregation is possible in the special case where all consumers have the same wealth effect, that is, $\frac{\partial x^i}{\partial m^i} = \frac{\partial x^j}{\partial m^j}$, for every two consumers i, j and p, m^i, m^j .

Chapter 3

Production Theory

3.1 Setups

First of all, we need the very assumptions for firms generating profits.

Assumption 3.1.1

1. *Perfect/complete* information: no uncertainty about input/output prices, production technology, etc.
2. *Perfectly competitive* input and output markets: firms are price-takers in both input and output markets.
3. Input/output prices are *linear*, justified by perfectly competitive markets.
4. Goods are perfectly *divisible*.
5. The technology is *exogenously given*.
6. The firm's managers are *perfectly controlled by the owners/shareholders*.

Remark.

1. Assumptions 1, 2 and 6 are crucial for the firm's objective of maximization:
 - Assumption 1: consider different risk preferences.
 - Assumption 2: consider a owner who also controls part of the input/output market.
 - Assumption 6: consider agency problem.
2. Firm's profit maximization object is determined by the sixth assumptions, instead of any other assumption.

Justification: Consider a firm jointly owned by I consumers, with consumer i 's share given $\theta_i \geq 0$ (notice that $\sum_{i \in I} \theta_i = 1$). Then consumer i 's utility maximiza-

tion problem is given by:

$$\begin{aligned} & \max_{\mathbf{x}_i \geq \mathbf{0}} u_i(\mathbf{x}_i) \\ & \text{s.t. } \mathbf{p} \cdot \mathbf{x}_i \leq m_i + \theta_i \mathbf{p} \cdot \mathbf{y} \end{aligned}$$

Recall that when the preference relation is locally non-satiated, the indirect utility function $v(\mathbf{p}, m)$ is strictly increasing in m , so every consumer i would unanimously agree that a higher profit $\mathbf{p} \cdot \mathbf{y}$ is strictly more preferred.

3.1.1 Production Set

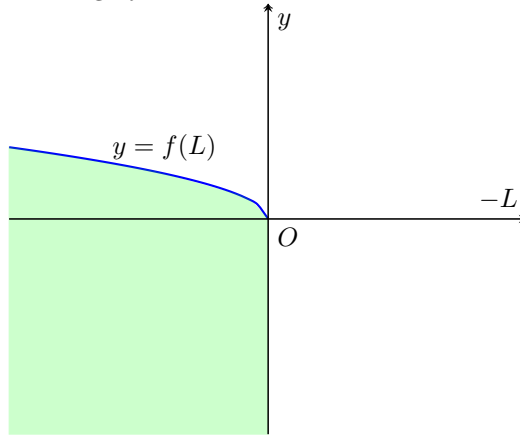
Definition 3.1.2: Production Plan; Production Set

A *production plan* is a vector $\mathbf{y} = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$, where y_i can be either positive or negative, with $y_i > 0$ standing for an output and $y_i < 0$ an input.

The *production set* of a firm is described by $Y \subset \mathbb{R}^n$, where any $\mathbf{y} \in Y$ is a *feasible* production plan, i.e., a production plan the firm can choose from.

Example.

For example, consider the example of one input and one output, suppose the production set is given by $Y = \{(-L, y) : y \leq f(L), L \geq 0\}$, where $f(\cdot)$ is the production function. The production set Y is the gray-shaded area:



Throughout our analysis, we will make the innocent technical assumptions that Y is *non-empty* (so as to have something to study), *closed* (to help ensure the existence of optimal production plans), and $Y \neq \mathbb{R}^n$ (so that there is some scarcity). In addition to those, some other assumptions are needed to make the problem more practical.

Assumption 3.1.3: Production Set

1. $Y \neq \emptyset$.
2. Y is *closed*.
3. *No free lunch* and the possibility of *shutdown*: $Y \cap \mathbb{R}_+^n = \{\mathbf{0}\}$.
4. *Free disposal*: $\mathbf{y} \in Y \implies \mathbf{y}' \in Y$, for any $\mathbf{y}' \leq \mathbf{y}$.

Remark.

Recall the distinction between the short run and the long run from intermediate microeconomics:

- Short run: some inputs are fixed.
- Long run: all inputs are variable.

In our discussion, we will mostly focus on the **long run** in advanced microeconomics. That is, all inputs are by default changeable.

3.1.2 Firm's Profit Maximization Problem

In the spirit of rational decision-making, the firm's problem can be framed as choosing the profit-maximizing production plan from its production set.

Definition 3.1.4: Profit Maximization Problem

$$\begin{aligned} & \max_{\mathbf{y}} \mathbf{p} \cdot \mathbf{y} \\ \text{s.t. } & \mathbf{y} \in Y \end{aligned}$$

Similarly, we define the firm's profit function and optimal supply correspondence, following the same logic with consumer theory.

Definition 3.1.5: Profit Function

Profit function is defined as the optimal value function of the firm's profit given $\mathbf{p}$:

$$\pi(\mathbf{p}) = \sup_{\mathbf{y} \in Y} \mathbf{p} \cdot \mathbf{y}.$$

Definition 3.1.6: Optimal Supply Correspondence

Optimal supply correspondence is defined as the firm's optimal choice(s) given $\mathbf{p}$:

$$\mathbf{y}^*(\mathbf{p}) = \arg \max_{\mathbf{y} \in Y} \mathbf{p} \cdot \mathbf{y} = \{\mathbf{y} \in Y : \mathbf{p} \cdot \mathbf{y} = \pi(\mathbf{p})\}.$$

Remark.

- We have not made sufficient assumptions to ensure that a maximum profit is achieved (i.e., $y^*(\mathbf{p}) \neq \emptyset$), so the sup in profit function cannot always be replaced with the max.

Example.

A firm uses labor and capital to produce its sole output. The production function is given by $f(L, K) = \sqrt{LK}$. Suppose $p = 4$ and $w = r = 1$. If the firm chooses $L = K = t$, the profit is given by $\pi = 2t$, which is unbounded in t .

- $y^*(\mathbf{p})$, the optimal supply correspondence, is a set-valued function, which maps an element from one set, the domain of the function, to a subset of another set.

3.2 Profit Maximization and Rationalizability

Recall our discussion in consumer theory. We will ask similar questions regarding the firm's profit maximization problem.

1. Suppose we observe some or all of the firm's supply decisions $y(\mathbf{p})$ (but not the production set Y), how do we know whether $y(\cdot)$ is rationalizable (i.e., consistent with profit maximization for some production set)?
2. Does the firm's profit maximization problem always have a solution?
3. Properties of the firm's profit function $\pi(\cdot)$ and the optimal supply correspondence $y^*(\cdot)$.
4. Derivation of the firm's profit maximization problem.

Note that these questions are parallel to those asked in "revealed preference" theory, with one important difference: In revealed preference theory, we observe the decision-maker's feasible sets and wish to infer his objective function, while here we do it reversely: we know the objective functions (profits for different prices) and want to infer the feasible set (production set), with the aim of rationalizing the firm's choice.

In practice, we do not know a firm's production set Y , but observe some of its supply choice $y(\mathbf{p})$ for $\mathbf{p} \in \mathbb{R}^n$. Hence, we define rationalizability on empirical meanings.

Definition 3.2.1: Rationalizability

Empirical supply correspondence $y : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ is *rationalized* by production set Y if $y(\mathbf{p}) \subset y^*(\mathbf{p}) = \arg \max_{\mathbf{y} \in Y} \mathbf{p} \cdot \mathbf{y}$ for all $\mathbf{p} \in \mathbb{R}^n$. Empirical supply correspondence $y(\cdot)$ is *rationalizable* if it is rationalized by some production set.

Intuitively, an observed supply correspondence $y(\cdot)$ is rationalizable if we can find a production set Y such that $y(\cdot)$ is consistent with rational decision-making.

Remark.

- We may only observe some of the optimal choice(s) at each price $\mathbf{p}$, so in the definition it writes " $y(\mathbf{p}) \subset y^*(\mathbf{p})$ ".

- Practically, it is not necessary that we observe all of the firm's optimal supply decisions at all prices. We can define without loss that $y(\mathbf{p}_0) = \emptyset$ if the firm's supply decision is not observed at $\mathbf{p}_0$.

We are naturally interested in what we can infer about the production set Y from the empirical observations if the supply choices are rationalizable. Suppose at price $\mathbf{p}$ the firm chooses production plan $y(\mathbf{p})$. Here are two plausible inferences:

1. Plan $y(\mathbf{p})$ must be feasible, i.e., $y(\mathbf{p}) \in Y$.
2. Any production plan $\mathbf{y}$ other than elements in $y(\mathbf{p})$ must generate no more profits than elements in $y(\mathbf{p})$ at price $\mathbf{p}$. Or equivalently, any production plan $\mathbf{y}$ that is more profitable than $y(\mathbf{p})$ at price $\mathbf{p}$ cannot be feasible.

We use the first idea to construct an “inner bound” on Y defined by all choices that the firm has actually made, as they must first be feasible to be chosen. We use the second idea to construct an “outer bound” on Y , which only includes plans that do not give the firm greater profits than its observed choices at any given price $\mathbf{p}$.

Definition 3.2.2: Inner Bound; Outer Bound

Given empirical supply correspondence $y(\cdot)$, we define the *inner bound* of the firm's production set as:

$$Y^I = \bigcup_{\mathbf{p} \in \mathbb{R}^n} y(\mathbf{p}),$$

and the *outer bound* of the firm's production set as:

$$Y^O = \{\mathbf{y} \in \mathbb{R}^n : \mathbf{p} \cdot \mathbf{y} \leq \mathbf{p} \cdot \mathbf{y}_p, \forall \mathbf{p} \in \mathbb{R}^n \text{ and } \mathbf{y}_p \in y(\mathbf{p})\}.$$

Intuitively, any optimal supply choice(s) must be feasible, so $Y^I \subset Y$; and any feasible production plan cannot yield a strictly higher profit at any price, so $Y \subset Y^O$. This intuition is formally characterized in the following proposition.

Proposition 3.2.3: Rationalizable Empirical Supply Correspondence

Production set Y rationalizes empirical supply correspondence $y(\cdot)$ if and only if

$$Y^I \subset Y \subset Y^O.$$

Proof for Proposition.

1. “Only if”
 - First consider any $\mathbf{z} \in Y^I$. By definition of Y^I , there exists a $\mathbf{p}$ such that $\mathbf{z} \in y(\mathbf{p})$. Since $y(\cdot)$ is rationalizable, $y(\mathbf{p}) \subset y^*(\mathbf{p}) \subset Y$. It follows that $\mathbf{z} \in Y$ and $Y^I \subset Y$.
 - Next consider any $\mathbf{y} \in Y$ and $\mathbf{p} \in \mathbb{R}^n$. Since $y(\cdot)$ is rationalizable, $y(\mathbf{p}) \subset y^*(\mathbf{p})$. By the definition of $y^*(\mathbf{p})$, for any $\mathbf{y}_p \in y^*(\mathbf{p})$, $\mathbf{p} \cdot \mathbf{y} \leq \mathbf{p} \cdot \mathbf{y}_p$. It follows that $\mathbf{y} \in Y^O$ and $Y \subset Y^O$.

2. “If”

- Fix $\mathbf{p} \in \mathbb{R}^n$ and consider any $\mathbf{y}_{\mathbf{p}} \in y(\mathbf{p})$.
- Since $y(\mathbf{p}) \subset Y^I \subset Y$, $\mathbf{y}_{\mathbf{p}} \in Y$.
- Moreover, for any $\mathbf{y} \in Y \subset Y^O$, $\mathbf{p} \cdot \mathbf{y} \leq \mathbf{p} \cdot \mathbf{y}_{\mathbf{p}}$. Consequently, $\mathbf{y}_{\mathbf{p}} \in y^*(\mathbf{p})$.

Remark.

For the “if” part, by definition of Y^O , if we fix any $\mathbf{p} \in \mathbb{R}^n$, and take $\mathbf{y}_{\mathbf{p}} \in y(\mathbf{p})$, $\mathbf{y}_{\mathbf{p}}$ maximizes $\mathbf{p} \cdot \mathbf{y}_{\mathbf{p}}$. However, with this condition we cannot simply conclude that $y(\cdot)$ is rationalizable, because $\mathbf{y}_{\mathbf{p}}$ has to be in the production set Y , though it sounds trivial; or equivalently speaking, $\mathbf{y}_{\mathbf{p}} \in Y^I$ may not fall in the outer bound Y^O .

The proposition indicates that, Y^I and Y^O carry all the information we have about the production set based on rational decision-making. The proposition immediately implies the following two corollaries:

Corollary 3.2.4: Weak Axiom of Profit Maximization (WAPM)

Let $P = \{\mathbf{p} \in \mathbb{R}^n : y(\mathbf{p}) \neq \emptyset\}$. Empirical supply correspondence $y(\cdot)$ is rationalizable if and only if $Y^I \subset Y^O$, that is, $\mathbf{p} \cdot \mathbf{y}_{\mathbf{p}} \geq \mathbf{p} \cdot \mathbf{y}_{\mathbf{p}'}$, for any $\mathbf{p} \in P$ and $\mathbf{y}_{\mathbf{p}} \in y(\mathbf{p})$, $\mathbf{y}_{\mathbf{p}'} \in y(\mathbf{p}')$.

One simple consequence of this characterization is that, when checking rationalizability, we can restrict attention to supply functions rather than correspondences. (Simply put, compared with the preceding proposition, the production set Y is “left out” here.)

WAPM directly implies “law of supply”.

Corollary 3.2.5: Law of Supply

Suppose empirical supply correspondence $y(\cdot)$ is rationalizable and let $P = \{\mathbf{p} \in \mathbb{R}^n : y(\mathbf{p}) \neq \emptyset\}$. Then for any $\mathbf{p} \in P$ and $\mathbf{y}_{\mathbf{p}} \in y(\mathbf{p})$, $\mathbf{y}_{\mathbf{p}'} \in y(\mathbf{p}')$,

$$(\mathbf{p} - \mathbf{p}') \cdot (\mathbf{y}_{\mathbf{p}} - \mathbf{y}_{\mathbf{p}'}) \geq 0.$$

Proof for Corollary.

Since $y(\cdot)$ is rationalizable, by WAPM, we have

$$\mathbf{p} \cdot \mathbf{y}_{\mathbf{p}} \geq \mathbf{p} \cdot \mathbf{y}_{\mathbf{p}'}.$$

Switching the role of $\mathbf{p}$ and $\mathbf{p}'$, we have

$$\mathbf{p}' \cdot \mathbf{y}_{\mathbf{p}'} \geq \mathbf{p}' \cdot \mathbf{y}_{\mathbf{p}}.$$

Adding the two equations above, we get the “law of supply”.

In particular, if there is a single output and $y(\cdot)$ is single-valued, then

$$(p - p')(y(p) - y(p')) \geq 0$$

In other words, any rationalizable supply function must be **(weakly) upward sloping**.

Corollary 3.2.6

Let $P = \{p \in \mathbb{R}^n : y(p) \neq \emptyset\}$. Empirical supply correspondence $y(\cdot)$ is rationalizable if and only if:

1. Any selection $\hat{y} : P \rightarrow \mathbb{R}^n$ is rationalizable.
2. For any two selections $\hat{y}$ and $\tilde{y}$ and any $p \in P$, $p \cdot \hat{y}(p) = p \cdot \tilde{y}(p)$. (Or equivalently, $\pi(p)$ is single-valued for each $p \in P$.)

Remark.

- The first statement of this corollary is equivalent to WAPM applied to $p' \neq p$,
- The second statement of this corollary is equivalent to WAPM applied to $p' = p$.
- Thus, when given a supply correspondence, we only need to check that
 1. Each selection from it is a rationalizable supply function, and
 2. The profit function $\pi(p) = p \cdot y$ is single-valued at any given $p \in P$; or equivalently speaking, $\pi(p)$ does not depend on which selection is chosen.

Since checking the second condition is trivial, we can focus on rationalizability of supply functions (single-valued correspondence).

Verifying rationalizability by checking all the WAPM inequalities is difficult when the set of observations is large. Fortunately, it turns out that when we have a continuum of observations, rationalizability can be verified much more easily using differential conditions. Specifically, we now suppose that we observe the firm's supply choices on an open convex set P of prices (e.g., P could be the set of all strictly positive price vectors).

Proposition 3.2.7: Rationalizability: Differentiable Case

Consider an empirical supply correspondence $y(\cdot)$ whose domain $P = \{p \in \mathbb{R}^n : y(p) \neq \emptyset\}$ is an open convex set. Suppose that $\pi(p) = p \cdot y(p)$ is a differentiable function on $p \in P$. Then $y(\cdot)$ is rationalizable if and only if:

1. (**Hotelling's Lemma**) $\nabla \pi(p) = y_p$, for any $p \in P$ and $y_p \in y(p)$.
2. $\pi(\cdot)$ is a convex function.

Proof for Proposition.

1. "If"

- Fix $\mathbf{q} \in P$ and take any $\mathbf{y}_{\mathbf{q}} \in y(\mathbf{q})$. Consider the “difference function”: $G(\mathbf{q}; \mathbf{p}) = \mathbf{p} \cdot \mathbf{y}_{\mathbf{q}} - \pi(\mathbf{p})$.
- It suffices to show that, $G(\mathbf{q}; \cdot)$ is maximized at $\mathbf{p} = \mathbf{q}$.
- Since $\pi(\cdot)$ is a convex function, then $G(\mathbf{q}; \cdot)$ is a concave function. Since $G(\mathbf{q}; \cdot)$ is differentiable in $\mathbf{p}$, the first-order condition is both necessary and sufficient.
- The F.O.C.: $\mathbf{y}_{\mathbf{q}} - \nabla \pi(\mathbf{p})|_{\mathbf{p}=\mathbf{q}} = 0$, which is precisely the Hotelling’s Lemma.

2. “Only if”

- The proof above also shows WAPM implies the Hotelling’s lemma. Indeed, it is just an application of envelope formula. It remains to show $\pi(\cdot)$ is a convex function.
- By rationalizability, there exists Y such that $\pi(\mathbf{p}) = \sup_{\mathbf{y} \in Y} \mathbf{p} \cdot \mathbf{y}$.
- Take any $\mathbf{p}, \mathbf{q} \in P$ and $t \in (0, 1)$. If $\pi(\mathbf{p}) = +\infty$ or $\pi(\mathbf{q}) = +\infty$, then clearly $\pi(t\mathbf{p} + (1-t)\mathbf{q}) \leq t\pi(\mathbf{p}) + (1-t)\pi(\mathbf{q})$. Otherwise,

$$\begin{aligned} \pi(t\mathbf{p} + (1-t)\mathbf{q}) &= \sup_{\mathbf{y} \in Y} (t\mathbf{p} + (1-t)\mathbf{q}) \cdot \mathbf{y} \\ &\leq t \cdot \sup_{\mathbf{y} \in Y} \mathbf{p} \cdot \mathbf{y} + (1-t) \cdot \sup_{\mathbf{y} \in Y} \mathbf{q} \cdot \mathbf{y} \\ &= t\pi(\mathbf{p}) + (1-t)\pi(\mathbf{q}) \end{aligned}$$

Proposition 3.2.8: Rationalizability: General Case

Consider an empirical supply function $y : P \rightarrow \mathbb{R}^n$, where $P \subset \mathbb{R}^n$ is a convex set. $y(\cdot)$ is rationalizable if and only if:

1. (**Producer Surplus Formula**): $\pi(\mathbf{p}) = \mathbf{p} \cdot y(\mathbf{p})$ satisfies, for any smooth path $\rho : [0, 1] \rightarrow P$, with $\rho(0) = \mathbf{p}$ and $\rho(1) = \mathbf{p}'$,

$$\pi(\mathbf{p}') - \pi(\mathbf{p}) = \int_0^1 y(\rho(t)) \cdot \rho'(t) dt.$$

2. (**Law of Supply**): For $\mathbf{p}, \mathbf{p}' \in P$,

$$(\mathbf{p} - \mathbf{p}') \cdot (y(\mathbf{p}) - y(\mathbf{p}')) \geq 0.$$

Proof for Proposition.

- “Only if” part

We have already shown the “law of supply”, so it suffices to show the “producer surplus formula”.

Let $\phi(t) = \pi(\rho(t))$. Consider the “difference function”:

$$\begin{aligned}\delta(\theta; t) &:= \rho(t) \cdot y(\rho(\theta)) - \pi(\rho(t)) \\ &= \rho(t) \cdot y(\rho(\theta)) - \phi(t)\end{aligned}$$

By rationalizability of $y(\cdot)$,

$$\begin{aligned}\delta(\theta; t) &\leq 0 = \delta(\theta; \theta) \\ \implies \left. \frac{\partial \delta(\theta; t)}{\partial t} \right|_{t=\theta} &= 0 \\ \iff \left. \frac{\partial \delta(\theta; t)}{\partial t} \right|_{t=\theta} &= y(\rho(\theta)) \cdot \rho'(\theta) - \phi'(\theta) = 0 \\ \iff \phi'(\theta) &= y(\rho(\theta)) \cdot \rho'(\theta) \\ \implies \phi(1) - \phi(0) &= \int_0^1 y(\rho(t)) \cdot \rho'(t) dt \\ \implies \pi(\mathbf{p}') - \pi(\mathbf{p}) &= \int_0^1 y(\rho(t)) \cdot \rho'(t) dt\end{aligned}$$

In fact in order to derive the integral result from the derivatives, we have to prove the continuity of $\phi(\cdot)$. It can be shown that $\phi(\cdot)$ is Lipschitz continuous and hence absolutely continuous. The proof is rather technical and thus omitted here.

- “If” part

In order to show the rationalizability of $y(\cdot)$, it suffices to show WAPM. Because the path integral is path-independent, take a straight line for math convenience:

$$\rho(t) = \mathbf{p} + t(\mathbf{p}' - \mathbf{p}).$$

We aim to prove

$$\pi(\mathbf{p}') - \mathbf{p}' \cdot y(\mathbf{p}) \geq 0.$$

Making tweaks to the difference in profit:

$$\begin{aligned}\pi(\mathbf{p}') - \mathbf{p}' \cdot y(\mathbf{p}) &= \pi(\mathbf{p}') - \pi(\mathbf{p}) + \pi(\mathbf{p}) - \mathbf{p}' \cdot y(\mathbf{p}) \\ &= (\pi(\mathbf{p}') - \pi(\mathbf{p})) - (\mathbf{p}' - \mathbf{p}) \cdot y(\mathbf{p}) \\ &= \int_0^1 y(\rho(t)) \cdot \rho'(t) dt - ((\mathbf{p}' - \mathbf{p}) \cdot y(\mathbf{p})) \\ &= \int_0^1 y(\mathbf{p} + t(\mathbf{p}' - \mathbf{p})) \cdot \rho'(t) dt - ((\mathbf{p}' - \mathbf{p}) \cdot y(\mathbf{p})) \\ &= \int_0^1 (\mathbf{p}' - \mathbf{p}) \cdot [y(\mathbf{p} + t(\mathbf{p}' - \mathbf{p})) - y(\mathbf{p})] dt\end{aligned}$$

Let

$$\begin{cases} \mathbf{q}' = \mathbf{p} + t(\mathbf{p}' - \mathbf{p}) \\ \mathbf{q} = \mathbf{p} \end{cases}$$

One direct observation is that $\mathbf{q}' - \mathbf{q} = t(\mathbf{p}' - \mathbf{p})$. Therefore, we can simplify the preceding integral as:

$$\int_0^1 (\mathbf{p}' - \mathbf{p}) \cdot [y(\mathbf{p} + t(\mathbf{p}' - \mathbf{p})) - y(\mathbf{p})] dt = \int_0^1 \frac{1}{t} (\mathbf{q}' - \mathbf{q}) \cdot (y(\mathbf{q}') - y(\mathbf{q})) dt.$$

By law of supply, $(\mathbf{q}' - \mathbf{q}) \cdot (y(\mathbf{q}') - y(\mathbf{q})) \geq 0$. Therefore,

$$\int_0^1 \frac{1}{t} (\mathbf{q}' - \mathbf{q}) \cdot (y(\mathbf{q}') - y(\mathbf{q})) dt \geq 0.$$

Example.

Consider a price-taking firm with m inputs and n outputs. Suppose we can observe the firm's input choices $\mathbf{z}$ for different input prices $\mathbf{w} \in \mathbb{R}_+^m$, but cannot observe its output choices or output prices (and may not even know the number of outputs n). We do know that output prices, whatever they are, do not change between the observations.

1. Suppose $m = 2$, and that at input prices $\mathbf{w}^1 = (1, 1)$ the firm chooses the input vector $\mathbf{z}^1 = (10, 15)$ and at input prices $\mathbf{w}^2 = (2, 3)$ it chooses the input vector $\mathbf{z}^2 = (13, 14)$. Is this pair of observations rationalizable (i.e., consistent with profit maximization for some production set and output prices)?
2. In general, give a necessary and sufficient condition for two input price-demand observations $\mathbf{z}^1, \mathbf{w}^1 \in \mathbb{R}_+^m$ and $\mathbf{z}^2, \mathbf{w}^2 \in \mathbb{R}^m$ to be rationalizable. Prove both necessity and sufficiency.
3. Now suppose instead that we have the following observations:
 - At prices $\mathbf{w}^1 = (1, 1)$, the firm chooses the input vector $\mathbf{z}^1 = (10, 15)$;
 - At prices $\mathbf{w}^2 = (2, 3)$, the firm chooses the input vector $\mathbf{z}^2 = (13, 13)$;
 - At prices $\mathbf{w}^3 = (4, 1)$, the firm chooses the input vector $\mathbf{z}^3 = (8, 9)$.

Are these three observations jointly rationalizable?

Solution.

1. Suppose instead we know the output price $\mathbf{p}$ and output vectors $\mathbf{y}^1, \mathbf{y}^2$, then by WAPM,

$$\begin{aligned} & \begin{cases} (\mathbf{w}^1, \mathbf{p}) \cdot (-\mathbf{z}^1, \mathbf{y}^1) \geq (\mathbf{w}^1, \mathbf{p}) \cdot (-\mathbf{z}^2, \mathbf{y}^2) \\ (\mathbf{w}^2, \mathbf{p}) \cdot (-\mathbf{z}^2, \mathbf{y}^2) \geq (\mathbf{w}^2, \mathbf{p}) \cdot (-\mathbf{z}^1, \mathbf{y}^1) \end{cases} \\ \implies & \begin{cases} \mathbf{p} \cdot \mathbf{y}^1 - 25 \geq \mathbf{p} \cdot \mathbf{y}^2 - 27 \\ \mathbf{p} \cdot \mathbf{y}^2 - 68 \geq \mathbf{p} \cdot \mathbf{y}^1 - 75 \end{cases} \\ \implies & -2 \leq \mathbf{p} \cdot \mathbf{y}^1 - \mathbf{p} \cdot \mathbf{y}^2 \leq -3, \end{aligned}$$

which apparently leads to a contradiction.

2. In the same way, rationalizability requires that

$$\begin{aligned} & \begin{cases} (\mathbf{w}^1, \mathbf{p}) \cdot (-\mathbf{z}^1, \mathbf{y}^1) \geq (\mathbf{w}^1, \mathbf{p}) \cdot (-\mathbf{z}^2, \mathbf{y}^2) \\ (\mathbf{w}^2, \mathbf{p}) \cdot (-\mathbf{z}^2, \mathbf{y}^2) \geq (\mathbf{w}^2, \mathbf{p}) \cdot (-\mathbf{z}^1, \mathbf{y}^1) \end{cases} \\ \implies & \mathbf{w}^1 \cdot (\mathbf{z}^1 - \mathbf{z}^2) \leq \mathbf{p} \cdot (\mathbf{y}^1 - \mathbf{y}^2) \leq \mathbf{w}^2 \cdot (\mathbf{z}^1 - \mathbf{z}^2) \\ \implies & (\mathbf{w}^1 - \mathbf{w}^2) \cdot (\mathbf{z}^1 - \mathbf{z}^2) \leq 0. \end{aligned}$$

So $(\mathbf{w}^1 - \mathbf{w}^2) \cdot (\mathbf{z}^1 - \mathbf{z}^2) \leq 0$ is a necessary condition. Then prove it is also sufficient.

Take any output price $\mathbf{p}$ and output vectors $\mathbf{y}^1, \mathbf{y}^2$ that satisfies $\mathbf{w}^1 \cdot (\mathbf{z}^1 - \mathbf{z}^2) \leq \mathbf{p} \cdot (\mathbf{y}^1 - \mathbf{y}^2) \leq \mathbf{w}^2 \cdot (\mathbf{z}^1 - \mathbf{z}^2)$. It is trivial that output price $\mathbf{p}$ and production set $Y = \{(-\mathbf{z}^1, \mathbf{y}^1), (-\mathbf{z}^2, \mathbf{y}^2)\}$ rationalize the pair of observations.

3. Again, when try to rationalize those choices, use necessary condition

$$\begin{cases} (\mathbf{w}^1, \mathbf{p}) \cdot (-\mathbf{z}^1, \mathbf{y}^1) \geq (\mathbf{w}^1, \mathbf{p}) \cdot (-\mathbf{z}^2, \mathbf{y}^2) \\ (\mathbf{w}^2, \mathbf{p}) \cdot (-\mathbf{z}^2, \mathbf{y}^2) \geq (\mathbf{w}^2, \mathbf{p}) \cdot (-\mathbf{z}^1, \mathbf{y}^1) \end{cases}$$

for pairwise checks. Then we can get

$$\begin{aligned} -1 & \leq \mathbf{p} \cdot (\mathbf{y}^1 - \mathbf{y}^2) \leq 0 \\ 22 & \leq \mathbf{p} \cdot (\mathbf{y}^2 - \mathbf{y}^3) \leq 24 \\ -14 & \leq \mathbf{p} \cdot (\mathbf{y}^3 - \mathbf{y}^1) \leq -8 \end{aligned}$$

Even though at first glance there is no apparent contradiction, if we try to take the sum of the first two inequalities:

$$\begin{aligned} & \begin{cases} -1 \leq \mathbf{p} \cdot (\mathbf{y}^1 - \mathbf{y}^2) \leq 0 \\ 22 \leq \mathbf{p} \cdot (\mathbf{y}^2 - \mathbf{y}^3) \leq 24 \end{cases} \\ \implies & 21 \leq \mathbf{p} \cdot (\mathbf{y}^1 - \mathbf{y}^3) \leq 24 \end{aligned}$$

which contradicts the third equation. Thus, the choices cannot be rationalized.

3.3 Profit Maximization Problem

Questions for analysis on firm's profit maximization problem are much of the same to those in consumer's utility maximization problem:

1. Does the firm's profit maximization problem always have a solution?
2. Properties of the firm's profit function $\pi(\cdot)$ and optimal supply correspondence $y^*(\cdot)$.
3. Derivation of the firm's profit maximization problem.

3.3.1 Returns to Scale

Recall the earlier counter-example for the non-existence of optimal supply correspondence. Intuitively, in some cases the firm can simply replicate its existing production plan infinitely many times and earns infinitely increasing profits. This motivates the following definitions.

Definition 3.3.1: Returns to Scale

The production set Y exhibits:

- *non-increasing returns to scale* if $\mathbf{y} \in Y \implies t\mathbf{y} \in Y, \forall t \in [0, 1]$.
- *non-decreasing returns to scale* if $\mathbf{y} \in Y \implies t\mathbf{y} \in Y, \forall t \in [1, +\infty)$.
- *constant returns to scale* if $\mathbf{y} \in Y \implies t\mathbf{y} \in Y, \forall t \geq 0$.

If a firm shows non-decreasing returns to scale, then intuitively there is no scarcity in its production ability. In the special case, there are exactly two possibilities: earning nothing or earning everything.

Proposition 3.3.2

If the production set $Y \neq \emptyset$ exhibits non-decreasing returns to scale, then for any $\mathbf{p} \in \mathbb{R}^n$, $\pi(\mathbf{p}) = 0$ or ∞ .

Proof for Proposition.

First fix any $\mathbf{p} \in \mathbb{R}^n$. By the possibility of inaction or shutdown, $\mathbf{0} \in Y$, so $\pi(\mathbf{p}) \geq \mathbf{p} \cdot \mathbf{0} = 0$. Now suppose instead that $0 < \pi(\mathbf{p}) < \infty$, then there must exist $\mathbf{y}_0 \in Y$ such that $\mathbf{p} \cdot \mathbf{y}_0 > \pi(\mathbf{p}) - \varepsilon > 0$, for any $\varepsilon > 0$ small enough. By non-decreasing returns to scale, $t\mathbf{y}_0 \in Y$, for any $t \geq 1$. Notice that $\mathbf{p} \cdot (t\mathbf{y}_0) = t(\mathbf{p} \cdot \mathbf{y}_0) > t\pi(\mathbf{p}) - t\varepsilon > \pi(\mathbf{p})$, for any $t > 1$ and $\varepsilon > 0$ small, which is a contradiction. It follows that $\pi(\mathbf{p}) = 0$ or ∞ . ■

3.3.2 Properties of Profit Function and Supply Correspondence

Proposition 3.3.3: Properties of Profit Function and Supply Correspondence

Suppose the production set Y is closed and satisfies the free disposal property. Let $\pi(\cdot)$ be the profit function and $y^*(\cdot)$ the associated optimal supply correspondence. Then for $\mathbf{p} \gg \mathbf{0}$,

- $\pi(\cdot)$ is homogeneous of degree 1.
- $\pi(\cdot)$ is a convex function.
- $y^*(\cdot)$ is homogeneous of degree 0.
- If Y is a convex set, then $y^*(\mathbf{p})$ is a convex set for all $\mathbf{p} \gg \mathbf{0}$. If Y is a strictly convex set, then $y^*(\mathbf{p})$ is either empty or single-valued.
- (Hotelling's Lemma) If $y^*(\mathbf{p})$ is single-valued, then $\pi(\cdot)$ is differentiable at $\mathbf{p}$ and $\nabla \pi(\mathbf{p}) = y^*(\mathbf{p})$, that is,

$$\frac{\partial \pi(\mathbf{p}_i)}{\partial p_i} = y_i^*(\mathbf{p}), \text{ for } i = 1, 2, \dots, n.$$

3.3.3 Derivation of Profit Maximization Problem

In preceding discussions, we seldom delve into the benchmark of judging if a production plan falls within the production set, i.e., being feasible. One convenient way to represent production possibility sets is using a transformation function $T : \mathbb{R}^n \rightarrow \mathbb{R}$, where $T(y) \leq 0$ implies that y is feasible, and $T(y) \geq 0$ implies that y is infeasible. The set of boundary points $\{y \in \mathbb{R}^n : T(y) = 0\}$ is called the transformation frontier.

Definition 3.3.4: Profit Maximization Problem

Suppose $T(\cdot)$ is the transformation function defining the production set:

$$\begin{aligned} & \max_{\mathbf{y}} \mathbf{p} \cdot \mathbf{y} \\ & \text{s.t. } T(\mathbf{y}) \leq 0 \end{aligned}$$

The Lagrangian is given by:

$$\mathcal{L}(\mathbf{y}, \lambda) = \mathbf{p} \cdot \mathbf{y} - \lambda T(\mathbf{y}) = \sum_{i=1}^n p_i y_i - \lambda T(\mathbf{y}).$$

The F.O.C.s are:

$$\lambda \cdot \nabla T(\mathbf{y}) = \mathbf{p}.$$

For most discussions in the course, we focus on single-output cases:

$$\begin{aligned} \max_z \quad & p \cdot f(\mathbf{z}) - \sum_{i=1}^m w_i z_i \\ \text{s.t.} \quad & \mathbf{z} \geq \mathbf{0} \end{aligned}$$

In this special case, the Lagrangian is given by:

$$\mathcal{L}(\mathbf{z}, \mu_i) = p \cdot f(\mathbf{z}) - \sum_{i=1}^m w_i z_i + \sum_{i=1}^m \mu_i z_i.$$

The F.O.C. are:

$$\begin{cases} p \cdot \frac{\partial f(\mathbf{z})}{\partial z_i} - w_i + \mu_i = 0 \\ \mu_i z_i = 0, \mu_i \geq 0 \end{cases}, \text{ for all } i = 1, 2, \dots, m.$$

Equivalently, the F.O.C. can be written as

$$p \cdot \frac{\partial f(\mathbf{z})}{\partial z_i} \leq w_i, \text{ with equality if } z_i > 0.$$

Interpretation of the first-order conditions is straightforward and intuitive. The LHS represents the firm's marginal benefit from using an additional unit of input i , while the RHS represents the marginal cost. The F.O.C. require that the marginal benefit cannot exceed the marginal cost, with equality if the input is ever used at the optimum.

Similarly, we should follow systematic procedures to solve the firm's profit maximization problem:

- Check returns to scale and whether the production technology is strictly convex and monotonic.
- If decreasing returns to scale, production technology being monotonic and strictly convex, apply the “tangency conditions”:

$$p \cdot MP_i = w_i, \forall i$$

- Check whether the inputs are non-negative, whether the profit is non-negative, and whether the profit is bounded.

Example.

A firm uses two inputs, labor (L) and capital (K), to produce a single output (Y). The production function is given by:

$$f(L, K) = L^{\frac{1}{4}} K^{\frac{1}{2}}$$

Suppose the input and output prices are $(w, r, p) \gg \mathbf{0}$. Solve the firm's profit maximization problem to derive the profit function $\pi(w, r, p)$ and optimal supply correspondence $y^*(w, r, p)$.

Solution.

- Step 1: The production function is Cobb-Douglas, and hence strictly convex and monotonic. Moreover, if $f(L, K) \geq y$, then $f(tL, tK) = t^{\frac{3}{4}} f(L, K) \geq t(L, K) \geq ty$, for any $0 \leq t \leq 1$, so decreasing returns to scale.
- Step 2: F.O.C. are given by

$$\begin{cases} [L] : p \cdot \frac{\partial f(L, K)}{\partial L} = w \\ [K] : p \cdot \frac{\partial f(L, K)}{\partial K} = r \end{cases} \implies \begin{cases} L^* = \frac{p^4}{64w^2r^2} \\ K^* = \frac{p^4}{32wr^3} \end{cases}$$

- Check non-negativity:

Clearly, $L^*, K^* > 0$. It follows that

$$\begin{cases} y^*(w, r, p) = \left(-\frac{p^4}{64w^2r^2}, -\frac{p^4}{32wr^3}, \frac{p^3}{16wr^2} \right) \\ \pi(w, r, p) = \frac{p^4}{64wr^2} > 0 \end{cases}$$

3.4 Cost Minimization Problem

Given that we already have a direct approach to solve for the firm's profit maximization problem, it is still useful to take a detour and analyze the cost minimization problem for the following two reasons:

- The cost minimization problem is more well-behaved than the profit maximization problem and has close connection with the expenditure minimization problem in consumer theory (e.g., recall the existence of solutions).
- As we shall see later, this indirect approach is more insightful if the firm has monopoly power in the output market (but is still a price-taker in the input market).

3.4.1 Setups and Properties

Definition 3.4.1: Cost Function; Conditional Factor Demand Correspondence

Let $Z(y) = \{z \in \mathbb{R}_+^n : f(z) \geq y\}$ be the firm's feasible set. We define the optimal (minimal) value function as the *cost function*:

$$c(w, y) = \inf_{z \in Z(y)} w \cdot z,$$

and the firm's optimal factor choice(s) as the *conditional factor demand correspondence*:

$$z(w, y) = \{z \in Z(y) : w \cdot z = c(w, y)\}.$$

Notice that $\min f(\mathbf{x})$ is equivalent to $\max(-f(\mathbf{x}))$, so the cost minimization problem can be viewed as profit maximization problem on the restricted production set

$$Y_y = \{(-\mathbf{z}, y) : \mathbf{z} \in \mathbb{R}_+^n, y \leq f(\mathbf{z})\}.$$

From this “nearly”-equivalent relationship between cost minimization problem and profit maximization problem, we immediately arrive at the following proposition of rationalizability in differentiable cases.

Proposition 3.4.2

Consider a conditional factor demand function $\mathbf{z} : W \times \mathbb{R} \rightarrow \mathbb{R}^m$ for a fixed output y on an open convex set $W \subset \mathbb{R}_+^m$ such that $c(\mathbf{w}, y) = \mathbf{w} \cdot \mathbf{z}(\mathbf{w}, y)$ is differentiable in $\mathbf{w}$. Then $\mathbf{z}$ is rationalizable by some production function if and only if,

1. (Shepard’s lemma) $\nabla_{\mathbf{w}} c(\mathbf{w}, y) = \mathbf{z}(\mathbf{w}, y)$.
2. $c(\cdot, y)$ is concave in $\mathbf{w}$.

When learning expenditure minimization problem and Hicksian demand, we skipped the discussions on rationalizability. In fact, expenditure minimization problem essentially works in the same logic with cost minimization problem, so same result holds for the expenditure function $e(\mathbf{p}, u)$.

Proposition 3.4.3: Properties of Cost Function and Conditional Factor Demand Correspondence

Suppose the production function $f(\cdot)$ is continuous and the production set Y satisfies the free disposal property. Then for $\mathbf{w} \gg \mathbf{0}$,

- **Existence of conditional factor demand:** If $Z(y) \neq \emptyset$, then the conditional factor demand correspondence $\mathbf{z}(\mathbf{w}, y) \neq \emptyset$.
- **Structure of conditional factor demand:** If the production technology is convex (i.e., the upper contour set $\{\mathbf{z} \geq \mathbf{0} : f(\mathbf{z}) \geq y\}$ is a convex set for any $y \geq 0$), then $\mathbf{z}(\mathbf{w}, y)$ is a convex set. If the production technology is strictly convex and $Z(y) \neq \emptyset$, then $\mathbf{z}(\mathbf{w}, y)$ is singleton.
- **Homogeneity:** $c(\mathbf{w}, y)$ is homogeneous of degree 1 in $\mathbf{w}$, and $\mathbf{z}(\mathbf{w}, y)$ is homogeneous of degree 0 in $\mathbf{w}$. **If the production function $f(\cdot)$ exhibits constant returns to scale, then $c(\mathbf{w}, y)$ and $\mathbf{z}(\mathbf{w}, y)$ are homogeneous of degree 1 in y .**
- **Monotonicity:** $c(\mathbf{w}, y)$ is non-decreasing in $\mathbf{w}$ and is strictly increasing in y for $y \geq 0$.
- **Binding production level:** For $y > 0$ and $Z(y) \neq \emptyset$, at any minimizer $\mathbf{z}^*$, $f(\mathbf{z}^*) = y$.
- **Convexity:** **If $f(\cdot)$ is a concave function, then $c(\mathbf{w}, \cdot)$ is a convex function of y .**
- **Shepard's lemma:** If $\mathbf{z}(\mathbf{w}, y)$ is single-valued, then $c(\mathbf{w}, y)$ is differentiable with respect to w_i and $\frac{\partial c(\mathbf{w}, y)}{\partial w_i} = z_i(\mathbf{w}, y)$.

Notice that there is no counterpart in EMP for properties in bold. The essential reason is that, in consumer theory we only care about utility representation, whose ordinal meaning matters; in producer theory, however, the production function has cardinal meanings, and that makes the difference.

3.4.2 Derivation of Cost Minimization Problem

Definition 3.4.4: Cost Minimization Problem

$$\begin{aligned} & \min_{\mathbf{z}} \mathbf{w} \cdot \mathbf{z} \\ \text{s.t. } & f(\mathbf{z}) \geq y \\ & z_i \geq 0, \forall i = 1, 2, \dots, m \end{aligned}$$

Notice that CMP is almost identical to EMP in consumer theory. The Lagrangian is

given by:

$$\mathcal{L}(\mathbf{z}; \lambda, \boldsymbol{\mu}) = \mathbf{w} \cdot \mathbf{z} - \lambda(f(\mathbf{z}) - y) - \sum_{i=1}^n \mu_i z_i.$$

The F.O.C.s are given by:

- w.r.t. z_i : $w_i - \lambda \frac{\partial f(\mathbf{z})}{\partial z_i} - \mu_i = 0$.
- Inequality constraints: $f(\mathbf{z}) \geq y$, $z_i \geq 0$, $\lambda \geq 0$, $\mu_i \geq 0$.
- Complementary slackness: $\lambda(f(\mathbf{z}) - y) = 0$, $\mu_i z_i = 0$.

For $y \geq 0$, we have binding production level (i.e., $f(\mathbf{z}) = y$), so the F.O.C. can be alternatively framed as

$$\lambda \frac{\partial f(\mathbf{z})}{\partial z_i} \leq w_i, \text{ with equality if } z_i > 0.$$

The economic intuition of $\lambda \frac{\partial f(\mathbf{z})}{\partial z_i} \leq w_i$ is that, the LHS corresponds to the marginal benefit, or shadow value of additional one unit of input z_i , and the RHS stands for the marginal cost of such investment. The F.O.C. state that at the optimum, the marginal benefit of inputs cannot exceed their marginal cost. Or more precisely, for those deployed inputs, their marginal benefit just equals marginal cost, while for inputs that are not ever invested, their marginal benefit must be no more than their marginal cost, otherwise the firm still have the room to cut down its cost, indicating the current solution has not reached the optimum.

Remark.

Apply envelope theorem to the Lagrangian of CMP, we have

$$\frac{\partial c(\mathbf{w}, y)}{\partial y} = \lambda^*.$$

Specifically, λ^* **measures the firm's marginal cost of producing an additional unit of output** (valued by the firm instead of the market). Hence, another interpretation for the F.O.C. is that, the firm's internal valuation of additional input cannot exceed that of the market. To be specific, the marginal cost of production using factor i , $\frac{w_i}{MP_i}$, should be weakly greater than the marginal cost of producing additional one unit of product, with equality if the factor is employed at the optimum.

With the cost function, we can restate the firm's profit maximization problem in an indirect approach as:

$$\max_{y \geq 0} py - c(\mathbf{w}, y)$$

Clearly, the F.O.C. is given by

$$p \leq \frac{\partial c(\mathbf{w}, y)}{\partial y}, \text{ with equality if } y > 0.$$

If we relax the assumption of perfect competition and instead assume that the firm is a monopolist in the output market but a price-taker in the input market, then the firm

no longer takes the output price as given. We can restate the firm's profit maximization problem as:

$$\max_{y \geq 0} p(y)y - c(\mathbf{w}, y)$$

The F.O.C. is given by

$$p'(y)y + p(y) \geq \frac{\partial c(\mathbf{w}, y)}{\partial y}, \text{ with equality if } y > 0.$$

The interpretation is similar to the perfectly competitive case. This indirect approach proves to be useful and attests to the power of cost minimization problem.

Chapter 4

Comparative Statics Analysis

A central question in economics is how an endogenous variable changes with an exogenous variable. For instance, in the indirect approach of single-product profit maximization:

$$\max_{y \geq 0} py - c(\mathbf{w}, y)$$

Holding the input prices $\mathbf{w}$ as given, we are interested in how the firm's optimal supply (correspondence) y^* changes with the output price p .

More generally, let $F : X \times \Theta \rightarrow \mathbb{R}$, where $X, \Theta \subset \mathbb{R}$, we are interested in how the maximizer

$$x^*(\theta) = \arg \max_{x \in X} F(x, \theta)$$

would change with the parameter θ .

4.1 Univariate Comparative Statics

The classical approach is to apply the implicit function theorem. We need at least two sets of assumptions:

- $F(\cdot, \cdot)$ is twice continuously differentiable.
- The maximizer $x^*(\theta)$ is unique and is characterized by the first-order condition.

Remark.

A set of sufficient conditions for the second assumption is needed:

- The choice set X is convex.
- $F(\cdot, \theta)$ is strictly concave in x .
- The solution is interior.

Under the two sets of assumptions, the first-order condition is:

$$F_x(x(\theta), \theta) = 0.$$

Differentiating on both sides with respect to θ , we have:

$$\begin{aligned} F_{xx}(x(\theta), \theta) \cdot x'(\theta) + F_{x\theta}(x(\theta), \theta) &= 0 \\ \implies x'(\theta) &= -\frac{F_{x\theta}(x(\theta), \theta)}{F_{xx}(x(\theta), \theta)} \end{aligned}$$

If $F(\cdot, \theta)$ is strictly concave in x , then $F_{xx}(\cdot, \cdot) < 0$ (assuming no reflection point), so $x(\cdot)$ is strictly increasing in θ if $F_{x\theta}(\cdot, \cdot) > 0$.

The advantage for the classical method is straightforward that, it provides explicit expression for $x'(\theta)$ and makes it possible for quantitative analysis. The disadvantages, however, grow out from its proposed advantage:

- Technical issue: Strong assumptions and tedious calculation in some examples.
- Substantive issue: The requirement of strict concavity in x counters our intuition. (Recall from consumer theory that the set of maximizers is unaffected by any positive monotonic transformation.)

Example.

Taking w as given, the profit maximization problem is given by

$$\max_{y \geq 0} py - c(y)$$

We use the classical approach to determine whether (and when) the firm's supply curve is (weakly) upward sloping.

Notice that the choice set $Y = [0, +\infty)$ is convex, and when $c(\cdot)$ is strictly convex, the objective function is strictly concave. Assuming interiority, the first-order condition is given by:

$$p = c'(y(p))$$

Differentiating on both sides with respect to p , we get

$$y'(p) = \frac{1}{c''(y(p))}$$

Thus, $c(\cdot)$ being strictly convex is a sufficient condition for the supply curve being upward sloping. However, the requirement of $c(\cdot)$ being strictly convex is not always a reasonable assumption. Moreover, this assumption is not necessary. Recall that we have *Law of Supply*, which states that the firm's supply curve is weakly upward sloping without any other assumption.

In conclusion, we argue that the differentiability and concavity of $F(\cdot, \theta)$ are not indispensable for comparative statics analysis. By comparison, $F_{x\theta}(\cdot, \cdot) > 0$ captures the *complementarity* between x and θ . Naturally, we would like to ask whether $F_{x\theta}(\cdot, \cdot) > 0$ alone is sufficient for $x^*(\theta)$ to be increasing in θ . Our goal in this part is to formulate a discrete analogue of this condition and show that $F_{x\theta}(\cdot, \cdot) > 0$ is sufficient for $x^*(\theta)$ to be increasing in θ .

That $F(\cdot, \cdot)$ is twice *continuously* differentiable is too strong an assumption. For the general case, we introduce the following definition of (strict) increasing differences.

Definition 4.1.1: Increasing Differences

Let $F : X \times \Theta \rightarrow \mathbb{R}$, where $X, \Theta \subset \mathbb{R}$. We say that $F(\cdot, \cdot)$ has *increasing differences* in (x, θ) if for any $x, x' \in X$ and $\theta, \theta' \in \Theta$ such that $x' > x$ and $\theta' > \theta$, we have

$$F(x', \theta') - F(x, \theta') \geq F(x', \theta) - F(x, \theta)$$

If the inequalities are strict for all such $x, x' \in X$ and $\theta, \theta' \in \Theta$, $F(\cdot, \cdot)$ has *strictly increasing differences* in (x, θ) .

Intuitively, *increasing differences* captures the case where the *marginal value* of x is higher at a higher value of θ . In other words, there is *complementarity* between x and θ .

Proposition 4.1.2: Increasing Differences for Smooth Functions

Suppose $X = [\underline{x}, \bar{x}]$ and $\Theta = [\underline{\theta}, \bar{\theta}]$, where $X, \Theta \subset \mathbb{R}$.

1. If $F(\cdot, \cdot)$ is continuously differentiable in both x and θ , $F(\cdot, \cdot)$ has increasing differences in (x, θ) if and only if either of the two conditions holds:
 - $F_x(x, \cdot)$ is non-decreasing in θ for all x .
 - $F_\theta(\cdot, \theta)$ is non-decreasing in x for all θ .
2. If $F(\cdot, \cdot)$ is twice continuously differentiable in both x and θ , $F(\cdot, \cdot)$ has increasing differences in (x, θ) if and only if $F_{x\theta}(\cdot, \cdot) \geq 0$ for all (x, θ) .

There is one technical question we need to address. When $F(\cdot, \theta)$ is not strictly concave in x , the maximizer $x^*(\theta)$ need not be unique. We shall introduce two natural ways for comparison of sets.

Definition 4.1.3: Comparison of Sets

For any two sets A and B , we say that:

- $A \leq B$ in the *strong set order* if for any $a \in A$ and $b \in B$, we have $\min\{a, b\} \in A$ and $\max\{a, b\} \in B$.
- $A \leq B$ *pointwise* if for any $a \in A$ and $b \in B$, we have $a \leq b$.

Intuitively, $A \leq B$ in the strong set order allows for a common range of A and B , while for the disjoint parts, any element in A is strictly less than that in B . $A \leq B$ pointwise requires every element in B be no less than any element in A .

Theorem 4.1.4: Univariate Topkis' Theorem

Let $F : X \times \Theta \rightarrow \mathbb{R}$, where $X, \Theta \subset \mathbb{R}$, and $x^*(\theta) = \arg \max_{x \in X} F(x, \theta)$. Then for any $\theta' > \theta$,

1. If $F(\cdot, \cdot)$ has increasing differences in (x, θ) , then $x^*(\theta) \leq x^*(\theta')$ in the strong set order.
2. If $F(\cdot, \cdot)$ has strictly increasing differences in (x, θ) , then $x^*(\theta) \leq x^*(\theta')$ pointwise.

Proof for Theorem

Take any $x \in x^*(\theta)$ and $x' \in x^*(\theta')$. Suppose $x > x'$, then by revealed preference,

$$\begin{aligned} F(x, \theta) &\geq F(x', \theta) \\ F(x', \theta') &\geq F(x, \theta') \end{aligned}$$

By increasing differences,

$$F(x, \theta) - F(x', \theta) \leq F(x, \theta') - F(x', \theta')$$

Jointly we have:

$$\begin{aligned} 0 &\leq F(x, \theta) - F(x', \theta) \leq F(x, \theta') - F(x', \theta') \leq 0 \\ \implies \begin{cases} F(x, \theta) = F(x', \theta) \\ F(x, \theta') = F(x', \theta') \end{cases} &\implies \begin{cases} x \in x^*(\theta') \\ x' \in x^*(\theta) \end{cases} \end{aligned}$$

If $F(\cdot, \cdot)$ has strictly increasing differences in (x, θ) , then the combined inequality cannot hold, so it must be that $x^*(\theta) \leq x^*(\theta')$ pointwise. In particular, if $F(\cdot, \cdot)$ is twice continuously differentiable in both x and θ and $x^*(\theta)$ is single-valued, then $F_{x\theta}(\cdot, \cdot) > 0$ is sufficient for $x^*(\theta)$ to be (weakly) increasing in θ . ■

Example.

Again consider the firm's profit maximization problem. How would the firm's supply (correspondence) change with the output price p ?

The objective function is $F(y, p) = py - c(\mathbf{w}, y)$. $Y = [0, +\infty)$ and $P = [0, +\infty)$ are both intervals in $\mathbb{R}$. Moreover, $F_p(y, p) = y$, which is strictly increasing in y , so $F(\cdot, \cdot)$ has strictly increasing differences. Therefore, the firm's supply increases with the output price p pointwise.

Example.

Consider a monopolist that faces a downward sloping demand curve $Q^D(p)$. Now suppose the government levies a unit tax t on the firm. How would the before-tax price p received by the firm change with the unit tax t ?

The objective function is $F(p, t) = (p - t)Q^D(p) - c(Q^D(p))$. $P = [0, +\infty)$ and $T = [0, +\infty)$ are both intervals in $\mathbb{R}$. Moreover, $F_t(p, t) = -Q^D(p)$, which is strictly

increasing in p (since the demand curve is downward sloping), so $F(\cdot, \cdot)$ has strictly increasing differences in (p, t) . Consequently, the firm's before-tax price p increases with t pointwise.

Notice that whether $x^*(\theta)$ increases/decreases with the parameter θ is an *ordinal* property, while (strictly) increasing differences is still a *cardinal* property. Indeed, we know from the discussion on consumer theory that $\max_x F(x, \theta)$ and $\max_x G(x, \theta)$ have the same set of maximizers if $G = \varphi \circ F$ for $\varphi(\cdot)$ strictly increasing. Nevertheless, G having increasing differences in (x, θ) does not necessarily mean F having (strictly) increasing differences in (x, θ) . In other words, the requirement of (strictly) increasing differences is still too strong for monotone comparative statics. For our purpose, if we can find a positive and monotonic transformation φ such that $G = \varphi \circ F$ has increasing differences or strictly increasing differences in (x, θ) , then we know $x^*(\theta)$ increases with θ in the strong set order or pointwise.

Example.

Consider the effects of an increase in the market size on monopoly quantity (and monopoly price). Each consumer in the market has an identical inverse function given by $p^D(q)$. Suppose the number of consumers N is exogenously given, and that the firm's cost function is $c(Q)$, where $Q = Nq$ is the total quantity sold (i.e., the number of consumers times per unit purchase). Discuss how the firm's cost function $c(\cdot)$ would affect the optimal *per-consumer* quantity $q^*(N)$ as the number of consumers N increases.

Solution.

The objective function is

$$F(q, N) = N \cdot p^D(q) \cdot q - c(Nq).$$

Notice that it is hard to check increasing differences of $F(\cdot, \cdot)$. (You may try it yourself and find the process blocked by some terms that need additional information to push forward the computation.) Consider $G(q, N) = \frac{F(q, N)}{N}$. Then if $G(\cdot, \cdot)$ is twice continuously differentiable (which indeed can be relaxed), we have

$$\begin{aligned} G(q, N) &= \frac{F(q, N)}{N} = p^D(q) \cdot q - \frac{c(Nq)}{N} \\ G_N(q, N) &= -\frac{q \cdot c'(Nq) \cdot N - c(Nq)}{N^2} \\ G_{Nq}(q, N) &= -qc''(Nq) \end{aligned}$$

If $c(\cdot)$ is concave, then $G_{Nq}(q, N) \geq 0$ and $G(q, N)$ has increasing differences in (q, N) , so $q^*(\cdot)$ weakly increases with N . If $c(\cdot)$ is convex, then $G_{Nq}(q, N) \leq 0$ and $G(q, N)$ has increasing differences in $(q, -N)$, so $q^*(\cdot)$ weakly decreases with N .

4.2 Multivariate Comparative Statics

Consider a two-variable maximization problem:

$$(x_1^*(\theta), x_2^*(\theta)) = \arg \max_{(x_1, x_2) \in X \subset \mathbb{R}^2} F(x_1, x_2, \theta).$$

If we know F has (strictly) differences in (x_1, θ) , we cannot conclude that $x_1^*(\theta)$ is weakly increasing in θ unless x_2^* is independent of θ . Intuitively, as θ varies, change in θ has direct effect on x_1 , but there is also the effect of θ on x_2 and consequently on x_1 , which constitutes the indirect effect of θ on x_1 .

Recall from univariate Topkis's theorem that $A \leq B$ in the strong set order if for any $a \in A$ and $b \in B$, we have $\min\{a, b\} \in A$ and $\max\{a, b\} \in B$. In the multivariate case, technically the definition must be properly extended to answer

- For two vectors $\mathbf{x}$ and $\mathbf{y}$, how should we define $\min\{\mathbf{x}, \mathbf{y}\}$ and $\max\{\mathbf{x}, \mathbf{y}\}$?
- Under the proposed definition, is there any guarantee that $\min\{\mathbf{x}, \mathbf{y}\}$ and $\max\{\mathbf{x}, \mathbf{y}\}$ fall within the choice set X ?

For the first question, we introduce the definition of *meet* and *join*, the first of which sets the greatest lower bound of the two vectors, and the second of which sets the smallest upper bound of the two vectors.

Definition 4.2.1: Meet; Join

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

- *meet* of $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$\mathbf{x} \wedge \mathbf{y} = (\min\{x_1, y_1\}, \min\{x_2, y_2\}, \dots, \min\{x_n, y_n\}).$$

- *join* of $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$\mathbf{x} \vee \mathbf{y} = (\max\{x_1, y_1\}, \max\{x_2, y_2\}, \dots, \max\{x_n, y_n\}).$$

where $\mathbf{x} \wedge \mathbf{y}$ is called the greatest lower bound of $\mathbf{x}$ and $\mathbf{y}$, and $\mathbf{x} \vee \mathbf{y}$ is called the smallest upper bound of $\mathbf{x}$ and $\mathbf{y}$.

For the second concern, we delineate the boundary of our discussion by defining a structure called *sublattice*.

Definition 4.2.2: Sublattice

A set $X \subset \mathbb{R}^n$ is a *sublattice* if for any $\mathbf{x}, \mathbf{y} \in X$, both $\mathbf{x} \wedge \mathbf{y} \in X$ and $\mathbf{x} \vee \mathbf{y} \in X$.

$\mathbb{R}^n$ itself is a lattice, so any set in $\mathbb{R}^n$ is called a sublattice.

Example.

- $X = X_1 \times X_2 \times \dots \times X_n$, where $X_i \subset \mathbb{R}$, for $i = 1, 2, \dots, n$.
- $X = \{\mathbf{x} \in \mathbb{R}_+^n : \mathbf{p} \cdot \mathbf{x} \leq m\}$, where $\mathbf{p} \gg \mathbf{0}$ is the price vector and $m > 0$ is the income.

(NOT a sublattice)

- $X = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 \leq g(x_1), \text{ for } g(\cdot) \text{ strictly increasing}\}$.

Similar to properties of complementarity of two variables, in the multivariate case we introduce the definition of *supermodularity*.

Definition 4.2.3: Supermodularity

Let $F : X \rightarrow \mathbb{R}$, where $X \subset \mathbb{R}^n$ is a sublattice. F is *supermodular* if for any $\mathbf{x}, \mathbf{y} \in X$,

$$F(\mathbf{x} \wedge \mathbf{y}) + F(\mathbf{x} \vee \mathbf{y}) \geq F(\mathbf{x}) + F(\mathbf{y}).$$

Remark.

- Note when $X = X_1 \times X_2 \subset \mathbb{R}^2$, supermodularity is equivalent to increasing differences in (x_1, x_2) .
- More generally, when $X = X_1 \times X_2 \times \cdots \times X_n \subset \mathbb{R}^n$, **supermodularity is equivalent to increasing differences in (x_i, x_j) for all pairs of $i \neq j$.**

Putting all together, we have the following multivariate Topkis's Theorem.

Proposition 4.2.4: Multivariate Topkis's Theorem

Let $F : X \times \Theta \rightarrow \mathbb{R}$, where $X \subset \mathbb{R}^n$ is a sublattice and $\Theta \subset \mathbb{R}$. Consider $\mathbf{x}^*(\theta) = \arg \max_{\mathbf{x} \in X} F(\mathbf{x}, \theta)$. If F is supermodular, then for any $\theta' > \theta$ and $\mathbf{x} \in \mathbf{x}^*(\theta)$ and $\mathbf{x}' \in \mathbf{x}^*(\theta')$, we have

$$\begin{aligned} \mathbf{x} \wedge \mathbf{x}' &\in \mathbf{x}^*(\theta), \\ \mathbf{x} \vee \mathbf{x}' &\in \mathbf{x}^*(\theta'). \end{aligned}$$

Proof for Proposition.

Since X is a sublattice, $\mathbf{x} \wedge \mathbf{x}' \in X$ and $\mathbf{x} \vee \mathbf{x}' \in X$. By revealed preference,

$$\begin{cases} F(\mathbf{x}, \theta) \geq F(\mathbf{x} \wedge \mathbf{x}', \theta) \\ F(\mathbf{x}', \theta') \geq F(\mathbf{x} \vee \mathbf{x}', \theta') \end{cases} \implies F(\mathbf{x}, \theta) + F(\mathbf{x}', \theta') \geq F(\mathbf{x} \wedge \mathbf{x}', \theta) + F(\mathbf{x} \vee \mathbf{x}', \theta')$$

Since F is supermodular,

$$\begin{aligned} F(\mathbf{x}, \theta) + F(\mathbf{x}', \theta) &\leq F((\mathbf{x}, \theta) \wedge (\mathbf{x}', \theta')) + F((\mathbf{x}, \theta) \vee (\mathbf{x}', \theta')) \\ &= F(\mathbf{x} \wedge \mathbf{x}', \theta) + F(\mathbf{x} \vee \mathbf{x}', \theta') \end{aligned}$$

Jointly, it must be that

$$F(\mathbf{x}, \theta) + F(\mathbf{x}', \theta') = F(\mathbf{x} \wedge \mathbf{x}', \theta) + F(\mathbf{x} \vee \mathbf{x}', \theta')$$

Consequently, we have

$$\begin{aligned} F(\mathbf{x} \wedge \mathbf{x}', \theta) &= F(\mathbf{x}, \theta) \\ F(\mathbf{x} \vee \mathbf{x}', \theta') &= F(\mathbf{x}', \theta') \end{aligned}$$

which indicates that $\mathbf{x} \wedge \mathbf{x}' \in \mathbf{x}^*(\theta)$ and $\mathbf{x} \vee \mathbf{x}' \in \mathbf{x}^*(\theta')$.

Example.

Suppose a firm uses two inputs (L and K) to produce a single output y , and the production function is given by

$$y = f(L, K)$$

where $f(\cdot, \cdot)$ is twice continuously differentiable.

1. Discuss how the optimal demand for capital $K^*(w, r, p)$ would be affected by an increase in the rental rate of the capital r .
2. Discuss how the optimal demand for labor $L^*(w, r, p)$ would be affected by an increase in the rental rate of the capital r .

Solution.

$$\pi(L, K, r) = pf(L, K) - wL - rK.$$

Immediately, $\pi_{Lr} = 0$ and $\pi_{Kr} = -1$, π is supermodular in $(-K, r)$, so $K^*(w, r, p)$ weakly decreases with r . Next consider the indirect path of r to L through K .

- If $f_{LK} > 0$, π is supermodular in $(-L, -K, r)$, so $L^*(w, r, p)$ weakly decreases with r .
- If $f_{LK} < 0$, π is supermodular in $(L, -K, r)$, so $L^*(w, r, p)$ weakly increases with r .

Example.

Consider the other form of market power, a market with a single buyer and competitive supply. Suppose the inverse supply curve $p^S(q) \geq 0$ is strictly increasing. Let $v(q)$ be the buyer's value for quantity q . Consequently, if there are q units of transaction in the market,

- The producer surplus is

$$S(q) = \int_0^q [p^S(q) - p^S(t)] dt.$$

- The buyer's gain is

$$B(q) = v(q) - p^S(q)q.$$

1. Formulate both the buyer's and the social planner's optimization problems (no need to solve either of them).
2. How does the single buyer's optimal quantity q^B compare with that of the social planner q^* ? Show your argument and explain intuitively.

Solution.

1. The buyer's optimization problem is

$$\max_{q \geq 0} v(q) - p^S(q)q.$$

The social planner's optimization problem is

$$\max_{q \geq 0} v(q) - \int_0^q p^S(t) dt.$$

2. Consider the following optimization problem:

$$\max_{q \geq 0} w(q, \lambda) = B(q) + \lambda S(q)$$

This becomes the buyer's optimization problem when $\lambda = 0$, and the social planner's when $\lambda = 1$. If we could show how q monotonically change with λ , then we are done with this question. This is the univariate comparative statics problem, so we check the increasing differences of $w(q, \lambda)$:

$$\frac{\partial w(q, \lambda)}{\partial \lambda} = S(q)$$

Take any $q' \geq q \geq 0$. We have

$$\begin{aligned} S(q') - S(q) &= \int_0^{q'} [p^S(q') - p^S(t)] dt - \int_0^q [p^S(q) - p^S(t)] dt \\ &= \int_q^{q'} [p^S(q') - p^S(q)] dt + \int_0^q [p^S(q') - p^S(q)] dt \\ &= \int_q^{q'} [p^S(q') - p^S(q)] dt + [p^S(q') - p^S(q)] q \\ &\geq 0 \end{aligned}$$

where the last inequality holds because $p^S(\cdot)$ is upward sloping.

Thus, $w(q, \lambda)$ has increasing differences in (q, λ) . From Topkis's Theorem, $q^*(\lambda) < q^*(\lambda')$.

Similar to increasing differences, supermodularity is a *cardinal* property, which is again too strong. Indeed, the weaker requirement, *quasi-supermodularity* serves our purpose.

Definition 4.2.5: Quasi-Supermodularity

Let $F : X \rightarrow \mathbb{R}$, where $X \subset \mathbb{R}^n$ is a sublattice. F is *quasi-supermodular* if for any $\mathbf{x}, \mathbf{y} \in X$,

$$\begin{aligned} F(\mathbf{x}) \geq F(\mathbf{x} \wedge \mathbf{y}) &\implies F(\mathbf{x} \vee \mathbf{y}) \geq F(\mathbf{y}), \\ F(\mathbf{x}) > F(\mathbf{x} \wedge \mathbf{y}) &\implies F(\mathbf{x} \vee \mathbf{y}) > F(\mathbf{y}). \end{aligned}$$

Under such extension, the analysis can be further extended beyond the case of $X \subset \mathbb{R}^n$ and X forming a lattice structure.

Chapter 5

Uncertainty

Example.

Consider an investor who must decide how much of their initial wealth w to put into a risky asset. The risky asset can have any of the positive or negative rates of return r_i with probabilities p_i , $i = 1, 2, \dots, n$. Suppose the investor is an expected utility maximizer and their utility for x amount of money for sure can be represented by a twice continuously differentiable, strictly increasing and strictly concave utility function $u(x)$. Let a^* be the investor's optimal amount of money to put in the risky asset. Give a necessary and sufficient condition for the investor to have strict incentives to invest in the risky asset, that is, $a^* > 0$ and a^* is strictly preferred to $a = 0$.

Solution.

Since the investor is an expected utility maximizer, optimization problem is given by:

$$\max_{0 \leq a \leq w} U(a) = \sum_{i=1}^n p_i \cdot u(w + ar_i)$$

A sufficient condition for $a^* > 0$ is $U'(0) = u'(w) \cdot \sum_{i=1}^n p_i r_i > 0$, that is, the expected rate of return $\sum_{i=1}^n p_i r_i > 0$.

Next suppose $\sum_{i=1}^n p_i r_i \leq 0$. Since $u(\cdot)$ is strictly concave, $U''(a) = \sum_{i=1}^n p_i \cdot r_i^2 \cdot u''(w + ar_i) \leq 0$, which implies $U(\cdot)$ is also strictly concave. Consequently, $U'(a) \leq U'(0) \leq 0$ for $a \geq 0$. It follows that the condition is also necessary.

Convenient as the expected utility representation is, it is rather ad-hoc. Similar to the essence of previous works, we want to understand what restrictions on the agent's preference relation induce an expected utility representation.

5.1 Setups

Definition 5.1.1: Simple Lottery; Compound Lottery

Let $X = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ denote a finite set of *sure* outcomes (i.e., without uncertainty).

- A *simple lottery* $\mathbf{p} = p_1 \circ \mathbf{x}_1 + p_2 \circ \mathbf{x}_2 + \dots + p_n \circ \mathbf{x}_n$ ($p_1 + p_2 + \dots + p_n = 1$) is a probability distribution over a finite number of sure outcomes $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$. When there is no confusion, we will also use $\mathbf{p} = \{p_1, p_2, \dots, p_n\}$ to denote such a simple lottery.
- A *compound lottery* $\sum_{j=1}^k \alpha_j \mathbf{p}^j$ ($\alpha_1 + \alpha_2 + \dots + \alpha_k = 1$) is a probability distribution over a finite number of simple lotteries $\{\mathbf{p}^1, \mathbf{p}^2, \dots, \mathbf{p}^k\}$.

To facilitate our analysis, we introduce two assumptions with regard to the way in which utility maximizer regard the lotteries.

Assumption 5.1.2

1. The probabilities are **objective**: the probabilities are regarded as objective or exogenously given facts by the decision-maker (in contrast to subjective evaluation).
2. The decision-maker is a **consequentialist**: decision-makers only care about the outcomes, instead of how lotteries are mixed.

Under this assumption, the compound lottery $\sum_{j=1}^k \alpha_j \mathbf{p}^j$ is identical to the simple lottery it induces, i.e.,

$$\sum_{j=1}^k \alpha_j \mathbf{p}^j \iff \left(\sum_{j=1}^k \alpha_j p_1^j, \sum_{j=1}^k \alpha_j p_2^j, \dots, \sum_{j=1}^k \alpha_j p_n^j \right)$$

The first assumption can be relaxed.

Given the two assumptions, we can restrict our focus on the space of simple lotteries $\mathcal{P} = \Delta(X)$.

Definition 5.1.3: Space of Simple Lotteries

The space of simple lotteries, $\Delta(X)$, is defined as

$$\Delta(X) = \{(p_1, \dots, p_n) : p_i \geq 0 \ \forall i, p_1 + \dots + p_n = 1\}.$$

Consider the agent's preference relation $\succsim$ over $\mathcal{P}$. To ensure a utility representation $U : \mathcal{P} \rightarrow \mathbb{R}$, we maintain the three basic assumptions as in the certainty benchmark.

- **Completeness**: For any $\mathbf{p}^1, \mathbf{p}^2 \in \mathcal{P}$, either $\mathbf{p}^1 \succsim \mathbf{p}^2$ or $\mathbf{p}^2 \succsim \mathbf{p}^1$ (or both).
- **Transitivity**: For any $\mathbf{p}^1, \mathbf{p}^2, \mathbf{p}^3 \in \mathcal{P}$, if $\mathbf{p}^1 \succsim \mathbf{p}^2$ and $\mathbf{p}^2 \succsim \mathbf{p}^3$, then $\mathbf{p}^1 \succsim \mathbf{p}^3$.

- **Continuity:** For any two sequences $(p^i)_{i=1}^n, (q^i)_{i=1}^n \in \mathcal{P}$, if $p^i \succsim q^i$ ($\forall i = 1, 2, \dots, n$) and $\lim_{n \rightarrow \infty} p^i = p^*$, $\lim_{n \rightarrow \infty} q^i = q^*$, then $p^* \succsim q^*$.

Remark.

There is an alternative definition of continuity. Let $\succsim$ be a complete, transitive preference relation on $\mathcal{P}$. $\succsim$ is *continuous* if for any $p^1, p^2, p^3 \in \mathcal{P}$ such that $p^1 \succsim p^2 \succsim p^3$, there exists $t \in [0, 1]$ such that $p^2 \sim tp^1 + (1-t)p^3$.

To ensure an expected utility representation, we need an additional restriction.

Definition 5.1.4: Independence

A preference relation $\succsim$ on $\mathcal{P}$ satisfies *independence* if for any $p^1, p^2, p^3 \in \mathcal{P}$ and $t \in (0, 1)$,

$$p^1 \succsim p^2 \iff tp^1 + (1-t)p^3 \succsim tp^2 + (1-t)p^3$$

Intuitively, this assumption requires that the relative ranking of two lotteries will not be affected by the mixture with a third lottery. Notice that both $tp^1 + (1-t)p^3$ and $tp^2 + (1-t)p^3$ are *compound* lotteries, which should be understood as the simple lotteries they induce.

Moreover, if the preference relation $\succsim$ on $\mathcal{P}$ satisfies independence, then for any $p^1, p^2, p^3 \in \mathcal{P}$ and $t \in (0, 1)$, we have

- $p^1 \succ p^2 \iff tp^1 + (1-t)p^3 \succ tp^2 + (1-t)p^3$
- $p^1 \sim p^2 \iff tp^1 + (1-t)p^3 \sim tp^2 + (1-t)p^3$
- If $p^1 \succsim p^2$ and $p^3 \succsim p^4$, then $tp^1 + (1-t)p^3 \succsim tp^2 + (1-t)p^4$.

In summary, in this chapter of uncertainty, we need the following four assumptions to guarantee expected utility representation.

Assumption 5.1.5

- **Completeness:** For any $p^1, p^2 \in \mathcal{P}$, either $p^1 \succsim p^2$ or $p^2 \succsim p^1$ (or both).
- **Transitivity:** For any $p^1, p^2, p^3 \in \mathcal{P}$, if $p^1 \succsim p^2$ and $p^2 \succsim p^3$, then $p^1 \succsim p^3$.
- **Continuity:** For any two sequences $(p^i)_{i=1}^n, (q^i)_{i=1}^n \in \mathcal{P}$, if $p^i \succsim q^i$ ($\forall i = 1, 2, \dots, n$) and $\lim_{n \rightarrow \infty} p^i = p^*$, $\lim_{n \rightarrow \infty} q^i = q^*$, then $p^* \succsim q^*$.
- **Independence:** For any $p^1, p^2, p^3 \in \mathcal{P}$ such that $p^1 \succsim p^2 \succsim p^3$, there exists $t \in [0, 1]$ such that $p^2 \sim tp^1 + (1-t)p^3$.

5.2 Expected Utility Representation

Before starting the representation theorem, we first formalize what we mean by an expected utility function.

Definition 5.2.1: Expected Utility Form

A utility function $U : \mathcal{P} \rightarrow \mathbb{R}$ has an *expected utility form* (or a *von Neumann-Morgenstern expected utility function*) if there is an assignment of numbers $(u_1, u_2, \dots, u_n)$ to each of the n sure outcomes $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ such that for any $\mathbf{p} \in \mathcal{P}$,

$$U(\mathbf{p}) = \sum_{i=1}^n p_i u_i.$$

A sure outcome $\mathbf{x}_i$ can be represented by a simple lottery $\mathbf{e}_i = (0, \dots, 0, 1, 0, \dots, 0)$ and we have $U(\mathbf{e}_i) = u(\mathbf{x}_i) = u_i$. Notice that an expected utility function can be written as

$$U(\mathbf{p}) = \sum_{i=1}^n p_i U(\mathbf{e}_i).$$

Thus, $U(\mathbf{p})$ is linear in the probabilities. The following proposition reveals that this observation rings true more generally.

Proposition 5.2.2

A linear function $U : \mathcal{P} \rightarrow \mathbb{R}$ has an expected utility form if and only if it is *linear*, that is, for any k (simple) lotteries $\mathbf{p}^j \in \mathcal{P}$, ($j = 1, 2, \dots, k$) and associated probabilities $\alpha_j \geq 0$ with $\sum_{j=1}^k \alpha_j = 1$, we have

$$U\left(\sum_{j=1}^k \alpha_j \mathbf{p}^j\right) = \sum_{j=1}^k \alpha_j U(\mathbf{p}^j).$$

Proof for Proposition.

- “If” part

Let $\mathbf{p}^j = \mathbf{e}_j$ for all $j = 1, 2, \dots, k$. Thus, $\mathbf{p} = \sum_{j=1}^k \alpha_j \mathbf{p}^j = (\alpha_1, \alpha_2, \dots, \alpha_k)$. The condition of U being linear is then transformed to

$$U(\mathbf{p}) = \sum_{j=1}^k p_j U(\mathbf{e}_j) = \sum_{j=1}^k \alpha_j u_j.$$

This directly implies that U has an expected utility form.

- “Only if” part

– Since the decision-maker is a consequentialist, the compound lottery $\sum_{j=1}^k \alpha_j \circ \mathbf{p}^j$ is identical to the simple lottery it induces,

$$\alpha_1 \circ \mathbf{p}^1 + \alpha_2 \circ \mathbf{p}^2 + \dots + \alpha_k \circ \mathbf{p}^k \iff \left(\sum_{j=1}^k \alpha_j p_1^j, \sum_{j=1}^k \alpha_j p_2^j, \dots, \sum_{j=1}^k \alpha_j p_n^j \right)$$

- Since the utility function has the expected utility form, we have

$$U \left(\sum_{j=1}^k \alpha_j p_1^j, \sum_{j=1}^k \alpha_j p_2^j, \dots, \sum_{j=1}^k \alpha_j p_n^j \right) = \sum_{i=1}^n \left(\sum_{j=1}^k \alpha_j p_i^j \right) u_i$$

- On the other hand, $U(\mathbf{p}^j) = \sum_{i=1}^n p_i u_i$, so

$$\sum_{j=1}^k \alpha_j U(\mathbf{p}^j) = \sum_{j=1}^k \alpha_j \left(\sum_{i=1}^n p_i^j u_i \right)$$

- Jointly, we have proved the result:

$$\begin{aligned} U \left(\sum_{j=1}^k \alpha_j \mathbf{p}^j \right) &= U \left(\sum_{j=1}^k \alpha_j p_1^j, \sum_{j=1}^k \alpha_j p_2^j, \dots, \sum_{j=1}^k \alpha_j p_n^j \right) \\ &= \sum_{i=1}^n \left(\sum_{j=1}^k \alpha_j p_i^j \right) u_i \\ &= \sum_{j=1}^k \alpha_j U(\mathbf{p}^j) \end{aligned}$$

Theorem 5.2.3

A complete and transitive preference relation $\succsim$ on $\mathcal{P}$ satisfies continuity and independence if and only if it admits an expected utility representation $U : \mathcal{P} \rightarrow \mathbb{R}$.

Proof for Theorem

The “if” part is easy to verify. We mainly focus on the “only if” part.

Suppose the complete and transitive preference relation $\succsim$ satisfies continuity and independence. We construct an expected utility function $U(\mathbf{p}) = \sum_{i=1}^n p_i u_i$ that represents $\succsim$ as follows:

- First, with a slight abuse of notation, interpret the consequences $x_1, \dots, x_n$ as degenerate lotteries, so that x_i puts probability 1 on x_i and probability 0 on all other consequences. Then any lottery p can be viewed as a mixture of those degenerate lotteries: $p = \sum_i p_i x_i$.
- Label the consequences so that (when viewed as degenerate lotteries) x_1 is the best and x_n is the worst, i.e., $x_1 \succsim x_i \succsim x_n$ for all i .
- By continuity, for each consequence x_i , there is some $\lambda_i \in [0, 1]$ such that $x_i \sim \lambda_i x_1 + (1 - \lambda_i) x_n$.
- By independence (iterated n times for each consequence), for any lottery $p = \sum_i p_i x_i$,

$$p \sim \sum_i p_i (\lambda_i x_1 + (1 - \lambda_i) x_n) = \left(\sum_i p_i \lambda_i \right) x_1 + \left(1 - \sum_i p_i \lambda_i \right) x_n$$

- If $x_1 \sim x_n$ then by independence, for all p we have

$$p \sim \left(\sum_i p_i \lambda_i \right) x_1 + \left(1 - \sum_i p_i \lambda_i \right) x_1 = x_1$$

Hence, letting $u_i = 0$ for all i gives an expected utility representation of $\succsim$.

- If $x_1 \succ x_n$, then for $1 \geq \lambda \geq \lambda' \geq 0$, letting $\delta = \lambda - \lambda' \in [0, 1]$ and $\hat{p} = \frac{\lambda'}{1-\delta} x_1 + \frac{1-\lambda}{1-\delta} x_n$ and using the Independence Axiom,

$$\begin{aligned} \lambda x_1 + (1 - \lambda) x_n &= \delta x_1 + (1 - \delta) \hat{p} \\ &\succ \delta x_1 + (1 - \delta) \hat{p} \\ &= \lambda' x_1 + (1 - \lambda') x_n \end{aligned}$$

Thus, letting $u_i = \lambda_i$ yields an expected utility representation of $\succsim$.

5.3 Measures of Risk

5.3.1 Risk Attitudes

Let the set of sure outcomes $X = \mathbb{R}$. For any sure outcome $x \in X$, its utility is determined by utility function $u : X \rightarrow \mathbb{R}$. A lottery is fully characterized by a corresponding distribution function F , whose expected utility is then $U(F) = \int_X u(x) dF(x)$. For simplicity, suppose $u(\cdot)$ is strictly increasing and continuous and $U(F) = \mathbb{E}_F[u(x)] < +\infty$.

Here we naturally extend the notion of lottery to $X = \mathbb{R}$, the infinite set, and avoid any technical issues related to the definition of lotteries on finite-many sure outcomes.

Definition 5.3.1: Risk Averse

For any non-degenerate lottery F , define its average payoff δ_{E_F} as

$$\delta_{E_F} = \int_X x dF(x).$$

A decision-maker is strictly *risk-averse* if for any non-degenerate lottery F , the sure outcome δ_{E_F} is strictly preferred to the lottery F , i.e., $\delta_{E_F} \succ F$. Or in utility terms,

$$u(\delta_{E_F}) = u\left(\int_X x dF(x)\right) > U(F) = \int_X u(x) dF(x)$$

Corollary 5.3.2

A decision-maker is (strictly) risk-averse if and only if $u(\cdot)$ is (strictly) concave.

Intuitively, the marginal utility of an additional unit of money is decreasing for a risk-averse individual.

Risk-loving and *risk-neutral* attitudes can be similarly defined.

We have defined risk attitudes based on how agents evaluate the utility of averaged payoff

against the averaged utility. Apart from determining risk attitudes, we develop a cursory measure, certainty equivalent, to capture such “deviation”.

Definition 5.3.3: Certainty Equivalent

The *certainty equivalent* $c(F, u)$ of a money lottery is defined as

$$u(c(F, u)) = U(F).$$

Intuitively, $c(F, u)$ is the sure outcome perceived as equivalent to the risk lottery.

Certainty equivalent depends on initial wealth. For simplicity, we assume $w_0 = 0$, i.e., the agent does not have initial wealth; then $c(F, u) \leq \mathbb{E}[F]$ if and only if the agent is risk-averse.

5.3.2 Measures of Risk Aversion

We know that an agent is (strictly) risk-averse if and only if $u(\cdot)$ is (strictly) concave. Recall the cardinal interpretation of vNM expected utility. Can we use $u(\cdot)$ to quantify and then compare different levels of risk aversion? Therefore, we introduce the measures of risk aversion.

Consider a risk-averse agent who has an initial wealth w and faces a (small) fair gamble.

- Scenario 1: The gamble $\tilde{\varepsilon}$ is measured in terms of monetary unit. Then consider how large a has to be to make

$$u(w - a) = U(w + \tilde{\varepsilon}).$$

- Scenario 2: The gamble $\tilde{\delta}$ is measured in terms of percentage of the initial wealth. Then consider how large r has to be to ensure

$$u((1 - r)w) = U((1 + \tilde{\delta})w).$$

Since the agent is risk-averse, $a, r > 0$. Intuitively, a measures how much money the agent is willing to give up to avoid the gamble; r measures the percentage of initial wealth the agent is willing to give up to avoid the gamble. If the agent were risk-neutral, $a = r = 0$. As a result, given initial wealth w and the small fair gamble, both a and r measure the agent's level of risk aversion.

Coefficient of Absolute Risk Aversion When $\tilde{\varepsilon}$ is small, a is small. By Taylor expansion and definition of expected utility:

$$\begin{aligned}
 u(w - a) &\approx u(w) - u'(w) \cdot a \\
 U(w + \tilde{\varepsilon}) &= \mathbb{E}_{\varepsilon} [u(w + \varepsilon)] = \int_{\varepsilon} u(w + \varepsilon) d\varepsilon \\
 &\approx \int_{\varepsilon} \left[u(w) + u'(w) \cdot \varepsilon + \frac{1}{2} u''(w) \cdot \varepsilon^2 \right] d\varepsilon \\
 &= u(w) \int_{\varepsilon} 1 d\varepsilon + u'(w) \int_{\varepsilon} \varepsilon d\varepsilon + \frac{1}{2} u''(w) \int_{\varepsilon} \varepsilon^2 d\varepsilon \\
 &= u(w) + \frac{1}{2} u''(w) \cdot \text{Var}[\tilde{\varepsilon}]
 \end{aligned}$$

It follows that

$$\begin{aligned}
 u(w) - u'(w) \cdot a &= u(w) + \frac{1}{2} u''(w) \cdot \text{Var}[\tilde{\varepsilon}] \\
 \implies a &\approx -\frac{1}{2} \cdot \frac{u''(w)}{u'(w)} \cdot \text{Var}[\tilde{\varepsilon}]
 \end{aligned}$$

Definition 5.3.4: Coefficient of Absolute Risk Aversion

Suppose the agent has initial wealth $x > 0$, and the utility function $u : \mathbb{R} \rightarrow \mathbb{R}$ is twice continuously differentiable. Then the *coefficient of absolute risk aversion*, $A(x, u)$, is defined as

$$A(x, u) = -\frac{u''(x)}{u'(x)}.$$

Coefficient of Relative Risk Aversion Similarly, we use Taylor expansion and definition of expected utility to compute r :

$$\begin{aligned}
 u(w(1 - r)) &\approx u(w) - u'(w) \cdot (wr) \\
 U(w + w\tilde{\delta}) &= \mathbb{E}_G [u(w + w\delta)] = \int_{\delta} u(w + w\delta) dG(\delta) \\
 &\approx \int_{\delta} \left[u(w) + u'(w) \cdot (w\delta) + \frac{1}{2} u''(w) \cdot (w\delta)^2 \right] dG(\delta) \\
 &= u(w) \int_{\delta} 1 dG(\delta) + u'(w) \int_{\delta} w\delta dG(\delta) + \frac{1}{2} u''(w) \int_{\delta} (w\delta)^2 dG(\delta) \\
 &= u(w) + \frac{1}{2} u''(w) \cdot w^2 \text{Var}[\tilde{\delta}]
 \end{aligned}$$

It follows that

$$\begin{aligned}
 u((1 - r)w) &= U((1 + \tilde{\delta})w) \\
 \implies r &\approx -\frac{wu''(w)}{2u'(w)} \text{Var}[\tilde{\delta}]
 \end{aligned}$$

Definition 5.3.5: Coefficient of Relative Risk Aversion

Suppose the agent has initial wealth $x > 0$, and the utility function $u : \mathbb{R} \rightarrow \mathbb{R}$ is twice continuously differentiable. Then the *coefficient of relative risk aversion*, $R(x, u)$, is defined as

$$A(x, u) = -\frac{xu''(x)}{u'(x)}.$$

Proposition 5.3.6: Equivalence of Different Measures of Risk Aversion

Suppose utility functions u and v are strictly increasing and twice differentiable. The following definitions of an agent characterized by u being more risk averse than another agent characterized by v are equivalent:

1. For any lottery F and sure outcome δ_X , if $F \succsim_u \delta_X$, then $F \succsim_v \delta_X$.
2. For any lottery F , $c(F, u) \leq c(F, v)$.
3. The function u is “more concave” than v , that is, there exists some increasing and concave function g such that $u = g \circ v$.
4. $r(x) = \frac{u'(x)}{v'(x)}$ is non-increasing in x .
5. For any x , $A(x, u) = -\frac{u''(x)}{u'(x)} \geq -\frac{v''(x)}{v'(x)} = A(x, v)$.

Proof for Proposition.

- (1) $\iff$ (2)
 - (1) $\implies$ (2): By definition of certainty equivalent, $\delta_{c(F, u)} \sim_u F$. By condition (1), we have $\delta_{c(F, v)} \sim_v F \succsim_v \delta_{c(F, u)}$. Recall that $v(\cdot)$ is strictly increasing, so we have $c(F, u) \leq c(F, v)$.
 - (2) $\implies$ (1): If $\delta_{c(F, u)} \sim F \succsim_u \delta_x$, then by strictly-increasing property of u , it follows that $c(F, u) \geq x$. So $c(F, v) \geq c(F, u) \geq x \implies c(F, v) \geq x$. Again by v being strictly increasing, we have $c(F, v) \sim F \succsim_v \delta_x$.
- (2) $\iff$ (3)
 - Since u and v are strictly increasing, v^{-1} is well defined and $g = u \circ v^{-1}$ is strictly increasing.
 - Again by strict monotonicity of u , we have $u(c(F, u)) \leq u(c(F, v))$ for any lottery F .

$$u(c(F, u)) = U(F) = \int_X u(x) dF(x) = \int_X g(v(x)) dF(x)$$

$$u(c(F, v)) = g(v(c(F, v))) = g\left(\int_X v(x) dF(x)\right)$$
 - By Jensen’s inequality, $\int_X g(v(x)) dF(x) \leq g\left(\int_X v(x) dF(x)\right)$ if and only if g is concave.

- (3) $\iff$ (4)
 - Since u and v are strictly increasing, v^{-1} is well defined and $g = u \circ v^{-1}$ is strictly increasing.
 - When u and v are both differentiable, $u'(x) = g'(v(x)) \cdot v'(x) \implies \frac{u'(x)}{v'(x)} = g'(v(x))$.
 - Since v is strictly increasing, $\frac{u'(x)}{v'(x)}$ is non-increasing in x if and only if g' is non-increasing, that is, g is concave.
- (4) $\iff$ (5)
 - When $r(x) > 0$, $r(x)$ is non-increasing in x if and only if $\ln r(x) = \ln u'(x) - \ln v'(x)$ is non-increasing in x .
 - When u and v are twice differentiable,

$$(\ln r(x))' = \frac{u''(x)}{u'(x)} - \frac{v''(x)}{v'(x)}$$
 - It follows that $r(x) = \frac{u'(x)}{v'(x)}$ is non-increasing in x if and only if $A(x, u) \geq A(x, v)$.

Example.

Consider an investor who must decide how much of their initial wealth w to put into a risky asset. The risky asset can have any of the positive or negative rates of return r_i with probabilities p_i , $i = 1, 2, \dots, n$. Suppose the investor is an expected utility maximizer and their utility for x amount of money for sure can be represented by a twice continuously differentiable, strictly increasing and strictly concave utility function $u(x)$. Let a^* be the investor's optimal amount of money to put in the risky asset. Suppose $\sum_{i=1}^n p_i r_i > 0$.

1. Give a sufficient condition for $a^* < w$.
2. Suppose $a^* < w$ and that $A(x, u)$ is strictly decreasing in x , how would $a^*(w)$ change with w ?

5.4 Comparison of Risky Prospects

The earlier proposition shows we can use the coefficient of absolute risk aversion $A(x, u)$ to compare the risk attitudes of two individuals, as well as the risk attitudes of the same individual at two wealth levels. But how about the comparisons of two lotteries? In particular, when can we unambiguously say that $F \succsim G$ for any (risk-averse) decision-maker?

When comparing monetary lotteries, an agent would naturally consider expected payoffs and variations in payoffs. Intuitively, $F \succsim G$ unambiguously in one of the following two cases:

1. F yields a higher return than G with a higher probability.
2. F and G have the same expected return, but there is less variation in F .

Roughly speaking, we introduce first-order stochastic dominance for the first possibility, and second-order stochastic dominance for the second.

5.4.1 First-Order Stochastic Dominance

Proposition 5.4.1: First-Order Stochastic Dominance

The following two conditions are equivalent:

1. For any non-decreasing $u : \mathbb{R} \rightarrow \mathbb{R}$,

$$\int_X u(t) dF(t) \geq \int_X u(t) dG(t).$$

2. $F(x) \leq G(x)$ for almost every x .

In this case, F *first-order stochastically dominates* G .

Proof for Proposition.

Assume that u is continuously differentiable, and $u'(\cdot) > 0$. Let $X = [\underline{x}, \bar{x}]$.

$$\begin{aligned} & \int_{\underline{x}}^{\bar{x}} u(t) dF(t) - \int_{\underline{x}}^{\bar{x}} u(t) dG(t) \\ &= \left[u(t) F(t) \right]_{\underline{x}}^{\bar{x}} - \int_{\underline{x}}^{\bar{x}} u'(t) F(t) dt - \left[u(t) G(t) \right]_{\underline{x}}^{\bar{x}} + \int_{\underline{x}}^{\bar{x}} u'(t) G(t) dt \\ &= \left[(u(\underline{x}) - u(\bar{x})) - \int_{\underline{x}}^{\bar{x}} u'(t) F(t) dt \right] - \left[(u(\underline{x}) - u(\bar{x})) - \int_{\underline{x}}^{\bar{x}} u'(t) G(t) dt \right] \\ &= \int_{\underline{x}}^{\bar{x}} u'(t) (G(t) - F(t)) dt \end{aligned}$$

It follows that $\int_{\underline{x}}^{\bar{x}} u(t) dF(t) \geq \int_{\underline{x}}^{\bar{x}} u(t) dG(t)$ if and only if $G(x) \geq F(x)$ almost everywhere. ■

Notice that the first condition says that any decision-maker who prefers more money would prefer lottery F over G , regardless of the agent's risk attitude, let alone the specific mathematical expression of utility function.

5.4.2 Second-Order Stochastic Dominance

Definition 5.4.2: Mean-Preserving Spread

Let X, Y be two random variables with distribution functions F and G , respectively. Then G is a *mean-preserving spread* of F if $Y = X + \tilde{\varepsilon}$, where $\mathbb{E}[\tilde{\varepsilon}|X] = 0$.

Intuitively, the two lotteries F and G yield the same expected return, but G involves more risk.

Proposition 5.4.3: Second-Order Stochastic Dominance

Suppose $\mathbb{E}[F] = \mathbb{E}[G]$. Then the following three conditions are equivalent:

1. For any (non-decreasing) concave $u : \mathbb{R} \rightarrow \mathbb{R}$,

$$\int_X u(t) dF(t) \geq \int_X u(t) dG(t)$$

2. For almost every $x \in X$,

$$\int_{-\infty}^x F(t) dt \leq \int_{-\infty}^x G(t) dt$$

3. G is a mean-preserving spread of F .

In this case, F second-order stochastically dominates G .

Proof for Proposition.

For simplicity, suppose $X = [\underline{x}, \bar{x}]$ and u is twice continuously differentiable.

- (1) $\iff$ (2)

- By earlier calculation, $\int_{\underline{x}}^{\bar{x}} u(t) dF(t) - \int_{\underline{x}}^{\bar{x}} u(t) dG(t) = \int_{\underline{x}}^{\bar{x}} u'(t) (G(t) - F(t)) dt$.
- Again by integration by parts and noticing $\mathbb{E}[F] = \mathbb{E}[G]$, we have:

$$\int_{\underline{x}}^{\bar{x}} u(t) dF(t) - \int_{\underline{x}}^{\bar{x}} u(t) dG(t) = \int_{\underline{x}}^{\bar{x}} u''(x) \left[\int_{\underline{x}}^x G(t) dt - \int_{\underline{x}}^x F(t) dt \right] dx$$

- Because u is concave, $u'' \leq 0$. It follows that $\int_{\underline{x}}^{\bar{x}} u(t) dF(t) \geq \int_{\underline{x}}^{\bar{x}} u(t) dG(t)$ if and only if $\int_{\underline{x}}^x F(t) dt \geq \int_{\underline{x}}^x G(t) dt$ almost everywhere.
- (1) $\iff$ (3)
- We will only show (3) $\implies$ (1).
- By the law of iterated expectation,

$$\mathbb{E}_G[u(Y)] = \mathbb{E}_F[\mathbb{E}_G[u(Y)|X]]$$

- Since $u(\cdot)$ is concave, by Jensen's inequality,

$$\mathbb{E}_G[u(Y)] = \mathbb{E}_F[\mathbb{E}_G[u(Y)|X]] \leq \mathbb{E}_F[u(\mathbb{E}_G[Y|X])] = \mathbb{E}_F[u(X)]$$

The first condition says that any risk-averse decision-maker would prefer lottery F over G . Additionally, when $\mathbb{E}[F] = \mathbb{E}[G]$, we actually do not need $u(\cdot)$ to be non-decreasing. If $\mathbb{E}[F] \neq \mathbb{E}[G]$, then the second and the third conditions are still equivalent, but we actually need $u(\cdot)$ to be non-decreasing and concave.

5.4.3 Likelihood Ratio Dominance

Definition 5.4.4: Likelihood Ratio Dominance

Let F and G be two distribution functions with common support $[\underline{x}, \bar{x}]$. Suppose the density functions exist and are given by f and g , respectively. F dominates G in the likelihood ratio order if $\frac{f(x)}{g(x)}$ is non-decreasing in x .

Intuitively, when F dominates G in the likelihood ratio order, F puts higher probabilities on higher returns compared with G .

Proposition 5.4.5: Likelihood Ratio Dominance Implies First-Order Stochastic Dominance

If F dominates G in the likelihood ratio order, then F first-order stochastically dominates G .

Proof for Proposition.

Since $F(\cdot)$ and $G(\cdot)$ are both non-decreasing, in order to show that $F(x) \leq G(x)$, it suffices to show that

$$\frac{F(x)}{1 - F(x)} \leq \frac{G(x)}{1 - G(x)}, \forall x \in (\underline{x}, \bar{x}).$$

We have

$$\begin{aligned} \frac{F(x)}{1 - F(x)} &= \frac{\int_{\underline{x}}^x f(t) dt}{\int_x^{\bar{x}} f(t) dt} \\ &= \frac{\int_{\underline{x}}^x \frac{f(t)}{g(t)} \cdot g(t) dt}{\int_x^{\bar{x}} \frac{f(t)}{g(t)} \cdot g(t) dt} \end{aligned}$$

Since $\frac{f(x)}{g(x)}$ is non-decreasing,

$$\begin{aligned} \forall t \in [\underline{x}, x], \quad \frac{f(t)}{g(t)} &\leq \frac{f(x)}{g(x)} \\ \forall t \in [x, \bar{x}], \quad \frac{f(t)}{g(t)} &\geq \frac{f(x)}{g(x)} \end{aligned}$$

Therefore, we have

$$\frac{F(x)}{1 - F(x)} = \frac{\int_{\underline{x}}^x \frac{f(t)}{g(t)} \cdot g(t) dt}{\int_x^{\bar{x}} \frac{f(t)}{g(t)} \cdot g(t) dt} \leq \frac{\frac{f(x)}{g(x)} \cdot \int_{\underline{x}}^x g(t) dt}{\frac{f(x)}{g(x)} \cdot \int_x^{\bar{x}} g(t) dt} = \frac{\int_{\underline{x}}^x g(t) dt}{\int_x^{\bar{x}} g(t) dt} = \frac{G(x)}{1 - G(x)}$$

5.5 Comparative Statics Under Risk

Our earlier discussions on comparative statics did not specifically take uncertainty into consideration. Whether and how would the comparative statics results carry over to choices under uncertainty?

The first result is that, if a random variable is “introduced” into a supermodular function, the expected utility function still preserves supermodularity.

Lemma 5.5.1: Expected Utility Function Preserves Supermodularity

Let $X \subset \mathbb{R}^n$ be a sublattice and $T \subset \mathbb{R}$. Suppose $u : X \times T \rightarrow \mathbb{R}$ is supermodular in $\mathbf{x}$. Then for any distribution function F on T , the function $U : X \rightarrow \mathbb{R}$ defined by

$$U(\mathbf{x}) = \int_t u(\mathbf{x}, t) dF(t)$$

is supermodular in $\mathbf{x}$.

Proof for Lemma

Take any $\mathbf{x}, \mathbf{x}' \in X$. we have

$$\begin{aligned} U(\mathbf{x} \wedge \mathbf{x}') + U(\mathbf{x} \vee \mathbf{x}') &= \int_t [u(\mathbf{x} \wedge \mathbf{x}', t) + u(\mathbf{x} \vee \mathbf{x}', t)] dF(t) \\ &\geq \int_t [u(\mathbf{x}, t) + u(\mathbf{x}', t)] dF(t) \\ &= U(\mathbf{x}) + U(\mathbf{x}') \end{aligned}$$

where the inequality in the second line follows from the supermodularity of $u(\mathbf{x}, t)$ in $\mathbf{x}$. Consequently by definition of supermodularity, $U(\cdot)$ is supermodular in $\mathbf{x}$. ■

The second result states that, first-order stochastic dominance preserves increasing differences.

Lemma 5.5.2: First-Order Stochastic Dominance Preserves Increasing Differences

Let $X \subset \mathbb{R}^n$ be a sublattice and $T, \Theta \subset \mathbb{R}$. Suppose $u : X \times T \rightarrow \mathbb{R}$ has increasing differences in $(\mathbf{x}, t)$, and $\{F_\theta\}_{\theta \in \Theta}$ is a family of distribution function on T such that $F_\theta \geq_{FOSD} F_{\theta'}$ if $\theta > \theta'$. Then the function $U : X \times \Theta \rightarrow \mathbb{R}$ defined by

$$U(\mathbf{x}, \theta) = \int_t u(\mathbf{x}, t) dF_\theta(t)$$

has increasing differences in $(\mathbf{x}, \theta)$.

Proof for Lemma

Take any $\mathbf{x} > \mathbf{x}'$ and $\theta > \theta'$, we have

$$\begin{aligned} U(\mathbf{x}, \theta) - U(\mathbf{x}', \theta) &= \int_t [u(\mathbf{x}, t) - u(\mathbf{x}', t)] dF_\theta(t) \\ &\geq \int_t [u(\mathbf{x}, t) - u(\mathbf{x}', t)] dF_{\theta'}(t) \\ &= U(\mathbf{x}, \theta') - U(\mathbf{x}', \theta') \end{aligned}$$

where the inequality holds because, $\delta(t) := u(\mathbf{x}, t) - u(\mathbf{x}', t)$ is non-decreasing in t by increasing differences of $u(\cdot, \cdot)$ in $(\mathbf{x}, t)$ and $F_\theta \geq_{FOSD} F_{\theta'}$. Consequently, $U(\mathbf{x}, \theta)$ has increasing differences in $(\mathbf{x}, \theta)$. ■

Intuitively, θ can be interpreted as a signal of t , indicating the distribution of t .

Proposition 5.5.3: Comparative Statics Under Risk

Let $X \subset \mathbb{R}^n$ be a sublattice and $T, \Theta \subset \mathbb{R}$. Suppose $u : X \times T \rightarrow \mathbb{R}$ is supermodular in $\mathbf{x}$ and has increasing differences in $(\mathbf{x}, t)$. Further suppose $\{F_\theta\}_{\theta \in \Theta}$ is a family of distribution function on T such that $F_\theta \geq_{FOSD} F_{\theta'}$ if $\theta > \theta'$. Then for the function $U : X \times \Theta \rightarrow \mathbb{R}$ defined by

$$U(\mathbf{x}, \theta) = \int_t u(\mathbf{x}, t) dF_\theta(t),$$

the set of maximizers, $\arg \max_{\mathbf{x} \in X} U(\mathbf{x}, \theta)$, is non-decreasing in θ in the strong set order.

Proof for Proposition.

- Fixing any θ , by the lemma of expected utility function preserving supermodularity, $U(\mathbf{x}, \theta)$ is supermodular in $\mathbf{x}$.
- By the lemma of first-order stochastic dominance preserving increasing differences, $U(\mathbf{x}, \theta)$ has increasing differences in $(\mathbf{x}, \theta)$.
- Since $X \times \Theta$ forms a product set, $U(\mathbf{x}, \theta)$ is supermodular in $(\mathbf{x}, \theta)$.
- By the multivariate Topkis' theorem, the set of maximizers, $\arg \max_{\mathbf{x} \in X} U(\mathbf{x}, \theta)$, is non-decreasing in θ in the strong set order.

Example.

Suppose a monopolist faces an uncertain demand and must make a production decision prior to learning the realized demand for its product. The monopolist does learn some information about demand prior to choosing its output. Formally, suppose the inverse demand function is $p(q, t) = \hat{p}(q) + t$, where $\hat{p}(q)$ is the estimated inverse demand, and t is a random noise. The ex-post profit of the firm is therefore

$$\pi(q, t) = p(q, t) \cdot q - c(q)$$

with $c(q)$ being the monopolist's cost function. The monopolist observes a signal θ that is informative about the parameter t . Specifically, suppose the distribution of t conditional on θ is $F_\theta(\cdot)$, and $F_\theta \geq_{FOSD} F_{\theta'}$ for $\theta > \theta'$. Determine how would the monopolist's optimal output $q^*(\theta)$ change with the observed signal θ .

Solution.

The monopolist solves the following maximization problem:

$$\max_{q \geq 0} \Pi(q, \theta) = \int_t \pi(q, t) dF_\theta(t)$$

Notice that $\frac{\partial \pi(q, t)}{\partial t} = q$, increasing in q . It follows that $\pi(q, t)$ has increasing differ-

ences in (q, t) . Since $F_\theta \geq_{FOSD} F_{\theta'}$ for $\theta > \theta'$, $\Pi(q, \theta)$ has increasing differences in (q, θ) and $q^*(\theta)$ is non-decreasing in θ in the strong set order.

Example.

Consider an agent with an initial wealth w , who faces a random loss $\tilde{l} \in [0, w]$. To counter the potential loss, the agent may purchase any fraction of insurance $y \in [0, 1]$. Specifically, if the agent purchases y unit of insurance, then they pay a total price of yp upfront, and get paid yl if they incur a loss of l . Suppose the agent is an expected utility maximizer and has a utility function $u(x)$ for x amount of money for sure. Further suppose $u(\cdot)$ is twice continuously differentiable with $u'(\cdot) > 0$ and $u''(\cdot) < 0$.

1. First suppose $p \leq \mathbb{E}[\tilde{l}]$. Show the optimal amount of insurance $y^* = 1$.
2. Next suppose $p > \mathbb{E}[\tilde{l}]$. Show the optimal amount of insurance $y^* < 1$.
3. Let $A(x, u)$ be the coefficient of absolute risk aversion. Write down its expression and show that if $A(x, u) = c_0$ (a constant), then the optimal amount of insurance y^* is independent of w .
4. Now suppose $A(x, u)$ strictly decreases with x . Show the optimal amount of insurance y^* is non-increasing in the agent's initial wealth w .

Solution.

1. Since the agent is an expected utility maximizer, their choice problem is given by

$$\max_{y \in [0, 1]} U(y) = \mathbb{E}[u(w - yp - (1 - y)l)]$$

Simple calculation shows

$$\begin{aligned} U'(y) &= \mathbb{E}[(l - p) \cdot u'(w - l + (l - p)y)] \\ U''(y) &= \mathbb{E}[(l - p)^2 \cdot u''(w - l + (l - p)y)] \leq 0 \end{aligned}$$

It follows that $U'(y) \geq U'(1) = u'(w - p) \left(\mathbb{E}[\tilde{l}] - p \right) \geq 0$, with strict inequality as long as $p < \mathbb{E}[\tilde{l}]$. Consequently, $y^* = 1$.

2. When $p > \mathbb{E}[\tilde{l}]$, $U'(1) = u'(w - p) \left(\mathbb{E}[\tilde{l}] - p \right) < 0$, and $U''(y) \leq 0$, we have $y^* < 1$.

3. By definition, $A(x, u) = -\frac{u''(x)}{u'(x)}$.

Take any $w_1 < w_2$, and let

$$\begin{aligned} v_1(x) &= u(w_1 + x) \\ v_2(x) &= u(w_2 + x) \end{aligned}$$

Then at initial wealth w_i ($i = 1, 2$), the agent's maximization problem is given by

$$\max_{y \in [0,1]} V_i(y) = \mathbb{E}[v_i(-l + (l-p)y)]$$

From the previous two questions, it suffices to focus on the case of $p > \mathbb{E}[\tilde{l}]$.

By earlier characterization, $A(x, u) = c_0$ implies $\frac{v'_1(x)}{v'_2(x)} = c_1$, which is also a constant.

Simple calculation shows

$$\begin{aligned} V'_i(y) &= \mathbb{E}[(l-p)v'_i(-l + (l-p)y)] \\ V''_i(y) &= \mathbb{E}[(l-p)^2 v''_i(-l + (l-p)y)] \leq 0 \end{aligned}$$

If $V'_1(0) \leq 0$, then $V'_2(0) \leq 0$, so $y_1^* = y_2^* = 0$. Otherwise, if $V'_1(y^*) = 0$, then $V'_2(y^*) = 0$; by $V''_i(\cdot) < 0$ for $i = 1, 2$, $V'_i(y)$ strictly decreases with y and achieves maximum at y , so $y_1^* = y_2^* = y^*$.

4. Suppose by contradiction that $y_2^* > y_1^* \geq 0$. By optimality of y_2^* ,

$$\int_0^w v'_2(-l + (l-p)y_2^*)(l-p) dF(l) = 0.$$

By our earlier characterization, $A(x, u)$ strictly decreases with x implies $\frac{v'_1(x)}{v'_2(x)}$ is non-increasing in x .

Consequently, we have

$$\begin{aligned} V'_1(y_2^*) &= \int_0^p v'_1(-l + (l-p)y_2^*)(l-p) dF(l) \\ &\quad + \int_p^w v'_1(-l + (l-p)y_2^*)(l-p) dF(l) \\ &= \int_0^p \frac{v'_1(-l + (l-p)y_2^*)}{v'_2(-l + (l-p)y_2^*)} v'_2(-l + (l-p)y_2^*)(l-p) dF(l) \\ &\quad + \int_p^w \frac{v'_1(-l + (l-p)y_2^*)}{v'_2(-l + (l-p)y_2^*)} v'_2(-l + (l-p)y_2^*)(l-p) dF(l) \\ &\geq \frac{v'_1(-p)}{v'_2(-p)} \int_0^w v'_2(-l + (l-p)y_2^*)(l-p) dF(l) \\ &= 0 \end{aligned}$$

Given $V''_i(\cdot) < 0$ and our assumption of $y_2^* > y_1^* \geq 0$, we have $V'_1(y_1^*) > V'_1(y_2^*) \geq 0$, which is a contradiction.

Chapter 6

General Equilibrium

Regardless what perspective we take to analyze the market, the central idea is invariant: the market should clear. Typically, we differentiate between partial equilibrium and general equilibrium. From the perspective of partial equilibrium, we suppose the prices of all other goods are held fixed, and then analyze the equilibrium in a single market. From the general equilibrium's perspective, in contrast, we extend our focus to all markets and require all markets clear simultaneously.

Superficial as the partial equilibrium may seem when juxtaposed with general equilibrium, the approach of partial equilibrium analysis is still powerful typically in cases where

- The prices of the goods in all other markets are held fixed; or more precisely, the effect on the prices of all other goods can be neglected.
- No wealth effect of the market of interest.

Example.

There are n towns in total, with n being a large number. Each town has an identical price-taking firm that produces a single consumption good with the production function $y = f(l)$, where $f'(l) > 0$ and $f''(l) < 0$. The single consumption good is traded in the national market. Suppose there are L units of inelastic labor supply in total. Workers can move freely across towns to seek the highest possible wage, which are regarded as complete information. Normalize $p_c = 1$ and let w_i denote the equilibrium wage in town i . Now suppose that town 1 levies a **small** tax $t > 0$ on firm 1 for each unit of labor hired. Which group of individual(s) will bear the tax burden?

Solution.

- Equilibrium without tax
 - Whether we adopt the partial or general equilibrium approach, the competitive equilibrium without the labor tax is identical.
 - Since workers can move freely and $f'(l) > 0$ and $f''(l) < 0$, we must have

$$\begin{cases} l^* = \frac{1}{n} \cdot L \\ w_1 = w_2 = \dots = w_n = f'(l^*) = f'\left(\frac{L}{n}\right) \end{cases}$$

- Partial equilibrium in town 1 with tax
 - Assuming the wage rate in other markets are unaffected, we must have $w_1(t) = w_0 + t$.
 - At the new equilibrium, $f'(l_1(t)) = w_1(t) = w_0 + t$.
 - Hence we can see that firm 1 bears all the tax burden.
- General equilibrium with town 1 levied with tax
 - Let $l_1(t)$ denote the equilibrium amount of labor in town 1 with a t unit tax, and $l_{-1}(t)$ denote the equilibrium amount of labor in any other town when town 1 imposes a t unit tax and $w(t)$ the equilibrium wage rate received by the worker. When all labor markets clear, we should have:

$$\begin{aligned} & \begin{cases} l_1(t) + (n-1)l_{-1}(t) = L \\ f'(l_1(t)) = w_1(t) = w(t) + t \\ f'(l_{-1}(t)) = w(t) \end{cases} \\ \Rightarrow & \begin{cases} f'(L - (n-1)l_{-1}(t)) = w(t) + t \\ f'(l_{-1}(t)) = w(t) \end{cases} \\ \Rightarrow & \begin{cases} -(n-1)l'_{-1}(t) \cdot f''(L - (n-1)l_{-1}(t)) = w'(t) + 1 \\ l'_{-1}(t) f''(l_{-1}(t)) = w'(t) \end{cases} \end{aligned}$$

- Set $t \rightarrow 0$, so we have $l_{-1}(0) = \frac{1}{n} \cdot L$. From the two equations above we obtain $w'(0) = -\frac{1}{n}$.
- Let $\Pi(w(t))$ denote the total profit of the firms when the wage rate is $w(t)$ and $\pi(w)$ the profit of a firm paying wage w . Naturally, $\Pi(w(t)) = \pi(w(t) + t) + (n-1)\pi(w(t))$, and

$$\begin{aligned} & \frac{\partial \Pi(w(t))}{\partial t} = (w'(t) + 1) \cdot \pi'(w(t) + t) + (n-1)w'(t) \cdot \pi'(w(t)) \\ \Rightarrow & \frac{\partial \Pi(w(t))}{\partial t} \Big|_{t \rightarrow 0} = (w'(0) + 1) \cdot \pi'(w(0)) + (n-1)w'(0) \cdot \pi'(w(0)) = 0 \end{aligned}$$

- From quantitative analysis we can see that, the firms as a whole do not bear the tax burden and the workers bear all the tax burden.
- Intuitively, it must be the workers that bear all the tax burden. Even though the labor supply for any firm is perfectly elastic, the **total** labor supply is **perfectly inelastic**. Small as the impact on the wages in other towns, it is not negligible. Indeed, the labor supply for any given firm is perfectly elastic, so we cannot ignore the general equilibrium effect.

6.1 Pure Exchange Economy

Traditionally in analysis of a market, there are three central activities of interest: consumption, production and trade. While in pure exchange economy, each agent has an endowment that they can trade with, and there is no production; we only care about the exchange of goods.

Example.

Consider a perfectly competitive economy with two agents ($i = A, B$) and two goods ($j = 1, 2$). Suppose the agents' preference relations and initial endowments are given by

$$u^A(x_1, x_2) = x_1^2 x_2 \text{ with } e^A = (1, 2)$$

$$u^B(x_1, x_2) = x_1 x_2^2 \text{ with } e^B = (2, 1)$$

Can we come up with a price vector (p_1, p_2) that clear both markets?

Solution.

The key idea is that, each agent has their own income, which is endogenously given by the equilibrium price vector.

Suppose there is an equilibrium price vector (p_1, p_2) , then agent A 's "income" is $m^A = p_1 + 2p_2$ and agent B 's "income" is $m^B = 2p_1 + p_2$. From this we can pin down the optimal individual demands

$$\begin{aligned} x^A &= \left(\frac{2(p_1 + 2p_2)}{3p_1}, \frac{1(p_1 + 2p_2)}{3p_2} \right) \\ x^B &= \left(\frac{1(2p_1 + p_2)}{3p_1}, \frac{2(2p_1 + p_2)}{3p_2} \right) \end{aligned}$$

In equilibrium, market demand equals market endowment for either good, respectively:

$$\frac{2(p_1 + 2p_2)}{3p_1} + \frac{1(2p_1 + p_2)}{3p_1} = 3$$

$$\frac{1(p_1 + 2p_2)}{3p_2} + \frac{2(2p_1 + p_2)}{3p_2} = 3$$

From either equation, we have $\frac{p_1^*}{p_2^*} = 1$. (Otherwise if there comes with a contradiction, the equilibrium cannot exist.)

From this example, we obtain four observations in this two-agent, two-good pure exchange economy:

1. There is an equilibrium price vector that clears both markets.
2. Only the relative price matters in equilibrium.
3. When one market clears, the other market clears simultaneously (and miraculously).
4. The equilibrium allocation is Pareto efficient.

6.1.1 Setups

Model Preliminaries

- $I = \{1, 2, \dots, n\}$ agents and $J = \{1, 2, \dots, m\}$ goods.
- $\succsim^i$: preference relation of agent i .
- $\mathbf{e}^i = \{e_1^i, e_2^i, \dots, e_m^i\}$ initial endowment of agent i (property rights are well-defined and no co-ownership).
- Denote the economy $\mathcal{E} = \{\succsim^i, \mathbf{e}^i\}_{i \in I}$.

Assumption 6.1.1: Market Structure

1. Perfect and complete information.
2. Perfectly competitive markets.
 - Agents are price-takers.
 - Prices are linear.
3. Goods are perfectly divisible.

Assumption 6.1.2: Agents' Preferences and Endowments

For any agent $i \in I$:

1. The preference relation $\succsim^i$ is rational (complete and transitive) and continuous.
2. The preference relation $\succsim^i$ is monotonic.
3. The preference relation $\succsim^i$ is (weakly) convex.
4. $e_j^i > 0$, for all $j \in J$.

Notice that by the first assumption, the utility representation of $\succsim^i$, say $u^i(\cdot)$, is guaranteed for any $i \in I$, so we can alternatively denote the economy as $\mathcal{E} = (u^i(\cdot), \mathbf{e}^i)_{i \in I}$.

6.1.2 Walrasian Equilibrium

Definition 6.1.3: Walrasian Equilibrium

A *Walrasian Equilibrium* for a perfectly competitive pure exchange economy $\mathcal{E} = (u^i(\cdot), \mathbf{e}^i)_{i \in \mathcal{I}}$ is a price vector $\mathbf{p} \geq \mathbf{0}$ and an allocation $(\mathbf{x}^i)_{i \in \mathcal{I}}$ such that:

1. Utility maximization: Agent $i \in \mathcal{I}$ solves:

$$\max_{\mathbf{x}^i \geq \mathbf{0}} u^i(\mathbf{x}^i) \text{ s.t. } \mathbf{p} \cdot \mathbf{x}^i \leq \mathbf{p} \cdot \mathbf{e}^i$$

2. Market clearing for all goods:

$$\sum_{i \in \mathcal{I}} \mathbf{x}^i(\mathbf{p}) = \sum_{i \in \mathcal{I}} \mathbf{e}^i$$

Under Walrasian equilibrium, all markets clear at the same time. A Walrasian equilibrium specifies both *equilibrium price vector* and *equilibrium allocation*.

For simplicity, suppose for any $i \in \mathcal{I}$, $\mathbf{x}^i(\mathbf{p})$ is always unique. This is guaranteed if we assume each $\succsim_i$ is strictly convex and $\mathbf{p} \gg \mathbf{0}$.

Definition 6.1.4: Excess Demand

The (aggregate) *excess demand* for good j is given by

$$z_j(\mathbf{p}) = \sum_{i \in \mathcal{I}} x_j^i(\mathbf{p}) - \sum_{i \in \mathcal{I}} e_j^i$$

With this formulation, a Walrasian equilibrium is achieved when each agent maximize their utility subject to budget constraint prescribed by their endowment, and

$$z_j(\mathbf{p}) = 0, \forall j \in \mathcal{J}$$

Proposition 6.1.5: Properties of Excess Demand

Let $\mathcal{E}$ be a perfectly competitive pure exchange economy with each agent's preference relation being rational, continuous and monotonic, then the aggregate excess demand function $\mathbf{z}(\mathbf{p})$ satisfies:

1. **Homogeneity of degree 0:** $z_j(\mathbf{p})$ is homogeneous of degree 0 in $\mathbf{p}$, for any $j \in \mathcal{J}$.
2. **Walras' law:** $\mathbf{p} \cdot \mathbf{z}(\mathbf{p}) = 0$.

Proof for Proposition.

- Homogeneous of degree 0
 - $\mathbf{x}^i(\mathbf{p})$ solves $\max_{\mathbf{x}^i \geq \mathbf{0}} u^i(\mathbf{x}^i)$ s.t. $\mathbf{p} \cdot \mathbf{x}^i \leq \mathbf{p} \cdot \mathbf{e}^i$.
 - For any $t > 0$, the solution must be the same as: $\max_{\mathbf{x}^i \geq \mathbf{0}} u^i(\mathbf{x}^i)$ s.t. $(t\mathbf{p}) \cdot \mathbf{x}^i \leq$

$(t\mathbf{p}) \cdot \mathbf{e}^i$. Hence, $\mathbf{x}^i(\mathbf{p}) = \mathbf{x}^i(t\mathbf{p})$.

– For any $t > 0$, $z_j(t\mathbf{p}) = \sum_{i=1}^n x_j^i(t\mathbf{p}) - \sum_{i=1}^n e_j^i = \sum_{i=1}^n x_j^i(\mathbf{p}) - \sum_{i=1}^n e_j^i = z_j(\mathbf{p})$.

• Walras' law

– Budget constraint must be binding for any agent $i \in \mathcal{I}$, $\mathbf{p} \cdot \mathbf{x}^i(\mathbf{p}) = \mathbf{p} \cdot \mathbf{e}^i$. (This must hold because of assumption of monotonicity of preference relation, which is stronger than locally non-satiation).

– It follows that

$$\begin{aligned} \mathbf{p} \cdot \mathbf{z}(\mathbf{p}) &= \mathbf{p} \cdot \left(\sum_{i=1}^n \mathbf{x}^i(\mathbf{p}) - \sum_{i=1}^n \mathbf{e}^i \right) \\ &= \sum_{i=1}^n (\mathbf{p} \cdot \mathbf{x}^i(\mathbf{p}) - \mathbf{p} \cdot \mathbf{e}^i) \\ &= 0 \end{aligned}$$

The following proposition, the existence of Walrasian equilibrium, is characterized as the cornerstone of modern economics.

Proposition 6.1.6: Existence of Walrasian Equilibrium

Let $\mathcal{E} = (u^i(\cdot), \mathbf{e}^i)_{i \in \mathcal{I}}$ be a perfectly competitive pure exchange economy that satisfies all the four assumptions on agents' preferences and endowments. Then a Walrasian equilibrium exists.

Proof for Proposition.

We give proof for the case of $m = 2$ and $\succsim^i$ being strictly convex for any $i \in \mathcal{I}$. Given rationality, continuity and strict convexity of $\succsim^i$, $\mathbf{x}^i(p_1, p_2)$ is always unique for any $\mathbf{p} \gg \mathbf{0}$. When $m = 2$, Walras' law states that

$$p_1 z_1(p_1, p_2) + p_2 z_2(p_1, p_2) = 0.$$

Then a Walrasian equilibrium is given by $z_1(p_1, p_2) = 0$ or $z_2(p_1, p_2) = 0$. By monotonicity and strictly convexity of preference relation, the only possibility is $p_1, p_2 > 0$ in equilibrium, so we can normalize $p_1 = 1$ or $p_2 = 1$.

By theorem of the maximum, given rationality, continuity, monotonicity and strict convexity of preference relation, $x_i^1(p_1, 1)$ is continuous for any $i \in \mathcal{I}$. Hence, $z_1(p_1, 1)$ is continuous on $(0, +\infty)$. Given monotonicity and strict convexity, $u^i(\cdot)$ is strongly monotone. Therefore, $\lim_{p_1 \rightarrow 0+} x_1^i(p_1, 1) = +\infty$ and $\lim_{p_1 \rightarrow 0+} z_1(p_1, 1) = +\infty$. Symmetrically, we have $\lim_{p_2 \rightarrow 0+} x_2^i(1, p_2) = +\infty$ and $\lim_{p_2 \rightarrow 0+} z_2(1, p_2) = +\infty$. By continuity, $\exists 0 < \underline{p} < \bar{p}$ such that $z_1(\underline{p}, 1) > 0 > z_1(\bar{p}, 1)$. By the intermediate value theorem, $\exists p_1^* \in (\underline{p}, \bar{p})$ such that $z_1(p_1^*, 1) = 0$.

6.2 Allocation

Recap on what we have learned in intermediate microeconomics:

- First theorem of welfare economics: Any Walrasian equilibrium is Pareto efficient.
- Second theorem of welfare economics: By adjusting the initial endowments, any Pareto efficient allocation can be supported by a Walrasian equilibrium.

Before we delve deep into discussions on allocation, a rigorous definition of feasible allocation is helpful.

Definition 6.2.1: Feasible Allocation

For a pure exchange economy $\mathcal{E}$, an allocation $(\mathbf{x}^i)_{i=1}^n$ with $\mathbf{x}^i \geq \mathbf{0}$ for all i is *feasible* if for any good j ,

$$\sum_{i=1}^n x_j^i \leq \sum_{i=1}^n e_j^i.$$

Feasible allocation requires that, the total allocation for any good cannot be more than the total initial endowment.

Definition 6.2.2: Pareto Efficient Allocation

Given an economy $\mathcal{E}$, a feasible allocation $(\mathbf{x}^i)_{i=1}^n$ is (strongly) *Pareto efficient* if there is no other feasible allocation $(\mathbf{w}^i)_{i=1}^n$ such that $\mathbf{w}^i \succsim^i \mathbf{x}^i$ for all $i \in \mathcal{I}$, with $\mathbf{w}^k \succ^k \mathbf{x}^k$ for some $k \in \mathcal{I}$.

Under Pareto efficient allocation, there is no way to make any agent strictly better off without harming some other agent(s).

Definition 6.2.3: Core Allocation

A feasible allocation $(\mathbf{x}^i)_{i=1}^n$ is *in the core* of $\mathcal{E}$ if there is no group $\mathcal{I}_0 \subseteq \mathcal{I}$ and an alternative allocation $(\mathbf{w}^i)_{i=1}^n$ such that:

1. For any $j \in \mathcal{J}$, $\sum_{i \in \mathcal{I}_0} w_j^i \leq \sum_{i \in \mathcal{I}_0} e_j^i$.
2. $\mathbf{w}^i \succsim^i \mathbf{x}^i$ for all $i \in \mathcal{I}_0$, with $\mathbf{w}^k \succ^k \mathbf{x}^k$ for some $k \in \mathcal{I}_0$.

Core allocation requires that, no group of agents would simultaneously form a blocking coalition in which none is made worse off and some strictly better off. Notice that an allocation being in the core implies it being Pareto efficient, but the converse is not necessarily true. Indeed, when examining Pareto efficiency, we are checking the possibility of grand blocking coalition.

Theorem 6.2.4: Strengthened First Theorem of Welfare Economics

Let $\mathcal{E}$ be a perfectly competitive pure exchange economy in which each agent's preference relation is rational, continuous and concave. Denote S_P as the set of Pareto efficient allocations, S_C the set of core allocations, and S_W the set of Walrasian equilibrium allocations. Then

$$S_W \subseteq S_C \subseteq S_P$$

Proof for Theorem

By definition, $S_C \subseteq S_P$, so it suffices to show $S_W \subseteq S_C$. Suppose not, then there exists a Walrasian equilibrium $(\mathbf{p}; (\mathbf{x}^i)_{i \in \mathcal{I}})$, a group $\mathcal{I}_0 \subseteq \mathcal{I}$ and an alternative allocation $(\mathbf{w}^i)_{i \in \mathcal{I}}$ such that:

$$\begin{cases} \mathbf{w}^i \succsim^i \mathbf{x}^i, & \text{for all } i \in \mathcal{I}_0 \\ \mathbf{w}^k \succ^k \mathbf{x}^k, & \text{for some } k \in \mathcal{K}_0 \subseteq \mathcal{I}_0 \end{cases}$$

By direct revealed preference,

$$\begin{cases} \mathbf{p} \cdot \mathbf{w}^k > \mathbf{p} \cdot \mathbf{x}^k, & \text{for all } k \in \mathcal{K}_0 \\ \mathbf{p} \cdot \mathbf{w}^i \geq \mathbf{p} \cdot \mathbf{x}^i, & \text{for all } i \in \mathcal{I}_0 \setminus \mathcal{K}_0 \end{cases}$$

Summing over $i \in \mathcal{I}_0$, we have

$$\begin{aligned} \mathbf{p} \cdot \sum_{i \in \mathcal{I}_0} \mathbf{w}^i &> \mathbf{p} \cdot \sum_{i \in \mathcal{I}_0} \mathbf{x}^i = \mathbf{p} \cdot \sum_{i \in \mathcal{I}_0} \mathbf{e}^i \\ \implies \sum_{i \in \mathcal{I}_0} \mathbf{w}^i &\not\leq \sum_{i \in \mathcal{I}_0} \mathbf{e}^i \end{aligned}$$

Remark.

- $S_W \subseteq S_P$ is indicated in *First Theorem of Welfare Economics*.
- $S_W \subseteq S_C$ implies that any Walrasian equilibrium is not only Pareto efficient, but also fair, which is a justification for the price mechanism.
- The strengthened First Theorem of Welfare Economics need very minimal requirements on the agents' preference relations, but assumptions on perfect competition, complete information, no externality and complete markets are indispensable.

Theorem 6.2.5: Second Theorem of Welfare Economics

Let $\mathcal{E}$ be a perfectly competitive pure exchange economy in which each agent's preference relation is rational, continuous and concave. If $\mathbf{x}^i \gg \mathbf{0}$ is Pareto efficient, for all i , then there exists an initial endowment $(\mathbf{e}^i)_{i \in \mathcal{I}}$ and a price vector $\mathbf{p} \geq \mathbf{0}$ such that $(\mathbf{p}; (\mathbf{x}^i)_{i \in \mathcal{I}})$ is a Walrasian equilibrium given these endowments.

However, the second theorem of welfare economics is less exciting, because the government can neither arbitrarily redistribute initial endowments, nor know each agent's preferences in the economy.

6.3 General Equilibrium with Production

6.3.1 Setups

Recall that in producer theory, we used production sets and ownership shares to describe the firms and their production technologies.

Suppose there are K firms ($\mathcal{K} = \{1, 2, \dots, K\}$) in the economy, each firm $k \in \mathcal{K}$ with its production set Y^k . Let $\alpha^{ki} \geq 0$ be agent i 's ownership share of firm k . The production economy can then be described as

$$\mathcal{E} = \left(\left(u^i, e^i, (\alpha^{ki})_{k \in \mathcal{K}} \right)_{i \in \mathcal{I}}, (Y^k)_{k \in \mathcal{K}} \right).$$

Definition 6.3.1: Walrasian Equilibrium with Production

A **Walrasian Equilibrium** for a perfectly competitive production economy $\mathcal{E} = \left(\left(u^i, e^i, (\alpha^{ki})_{k \in \mathcal{K}} \right)_{i \in \mathcal{I}}, (Y^k)_{k \in \mathcal{K}} \right)$ is a vector $(p, (x^i)_{i \in \mathcal{I}}, (y^k)_{k \in \mathcal{K}})$ such that:

1. **Profit maximization:** Firm k solves

$$\max_{y^k \in Y^k} p \cdot y^k$$

2. **Utility maximization:** Agent i solves

$$\max_{x^i \geq 0} u^i(x^i) \text{ s.t. } p \cdot x^i \leq p \cdot e^i + \sum_{k \in \mathcal{K}} \alpha^{ik} p \cdot y^k$$

3. **Market clearing for all goods:**

$$\sum_{i \in \mathcal{I}} x^i(p) = \sum_{i \in \mathcal{I}} e^i + \sum_{k \in \mathcal{K}} y^k(p)$$

That is, in Walrasian equilibrium all markets clear at the same time, with all consumers (also shareholders of firms) and firms optimize their goals simultaneously.

The essence of Walrasian equilibrium with production is the same as that without production. But for concerns of technical issues, we introduce three additional assumptions on production technology to pave way for the theorem of existence of Walrasian equilibrium with production.

Assumption 6.3.2: Production Technology

For each firm $k \in \mathcal{K}$,

1. $Y^k \neq \emptyset$ is **closed and convex**.
2. **Shutdown and free disposal**: that is, $\mathbf{0} \in Y^k$, and $\mathbf{y} \in Y^k$ implies $\mathbf{y}' \in Y^k$, for all $\mathbf{y}' \leq \mathbf{y}$.
3. **Irreversibility**: Let $Y = \bigcup_{k \in \mathcal{K}} Y^k$, then $Y \cap (-Y) = \{\mathbf{0}\}$.

Remark.

- $Y^k \neq \emptyset$ is innocuous, otherwise since firm k is allowed to produce nothing, we can just discard Y^k if $Y^k = \emptyset$.
- Irreversibility says that the production cannot be completely reversed.

6.3.2 Existence of Walrasian Equilibrium with Production**Theorem 6.3.3: Existence of Walrasian Equilibrium with Production**

Let $\mathcal{E} = \left((u^i, e^i, (\alpha^{ki})_{k \in \mathcal{K}})_{i \in \mathcal{I}}, (Y^k)_{k \in \mathcal{K}} \right)$ be a perfectly competitive production economy satisfying all the assumptions on preference relation and endowments, and those on production technology. Then a Walrasian equilibrium $(\mathbf{p}, (\mathbf{x}^i)_{i \in \mathcal{I}}, (\mathbf{y}^k)_{k \in \mathcal{K}})$ exists.

Remark.

- This proposition of existence is the extension of the existence result on pure exchange economies.
- The first and second theorem of welfare economics also generalize to economics with production.

Recall that when a production technology exhibits constant returns to scale, the firm's profit must be either 0 or $+\infty$. For market clearing to make sense, it must be that each firm is making 0 profit.

For simplicity, consider an economy with two goods. The production technology can linearly transform b units of good 1 into (at most) c units of good 2. The production set can then be depicted as

$$Y = \left\{ (y_1, y_2) \mid y_1 \leq 0, y_2 \geq 0, y_2 \leq -\frac{c}{b}y_1 \right\}.$$

Alternatively, we can present the production technology with the vector $a = \left(-1, \frac{c}{b} \right)$.

If a linear activity production $a = (a_1, a_2)$ is ever used in a Walrasian equilibrium (i.e., $\lambda^* > 0$), then it must be that (because the firm is making zero profit)

$$(p_1^*, p_2^*) \cdot (a_1, a_2) = 0.$$

Example.

Consider a perfectly competitive economy $\mathcal{E}$ with two goods (1 and 2) and two agents (A and B). The agents' utility functions and initial endowments are as follows:

$$u^A(x_1, x_2) = x_1 x_2, \mathbf{e}^A = (1, 0)$$

$$u^B(x_1, x_2) = x_1 x_2^2, \mathbf{e}^B = (0, 1)$$

First suppose the economy is pure exchange with no production.

1. Derive the set of Pareto efficient allocations.
2. Derive the set of core allocations.
3. Derive a Walrasian equilibrium $((p_1, p_2); (x_1^A, x_2^A, x_1^B, x_2^B))$.

Next we introduce production: Suppose there is a perfectly competitive firm which can transform one unit of good 1 into (at most) one unit of good 2, i.e., $\mathbf{a} = (-1, 1)$.

4. Derive a Walrasian equilibrium with production $((p_1, p_2, \lambda); (x_1^A, x_2^A, x_2^B, x_2^B), \lambda)$.

Solution.

1. Since both agents' preference relations are monotonic, any Pareto efficient allocation must satisfy

$$\mathbf{x}^A + \mathbf{x}^B = \mathbf{e}^A + \mathbf{e}^B$$

Clearly, $((1, 1), (0, 0))$ and $((0, 0), (1, 1))$ are Pareto efficient.

$((1, 0), (0, 1))$ or $((0, 1), (1, 0))$ are not necessarily Pareto efficient, though we cannot make any agent better off material-wise without making the other worse off material-wise. However, agents' well-being is measured in terms of magnitude of utilities.

Apart from corner solutions, by the idea of no gain from trade in equilibrium, we must have

$$|MRS^A| = |MRS^B|$$

Therefore, the set of efficient allocations is given by

$$S_P = \{(x_1^A, x_2^A, x_1^B, x_2^B) \geq \mathbf{0} : x_1^A x_2^B = 2x_2^A x_1^B, x_1^A + x_1^B = x_2^A + x_2^B = 1\}.$$

It has been checked that $((1, 1), (0, 0))$ and $((0, 0), (1, 1))$ are included in the set S_P .

2. When there are only two agents, the only subgroups other than the grand coalition are $\{1\}$ and $\{2\}$. In other words, on the basis of Pareto efficiency, the additional

requirement for an allocation to be in the core is that no agent is made worse in any allocation compared to their own endowment, which is also termed *individual rationality*. Since the initial endowment is the worst possible allocation for each agent, the set of core allocations $S_C = S_P$.

3. Suppose there is an equilibrium price vector (p_1, p_2) . Then agent A's "income" $m^A = p_1$ and agent B's "income" $m^B = p_2$. The optimal individual demands are

$$\begin{aligned}(x_1^A, x_2^A) &= \left(\frac{1}{2}, \frac{p_1}{2p_2}\right) \\ (x_1^B, x_2^B) &= \left(\frac{p_2}{3p_1}, \frac{2}{3}\right)\end{aligned}$$

In equilibrium, market demand equals total endowments for each good.

$$\begin{cases} \frac{1}{2} + \frac{p_2}{3p_1} = 1 \\ \frac{p_1}{2p_2} + \frac{2}{3} = 1 \end{cases} \implies \frac{p_1^*}{p_2^*} = \frac{2}{3}$$

Hence, a Walrasian equilibrium is $(\mathbf{p} = (2, 3), \mathbf{x}^A = (\frac{1}{2}, \frac{1}{3}), \mathbf{x}^B = (\frac{1}{2}, \frac{2}{3}))$.

Here "a Walrasian equilibrium" is emphasized because only relative prices matter in Walrasian equilibrium; theoretically there could be infinitely-many equilibria in terms of absolute prices.

4. Suppose the production technology is used in equilibrium, then we must have $(p_1, p_2) \cdot \mathbf{a} = \mathbf{0}$, that is, $p_1 = p_2 \iff \frac{p_1}{p_2} = 1$.

In equilibrium, market demand equals market supply for each good.

$$\begin{cases} \frac{1}{2} + \frac{1}{3} = 1 - \lambda \\ \frac{1}{2} + \frac{2}{3} = 1 + \lambda \end{cases} \implies \lambda^* = \frac{1}{6}$$

A Walrasian equilibrium is given by $(\mathbf{p}^* = (1, 1), \mathbf{x}^A = (\frac{1}{2}, \frac{1}{2}), \mathbf{x}^B = (\frac{1}{3}, \frac{2}{3}), \lambda^* = \frac{1}{6})$.